GRAND CANYON NATIONAL PARK
WORLD HERITAGE SITE

REHABILITATE NORTH RIM INFRASTRUCTURE

VOLUME 9 – POWER DISTRIBUTION – NORTH RIM

VOLUME #	DESCRIPTION
VOLUME 0	GENERAL
VOLUME 1	PIPELINES - NORTH RIM
VOLUME 2	PIPELINES - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 3	PIPELINES - NORTH RIM TO SUPAI TUNNEL
VOLUME 4	PIPELINES - ROARING SPRINGS CAVE TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 5	ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 6	NORTH RIM WATER TREATMENT PLANT
VOLUME 7	NOT USED
VOLUME 8	NORTH RIM WASTEWATER COLLECTION SYSTEM
VOLUME 9	POWER DISTRIBUTION - NORTH RIM
VOLUME 10	POWER DISTRIBUTION - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 11	POWER DISTRIBUTION - ROARING SPRINGS PUMPHOUSE (RSPH) TO MANZANITA
VOLUME 12	POWER SYSTEM - COTTONWOOD CAMPGROUND

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
1	090G001	GENERAL	SHEET INDEX
2	091X101	POWER DISTRIBUTION	SITE DEMOLITION PLAN
3	091X102	POWER DISTRIBUTION	DEMOLITION PLAN NORTH RIM LODGE AREA
4	091X103	POWER DISTRIBUTION	DEMOLITION PLAN CONCESSIONAIRE AREA
5	091X104	POWER DISTRIBUTION	DEMOLITION PLAN GAS STATION AREA
6	091X105	POWER DISTRIBUTION	DEMOLITION PLAN ENTRANCE STATION AREA
7	091X601	POWER DISTRIBUTION	ONE-LINE DIAGRAM DEMO
8	091X602	POWER DISTRIBUTION	ONE-LINE DIAGRAM DEMO
9	091X603	POWER DISTRIBUTION	ONE-LINE DIAGRAM DEMO
10	091E101	POWER DISTRIBUTION	SITE PLAN
11	091E102	POWER DISTRIBUTION	REPLACEMENT (ESB-15, SR7, T-16)
12	091E103	POWER DISTRIBUTION	REPLACEMENT (T-62, T-63, ESB-22, ESB-23)
13	091E104	POWER DISTRIBUTION	REPLACEMENT (ESB-17, ESB-18)
14	091E105	POWER DISTRIBUTION	REPLACEMENT (T-38)
15	091E501	POWER DISTRIBUTION	SHUNT REACTOR DETAILS
16	091E502	POWER DISTRIBUTION	MEDIUM VOLTAGE EQUIPMENT INSTALLATION TABLES
17	091E601	POWER DISTRIBUTION	ONE-LINE DIAGRAM MODIFIED
18	091E602	POWER DISTRIBUTION	ONE-LINE DIAGRAM MODIFIED
19	091E603	POWER DISTRIBUTION	ONE-LINE DIAGRAM MODIFIED
20	092X101	POWER DISTRIBUTION CENTER	DEMOLITION PLAN POWER DELIVERY CENTER AREA
21	092X102	POWER DISTRIBUTION CENTER	DEMOLITION PLAN
22	092X601	POWER DISTRIBUTION CENTER	GENERATOR BUILDING DEMOLITION ONE-LINE DIAGRAM
23	092C101	POWER DISTRIBUTION CENTER	SITE PLAN
24	092S001	POWER DISTRIBUTION CENTER	GENERAL NOTES
25	092S101	POWER DISTRIBUTION CENTER	GROUND LEVEL PLAN
26	092S102	POWER DISTRIBUTION CENTER	ROOF FRAMING PLAN
27	092S301	POWER DISTRIBUTION CENTER	SECTIONS
28	092A101	POWER DISTRIBUTION CENTER	ARCHITECTURAL PLANS AND DETAILS
29	092D101	POWER DISTRIBUTION CENTER	FUEL BULK TANK PLAN
30	092D102	POWER DISTRIBUTION CENTER	GENERATOR ROOM PIPING PLAN
31	092D301	POWER DISTRIBUTION CENTER	FUEL BULK TANK SECTION
32	092D302	POWER DISTRIBUTION CENTER	FUEL SYSTEM MECHANICAL DETAILS
33	092D501	POWER DISTRIBUTION CENTER	FUEL SYSTEM MECHANICAL DETAILS
34	092D502	POWER DISTRIBUTION CENTER	FUEL SYSTEM TANK AREA LABELS
35	092D503	POWER DISTRIBUTION CENTER	FUEL SYSTEM ELECTRICAL DETAILS
36	092D504	POWER DISTRIBUTION CENTER	FUEL SYSTEM HAZARDOUS CLASSIFIED AREA DETAIL
37	092D601	POWER DISTRIBUTION CENTER	FUEL SYSTEM DAY TANK & GENSET FLOW SCHEMATIC
38	092D602	POWER DISTRIBUTION CENTER	FUEL SYSTEM TANK LAYOUT FLOW SCHEMATIC
39	092M101	POWER DISTRIBUTION CENTER	HVAC PLAN
40	092M301	POWER DISTRIBUTION CENTER	MECHANICAL SECTIONS
41	092E101	POWER DISTRIBUTION CENTER	ELECTRICAL SITE PLAN
42	092E102	POWER DISTRIBUTION CENTER	GENERATOR BUILDING FLOOR PLAN
43	092E600	POWER DISTRIBUTION CENTER	GENERATOR BUILDING POWER DISTRIBUTION FUNCTIONAL DIAGRAM
44	092E601	POWER DISTRIBUTION CENTER	GENERATOR BUILDING ONE-LINE DIAGRAM
45	092E602	POWER DISTRIBUTION CENTER	GENERATOR BUILDING SWITCHGEAR ELEVATION
46	092E603	POWER DISTRIBUTION CENTER	GENERATOR BUILDING PLC ONE-LINE DIAGRAM
47	092E604	POWER DISTRIBUTION CENTER	GENERATOR BUILDING PANEL SCHEDULES
48	092E605	POWER DISTRIBUTION CENTER	GENERATOR BUILDING PANEL SCHEDULES
49	092E701	POWER DISTRIBUTION CENTER	GENERATOR BUILDING PHOTOGRAPHS
50	092Y601	POWER DISTRIBUTION CENTER	SCADA NETWORK BLOCK DIAGRAM
51	092Y620	POWER DISTRIBUTION CENTER	GENERATOR NETWORK ENCLOSURE LAYOUT

REGISTERED PROFESSIONAL ENGINEER (ELECTRONICS) CERTIFICATE NO. 75014 TOU VANG Date Signed

REGISTERED PROFESSIONAL ENGINEER (CIVIL) CERTIFICATE NO. 41809 RYAN RHOADES Date Signed ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD N. LYNCH TECH. REVIEW: M. GOUGH DATE: AUG 2026	SUB SHEET NO. 090 G001	TITLE OF SHEET GENERAL SHEET INDEX NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 1 OF 51

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GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. REFER TO OVERALL NORTH RIM POWER DISTRIBUTION SYSTEM ONE-LINE DRAWINGS 091E601 – 091E603 FOR ADDITIONAL INFORMATION ON POWER SYSTEM IMPROVEMENTS.

C. TRANSFORMER T-67 IS LOCATED AT THE ROARING SPRINGS PUMPHOUSE, AND IS NOT SHOWN ON THIS DRAWING. SEE VOLUME 5 FOR T-67 TRANSFORMER REPLACEMENT DETAILS.

D. SEE SURVEY CONTROL PLAN IN VOLUME 0.

E. COORDINATE MEDIUM VOLTAGE POWER INSTALLATION AND DEMO SEQUENCING TO MINIMUM SYSTEM POWER OUTAGES.

300 0 300 600 1" = 300'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: MAY 2026	SUB SHEET NO. 091 X101	TITLE OF SHEET POWER DISTRIBUTION SITE DEMOLITION PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 2 OF 51

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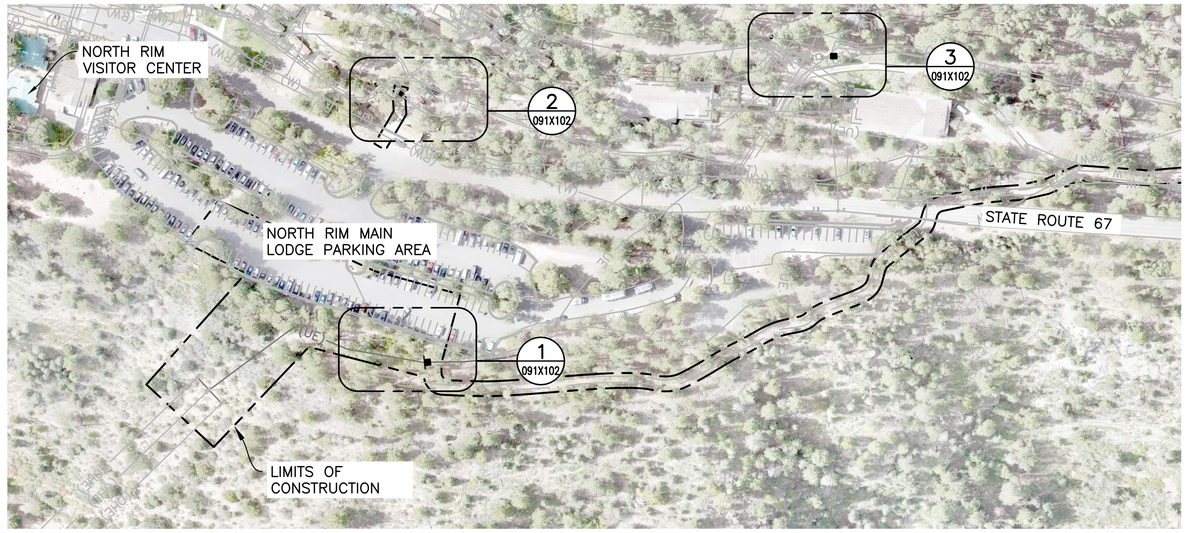
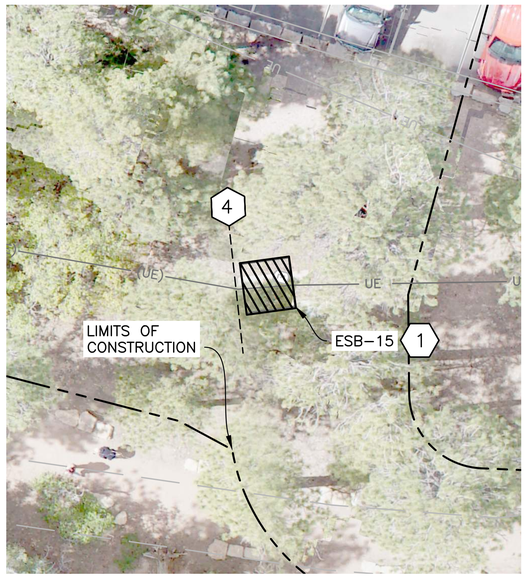


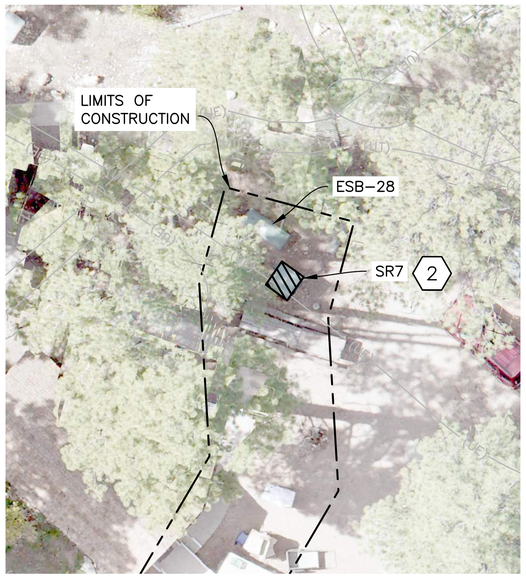
NORTH RIM LODGE AREA PLAN

SCALE (A)



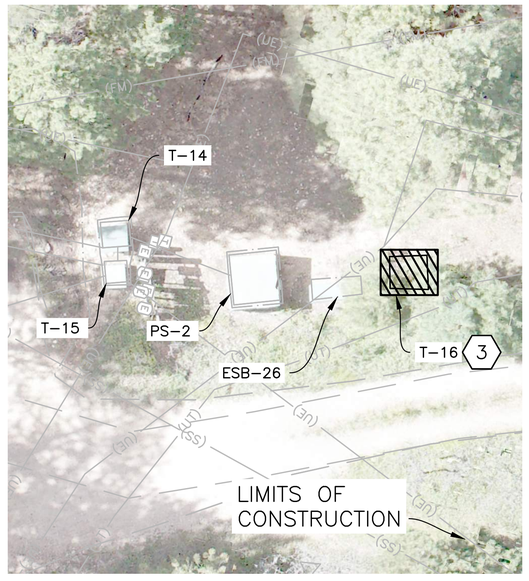
1 / 091X102 ESB-15

SCALE (B)



2 / 091X102 SR7

SCALE (B)



3 / 091X102 T-16

SCALE (B)

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. REFER TO OVERALL NORTH RIM POWER DISTRIBUTION SYSTEM ONE-LINE DRAWINGS 091E601 TO 091E603 FOR ADDITIONAL INFORMATION ON POWER SYSTEM IMPROVEMENTS.

C. LOCATION OF ALL UNDERGROUND UTILITIES SHOWN ARE APPROXIMATE AND TO BE VERIFIED IN FIELD BY CONTRACTOR.

D. PROVIDE ROOT PRESERVATION AND TREE/PLANT PROTECTION.

E. CABLE LENGTHS, ENCLOSURE LOCATIONS, AND NORTHING/EASTING COORDINATES ARE APPROXIMATE. FINAL EQUIPMENT LOCATIONS AND CABLE TRENCH LENGTHS TO BE VERIFIED IN FIELD BY CONTRACTOR.

F. PROVIDE TEMPORARY PROTECTION FOR REUSED EQUIPMENT PRIOR TO EQUIPMENT INSTALLATIONS. PROTECT EXIST MEDIUM VOLTAGE POWER CABLE AND ELBOW TERMINATIONS AT ALL MEDIUM VOLTAGE POWER EQUIPMENT NOTED FOR REPLACEMENT. CABLES AND ELBOWS TO BE REUSED.

G. SEE VOLUME 10 FOR DEMOLITION AREAS OF MEDIUM VOLTAGE POWER CABLE AND FIBER OPTIC BETWEEN THE NORTH RIM OVERHEAD STRUCTURES, ESB-15 AND FIBER PEDESTAL ENCLOSURE.

H. SEE VOLUME FOR FIBER OPTIC ROUTING AT THE NORTH RIM.

I. SEE SHEET 091E102 FOR EQUIPMENT IMPROVEMENTS.

J. ACCESS FOR ALL VOLUME 9 AREAS SHALL BE THROUGH PREVIOUSLY DISTURBED AREAS. CONTRACTOR SHALL NOT DAMAGE ANY ROCK CURBING, EDGING, SIDEWALKS, ASPHALT. AREAS SHALL BE PROTECTED IF NEAR A CONSTRUCTION LOCATION.

KEYNOTES: (X)

1. REMOVE EXIST PADMOUNT JUNCTION ENCLOSURE ESB-15 AND EQUIPMENT PAD.
2. REMOVE EXIST PADMOUNT SHUNT REACTOR SR7 AND EQUIPMENT SLEEVE.
3. REMOVE EXIST PADMOUNT TRANSFORMER AND EQUIPMENT PAD.
4. SEE VOLUME 10 FOR CONTINUATION OF DEMOLITION PLAN TO NORTH RIM POLES.

SCALE (A) 100 0 100 200 1" = 100'
SCALE OF FEET

SCALE (B) 10 0 10 20 1" = 10'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CALIFORNIA
CERTIFICATE NO 75014
TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: MAY 2026	SUB SHEET NO. 091 X102	TITLE OF SHEET POWER DISTRIBUTION DEMOLITION PLAN NORTH RIM LODGE AREA NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 3 OF 51

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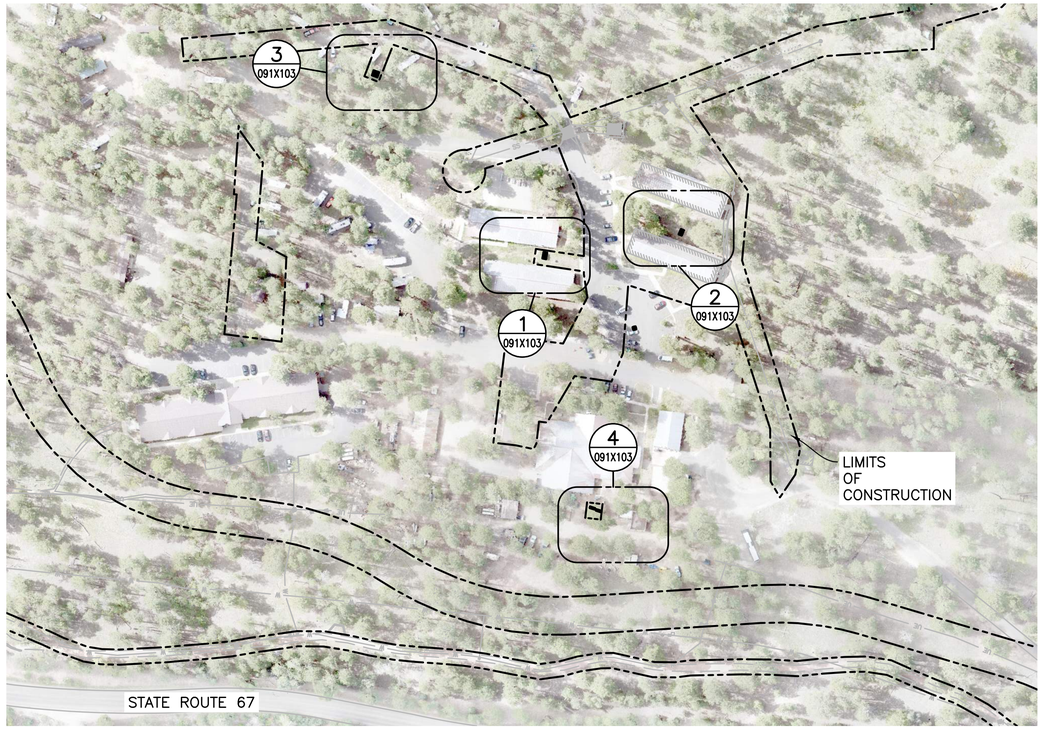
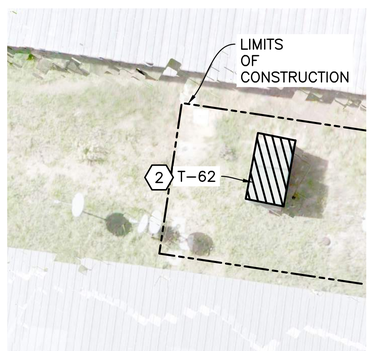
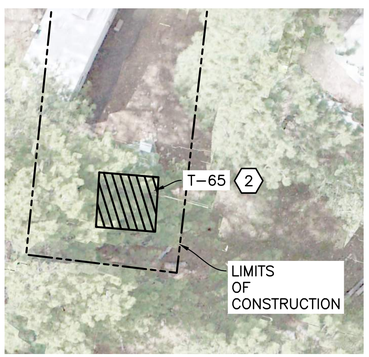


NORTH RIM CONCESSIONAIRE AREA DEMOLITION PLAN
SCALE (A)

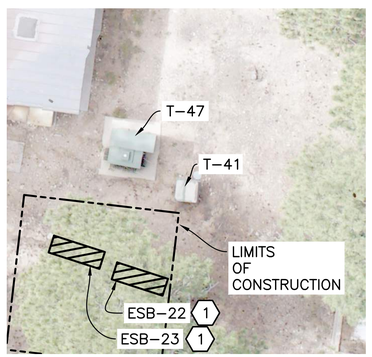


1 091X103 T-62
SCALE (B)

2 091X103 T-63
SCALE (B)



3 091X103 T-65
SCALE (B)



4 091X103 ESB-22 & ESB-23
SCALE (B)

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. REFER TO OVERALL NORTH RIM POWER DISTRIBUTION SYSTEM ONE-LINE DRAWINGS 091E601 TO 091E603 FOR ADDITIONAL INFORMATION ON POWER SYSTEM IMPROVEMENTS.

C. LOCATION OF ALL UNDERGROUND UTILITIES SHOWN ARE APPROXIMATE AND TO BE VERIFIED IN FIELD BY CONTRACTOR.

D. PROVIDE ROOT PRESERVATION AND TREE/PLANT PROTECTION.

E. CABLE LENGTHS, ENCLOSURE LOCATIONS, AND NORTHING/EASTING COORDINATES ARE APPROXIMATE. FINAL EQUIPMENT LOCATIONS AND CABLE TRENCH LENGTHS TO BE VERIFIED IN FIELD BY CONTRACTOR.

F. PROVIDE TEMPORARY PROTECTION FOR REUSED EQUIPMENT PRIOR TO NEW EQUIPMENT INSTALLATIONS.

G. SEE SHEET 091E103 FOR MEDIUM VOLTAGE EQUIPMENT IMPROVEMENT PLANS.

H. ACCESS FOR ALL VOLUME 9 AREAS SHALL BE THROUGH PREVIOUSLY DISTURBED AREAS. CONTRACTOR SHALL NOT DAMAGE ANY ROCK CURBING, EDGING, SIDEWALKS, ASPHALT. AREAS SHALL BE PROTECTED IF NEAR A CONSTRUCTION LOCATION.

KEYNOTES: (X)

1. REMOVE EXIST PADMOUNT JUNCTION ENCLOSURE ESB-22 AND ESB-23. REMOVE EXIST JUNCTION ENCLOSURE SLEEVES.

2. REMOVE EXIST PADMOUNT TRANSFORMER T-62, T-63, AND T-65. REMOVE EXIST CONCRETE TRANSFORMER PADS.

SCALE (A) 100 0 100 200 1" = 100'
SCALE OF FEET

SCALE (B) 10 0 10 20 1" = 10'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO. 75014 TOU VANG
Bob Stewart

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: MAY 2026	SUB SHEET NO. 091 X103	TITLE OF SHEET POWER DISTRIBUTION DEMOLITION PLAN CONCESSIONAIRE AREA NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 4 OF 51

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GAS STATION AREA PLAN

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. REFER TO OVERALL NORTH RIM POWER DISTRIBUTION SYSTEM ONE–LINE DRAWINGS 091E601 TO 091E603 FOR ADDITIONAL INFORMATION ON POWER SYSTEM IMPROVEMENTS.

C. LOCATION OF ALL UNDERGROUND UTILITIES SHOWN ARE APPROXIMATE AND TO BE VERIFIED IN FIELD BY CONTRACTOR.

D. PROVIDE ROOT PRESERVATION AND TREE/PLANT PROTECTION.

E. CABLE LENGTHS, ENCLOSURE LOCATIONS, AND NORTHING/EASTING COORDINATES ARE APPROXIMATE. FINAL EQUIPMENT LOCATIONS AND CABLE TRENCH LENGTHS TO BE VERIFIED IN FIELD BY CONTRACTOR.

F. PROVIDE TEMPORARY PROTECTION FOR REUSED EQUIPMENT PRIOR TO NEW EQUIPMENT INSTALLATIONS.

G. ACCESS FOR ALL VOLUME 9 AREAS SHALL BE THROUGH PREVIOUSLY DISTURBED AREAS. CONTRACTOR SHALL NOT DAMAGE ANY ROCK CURBING, EDGING, SIDEWALKS, ASPHALT. AREAS SHALL BE PROTECTED IF NEAR A CONSTRUCTION LOCATION.

KEYNOTES: (X)

1. REMOVE EXIST PADMOUNT JUNCTION ENCLOSURE ESB–18 AND ESB–19. REMOVE EXIST JUNCTION ENCLOSURE SLEEVES. SEE SHEET 091E104 FOR MEDIUM VOLTAGE EQUIPMENT IMPROVEMENT PLANS.

SCALE (B) 10 0 10 20 1" = 10'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: MAY 2026	SUB SHEET NO. 091 X104	TITLE OF SHEET POWER DISTRIBUTION DEMOLITION PLAN GAS STATION AREA NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 5 OF 51

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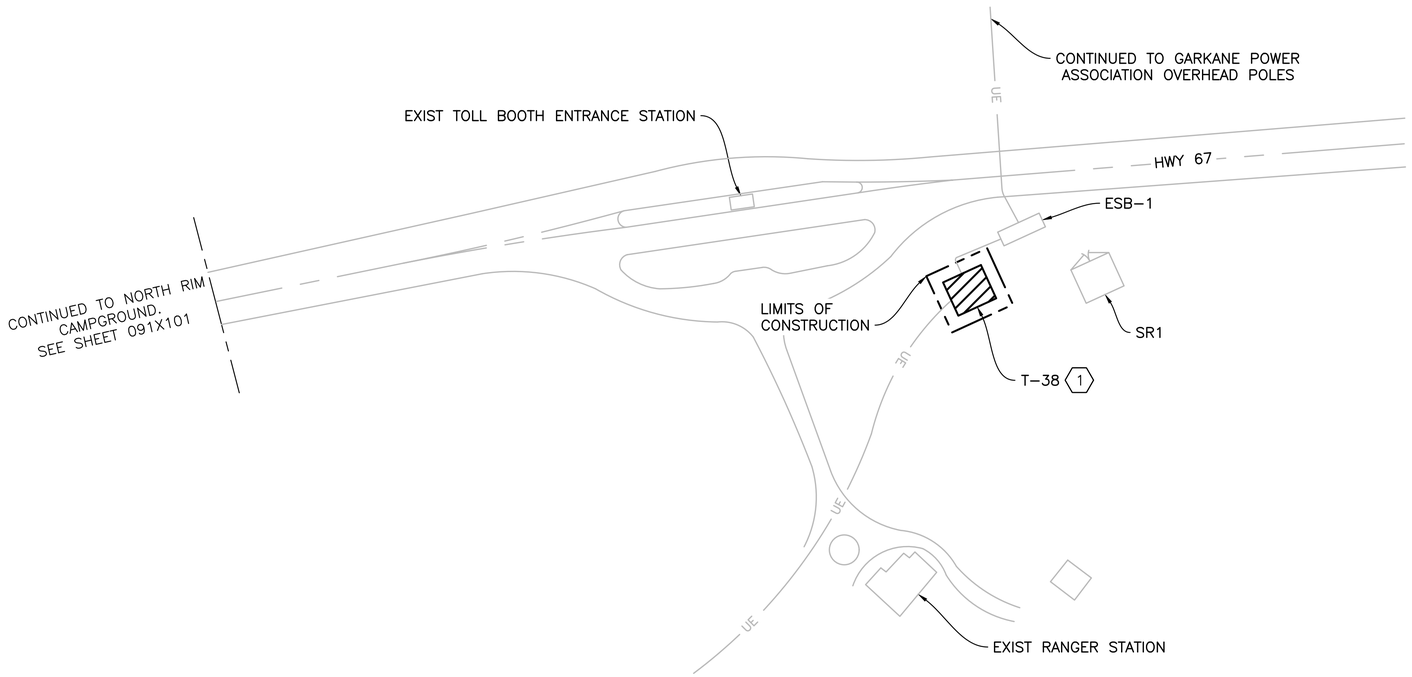

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. REFER TO OVERALL NORTH RIM POWER DISTRIBUTION SYSTEM ONE-LINE DRAWINGS 091E601 TO 091E603 FOR ADDITIONAL INFORMATION ON POWER SYSTEM IMPROVEMENTS.

C. LOCATION OF ALL UNDERGROUND UTILITIES SHOWN ARE APPROXIMATE AND TO BE VERIFIED IN FIELD BY CONTRACTOR.

D. PROVIDE ROOT PRESERVATION AND TREE/PLANT PROTECTION.

E. CABLE LENGTHS, ENCLOSURE LOCATIONS, AND NORTHING/EASTING COORDINATES ARE APPROXIMATE. FINAL EQUIPMENT LOCATIONS AND CABLE TRENCH LENGTHS TO BE VERIFIED IN FIELD BY CONTRACTOR.

F. ACCESS FOR ALL VOLUME 9 AREAS SHALL BE THROUGH PREVIOUSLY DISTURBED AREAS. CONTRACTOR SHALL NOT DAMAGE ANY ROCK CURBING, EDGING, SIDEWALKS, ASPHALT. AREAS SHALL BE PROTECTED IF NEAR A CONSTRUCTION LOCATION.

KEYNOTES: (X)

1. REMOVE EXIST PADMOUNT TRANSFORMER T-38. REFER TO SHEET 091E105 FOR ELECTRICAL DISTRIBUTION IMPROVEMENT PLANS AT THE NORTH RIM ENTRANCE.

60 0 60 120 1" = 60'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO. 75014 TOU VANG
STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: MAY 2026	SUB SHEET NO. 091 X105	TITLE OF SHEET POWER DISTRIBUTION DEMOLITION PLAN ENTRANCE STATION AREA NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 6 OF 51

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GENERAL NOTES:

A. LINES A AND C ARE OPERATED BY THE NATIONAL PARK SERVICES. LINES B AND D ARE OPERATED BY CONCESSIONS.

B. SEE VOLUME 0 FOR ONE-LINE LEGEND.

BOOTH & LOG CABIN

METER ON POWER PEDESTAL NEAR T-38

GARKANE ENERGY COOPERATIVE (24.9Y/14.4kV)

T-38
14.4 kV - 240/120 V
25 kVA
1-PH (C)

SR1
250 KVAR
OIL SHUNT REACTOR

LINDBERG HILL SITES

T-60
14.4 kV - 240/120 V
1.7% Z
10 kVA
1-PH (B)

T-50
14.4 kV - 240/120 V
1.7% Z
50 kVA
1-PH (A)

T-51
14.4 kV - 240/120 V
1.7% Z
50 kVA
1-PH (A)

T-52
14.4 kV - 240/120 V
1.7% Z
50 kVA
1-PH (A)

PS-3
N.O. 12K
N.C. 50K
AT LINDBERGH HILL

ESB-1 ESB-2 ESB-3 ESB-4 ESB-6 ESB-7 ESB-8 ESB-9 ESB-10

PS-1
AT NORTH KAIBAB TRAINHEAD PARKING LOT

TO ESB-30 CONTINUED BOTTOM LEFT

T-39
24.9 kV - 208/120 V
3.3% Z
45 kVA
3-PH

TO C.C. HILL DEVELOPMENT

GENERATOR BUILDING SITE (SEE 092X SERIES)

MC-1
G.E.C.
CONCESS. FEEDERS METERING
3PT'S
3CT'S

ESB-30

PTS-1
PTS-2
SOURCE TRANSFER CONTROL
N.C.
N.C. 25K
N.O.
SPARE

FROM PS-1 CONTINUED FROM TOP RIGHT

MC-2
N.P.S. FEEDERS METERING

SR3
250 KVAR
OIL SHUNT REACTOR

MC-3
N.P.S
CONCESS. FEEDERS METERING

T-36
24.9GRDY/14.4kV-480V
500 kVA
3-PH
GEN. BLDG.

T-45
24.9GRDY/14.4kV-480V
1500 kVA
3-PH
GEN. BLDG.

T-35
24.9GY/14.4 kV - 480GY/277 V
112.5 kVA
3-PH
GEN. BLDG.

LINE B (CONCESSION)
LINE A (N.P.S)
LINE C (N.P.S)

ESB-12
ESB-13
ESB-14

T-37
24.9GY/14.4 kV - 480Y/277 V
1.9% Z
112.5 kVA
3-PH
TO WWTP BLDG.

TO HELIBASE TRAILERS

T-34
14.4 kV - 240/120 V
1.6% Z
50 kVA
1-PH (B)

TO ADMIN. SHOPS

T-33
24.9GY/14.4 kV - 208GY/120 V
1.9% Z
30 kVA
3-PH

TO RESIDENCES, ADMIN. SHOPS, & LIFT STA. NO. 4

T-32
24.9GY/14.4 kV - 208GY/120 V
2.3% Z
45 kVA
3-PH

TO COCKREY COURT APARTMENTS

T-31
14.4 kV - 240/120 V
1.6% Z
50 kVA
1-PH (A)

TO HELIBASE TRAILERS

T-30
14.4 kV - 240/120 V
2.1% Z
25 kVA
1-PH (A)

TO WATER TANKS

T-27
24.9GY/14.4 kV - 208GY/120 V
2.3% Z
45 kVA
3-PH

TO EMS BLDG.

T-48
24.9GY/14.4 kV - 208GY/120 V
1.47% Z
112.5 kVA
3-PH

TO N.P.S CABINS

T-42
24.9GY/14.4 kV - 208GY/120 V
3.3% Z
45 kVA
3-PH

TO ADMIN CABINS

T-29
14.4 kV - 240/120 V
2.5% Z
15 kVA
1-PH (B)

TO ADMIN OFFICE & CABINS

T-28
14.4 kV - 240/120 V
1.6% Z
50 kVA
1-PH (A)

SR5
100 KVAR
OIL SHUNT REACTOR

ESB-16 ESB-17 ESB-18 ESB-18A

24.9 kV
LINE C CONTINUED TO ESB-21
(SEE SHEET 091X603)

24.9 kV
LINE B CONTINUED TO ESB-19
(SEE SHEET 091X602)

REGISTERED PROFESSIONAL ENGINEER COLORADO
CERTIFICATE NO. 75014
TOU VANG

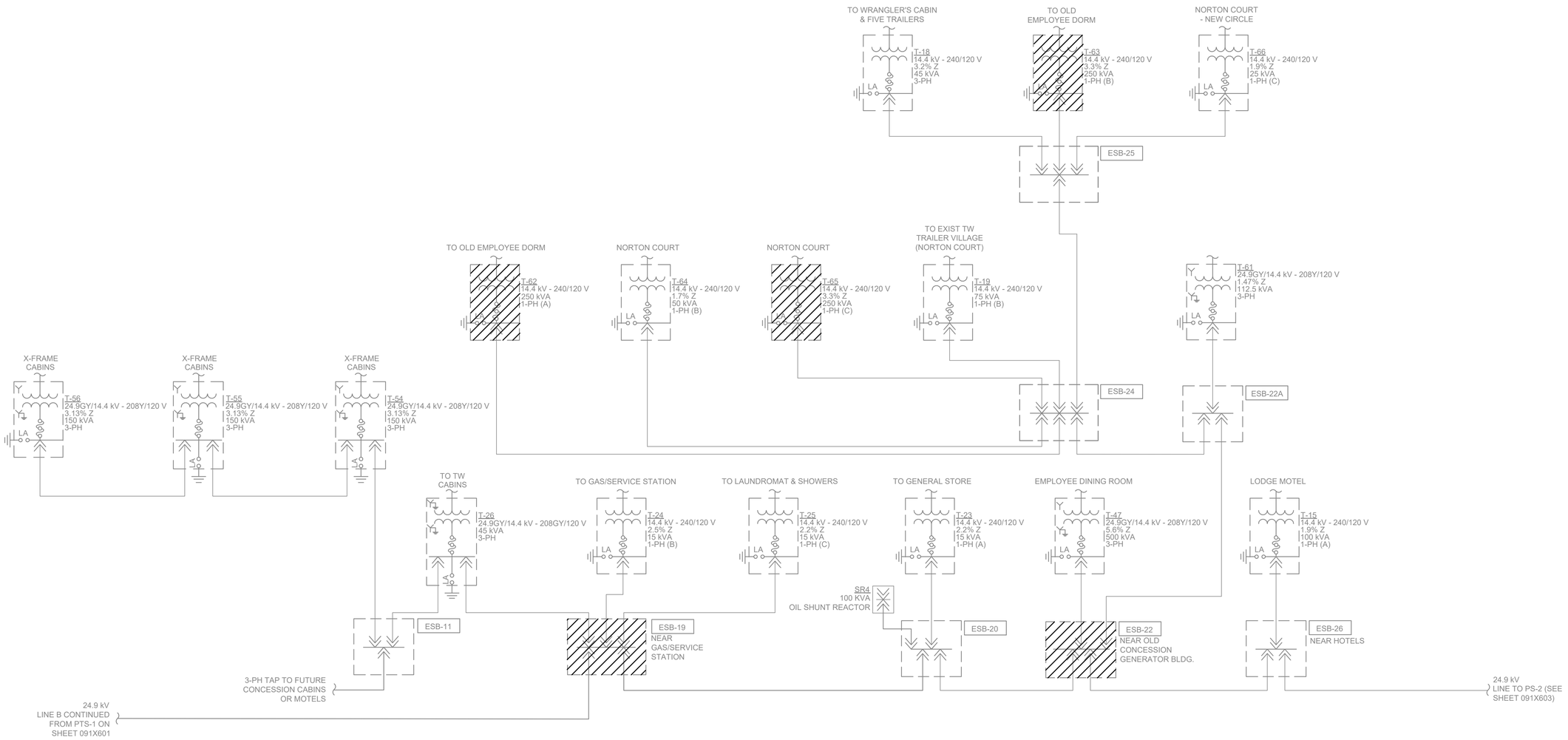
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: MAY 2026	SUB SHEET NO. 091 X601	TITLE OF SHEET POWER DISTRIBUTION ONE-LINE DIAGRAM DEMO NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 7 OF 51

4/17/2026 Vang, Tou C:\Users\TVANG\DC\ACCDocs\HDR\103505545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\091E6 & 091X6 SHEETS.dwg

GENERAL NOTES:

A. LINES A AND C ARE OPERATED BY THE NATIONAL PARK SERVICES. LINES B AND D ARE OPERATED BY CONCESSIONS.

B. SEE VOLUME 0 FOR ONE-LINE LEGEND.



A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: MAY 2026	SUB SHEET NO. 091 X602	TITLE OF SHEET POWER DISTRIBUTION ONE-LINE DIAGRAM DEMO NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 8 OF 51

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF ARIZONA

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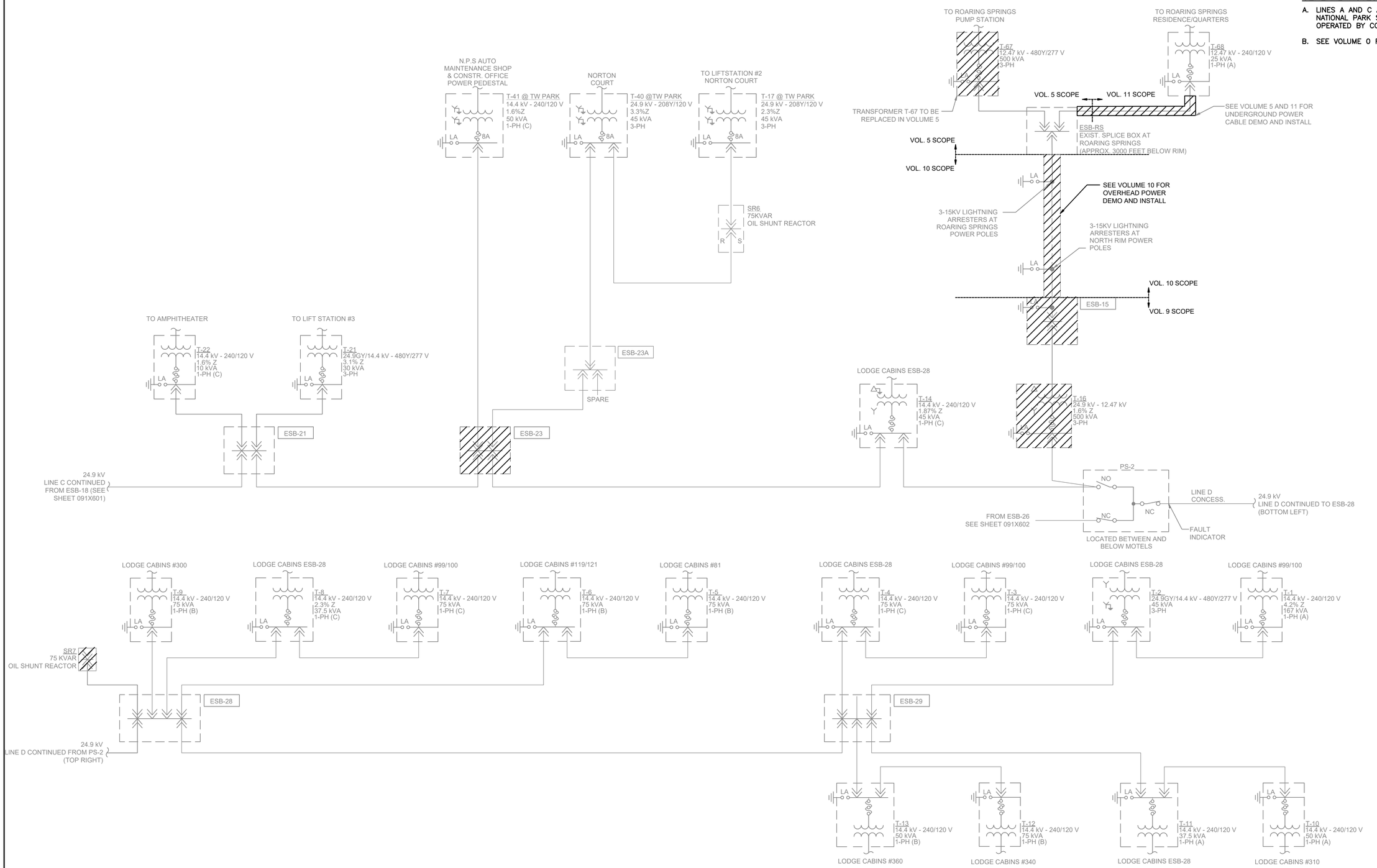
GENERAL NOTES:

A. LINES A AND C ARE OPERATED BY THE NATIONAL PARK SERVICES. LINES B AND D ARE OPERATED BY CONCESSIONS.

B. SEE VOLUME 0 FOR ONE-LINE LEGEND.



REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: MAY 2026	SUB SHEET NO. 091 X603	TITLE OF SHEET POWER DISTRIBUTION ONE-LINE DIAGRAM DEMO NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 9 OF 51

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GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. REFER TO OVERALL NORTH RIM POWER DISTRIBUTION SYSTEM ONE–LINE DRAWINGS 091E601 TO 091E603 FOR ADDITIONAL INFORMATION ON POWER SYSTEM IMPROVEMENTS.

C. TRANSFORMER T–67 IS LOCATED AT THE ROARING SPRINGS PUMPHOUSE, AND IS NOT SHOWN ON THIS DRAWING. SEE VOLUME 5 FOR T–67 TRANSFORMER REPLACEMENT DETAILS.

D. SEE VOLUME 1 FOR ROUTING OF FIBER OPTIC AND UNDERGROUND MEDIUM VOLTAGE FROM POWER DISTRIBUTION CENTER TO WASTE WATER TREATMENT PLANT AND WATER TREATMENT PLANT.

E. SEE SURVEY CONTROL PLAN IN VOLUME 0.

F. SEE SHEET 091E501 FOR PADMOUNTED SHUNT REACTOR DETAILS.

G. SEE VOLUME 0 SHEET 008E509 FOR PADMOUNTED JUNCTION ENCLOSURE AND TRANSFORMER DETAILS.

H. SEE VOLUME 0 SHEET 008E510 FOR CONDUCTOR TRENCHING, ELBOWS AND CONNECTORS, VAULT GROUNDING DETAILS, AND PADMOUNTED SWITCHGEAR DETAIL.

I. LOCATION OF ALL UNDERGROUND UTILITIES SHOWN ARE APPROXIMATE AND TO BE VERIFIED IN FIELD BY CONTRACTOR.

J. PROVIDE ROOT PRESERVATION AND TREE/PLANT PROTECTION.

K. CABLE LENGTHS, ENCLOSURE LOCATIONS, AND NORTHING/EASTING COORDINATES ARE APPROXIMATE. FINAL EQUIPMENT LOCATIONS AND CABLE TRENCH LENGTHS TO BE VERIFIED IN FIELD BY CONTRACTOR.

L. MEDIUM VOLTAGE EQUIPMENT CLEARANCES IN FRONT OF EQUIPMENT DOORS OR ACCESS POINTS SHALL BE AT MINIMUM 10 FT AND 3 FT ON THE SIDES.

M. EQUIPMENT DIMENSIONS ARE TYPICAL AND SHOWN FOR REFERENCE ONLY.

N. INSTALL 144–STRAND SM FIBER IN ORANGE TUBE OF NEW 7–WAY MICRODUCT.

300 0 300 600 1" = 300'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: MAY 2026	SUB SHEET NO. 091 E101	TITLE OF SHEET POWER DISTRIBUTION SITE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 10 OF 51

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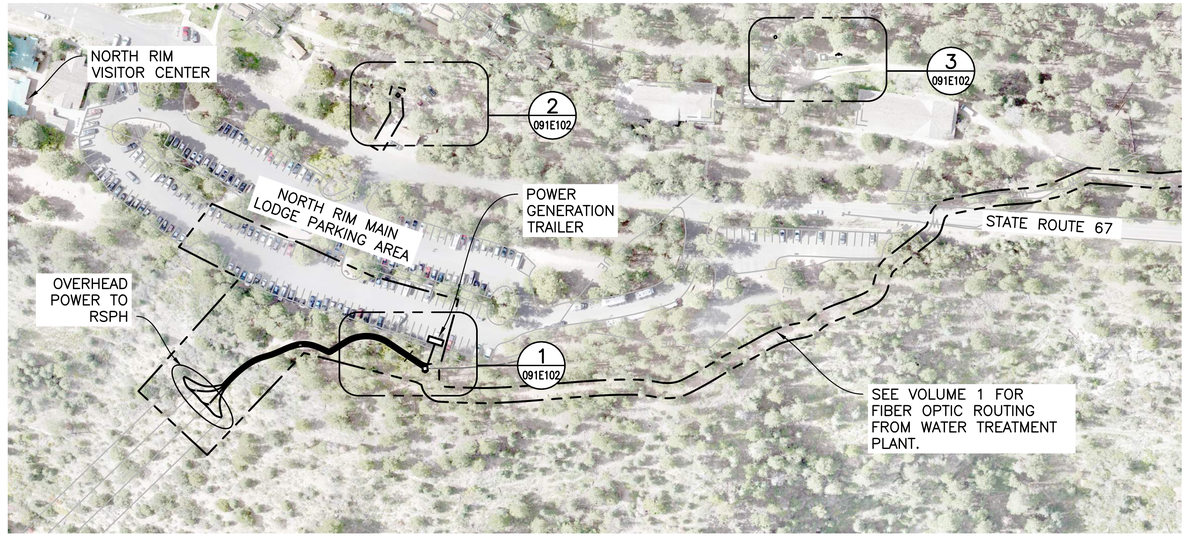

NORTH RIM LODGE AREA PLAN

SCALE (A)

1 091E102 ESB-15

SCALE (B)

2 091E102 SR7

SCALE (B)

3 091E102 T-16

SCALE (B)

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. SEE SHEET 091E101 FOR ADDITIONAL GENERAL NOTES.

C. REUSE EXIST ELBOW TERMINATIONS UNLESS OTHERWISE NOTATED. IF EXIST MEDIUM VOLTAGE POWER CABLE AND ELBOW TERMINATIONS CANNOT REACH REPLACED EQUIPMENT TERMINATION BUSHINGS, ADDITIONAL CABLE MAY BE SPLICED AT REPLACED EQUIPMENT AS NEEDED TO REACH EQUIPMENT CABLE TERMINATIONS. IN THESE INSTANCES, INSTALL ADDITIONAL 5 FT OF CABLE SLACK AT EQUIPMENT BASEMENT. RETERMINATE CABLE WITH APPROPRIATE ELBOW TERMINATIONS. CONTRACTOR TO MATCH ADDITIONAL MEDIUM VOLTAGE POWER CABLE VOLTAGE, SIZE, AND TYPE WITH EXIST MEDIUM VOLTAGE POWER CABLE. CONTRACTOR TO RECEIVE APPROVAL FROM CONTRACTING OFFICER PRIOR TO SPLICING AND TERMINATING MEDIUM VOLTAGE POWER CABLES AT REPLACED EQUIPMENT.

D. ACCESS FOR ALL VOLUME 9 AREAS SHALL BE THROUGH PREVIOUSLY DISTURBED AREAS. CONTRACTOR SHALL NOT DAMAGE ANY ROCK CURBING, EDGING, SIDEWALKS, ASPHALT. AREAS SHALL BE PROTECTED IF NEAR A CONSTRUCTION LOCATION.

E. SEE SHEET 091E501 DETAIL 1 FOR TYPICAL THREE-PHASE PADMOUNT SHUNT REACTOR DETAIL.

F. SEE VOLUME 0, SHEET 009Y501, FOR TYPICAL FIBER ENCLOSURE DETAILS.

G. SEE SHEET 091E502 FOR MEDIUM VOLTAGE POWER CABLE (X) SCHEDULE.

KEYNOTES: (X)

1. REPLACE EXIST JUNCTION ENCLOSURE ESB-15 WITH 15kV 4 WAY (1 SOURCE – 3 PROTECTED TAP) PADMOUNT SWITCHGEAR AND EQUIPMENT VAULT AND PAD. RECONNECT EXIST MEDIUM VOLTAGE POWER CABLE FROM T-16 ON SOURCE POSITION AND UTILIZE EXIST ELBOW TERMINATIONS. TERMINATE MEDIUM VOLTAGE POWER CABLE TO NORTH RIM OVERHEAD POLES WITH 200 AMP LOADBREAK ELBOWS ON PROTECTED TAP POSITIONS. INSTALL 200 AMP PROTECTIVE CAPS ON UNUSED BUSHINGS.
2. REPLACE EXIST SHUNT REACTOR SR7 WITH 25kV 75KVAR OIL-FILLED SHUNT REACTOR. INSTALL EQUIPMENT VAULT AND RECONNECT EXIST CONDUCTORS FROM ESB-28. UTILIZE EXIST ELBOW TERMINATIONS. INSTALL 200 AMP PROTECTIVE CAPS ON UNUSED BUSHINGS.
3. REPLACE EXIST TRANSFORMER T-16 WITH 750KVA, 24.9GY/14.4kV – 12.47GY/7.2kV PADMOUNT STEP-DOWN TRANSFORMER. INSTALL EQUIPMENT VAULT AND PAD. RECONNECT EXIST CONDUCTORS FROM ESB-15 AND PS-2. UTILIZE EXIST ELBOW TERMINATIONS.
4. INSTALL FOPP61 FIBER CABINET, FOR LAYOUT SEE DETAIL 1-3 009Y501. FOR FIBER NETWORK DIAGRAM, REFER TO 009Y101. SEE VOLUME 1 FOR FIBER ROUTING FROM WATER TREATMENT PLANT.
5. SEE VOLUME 10 FOR ROUTING OF FIBER OPTIC AND UNDERGROUND MEDIUM VOLTAGE POWER CABLE BETWEEN ESB-15 AND NORTH RIM OVERHEAD POLES.
6. CONTRACTOR TO SET TEMPORARY 15kV (MINIMUM 500kW) POWER GENERATION TRAILER FOR SERVICE AS NEEDED TO ESB-15. SET 25kV (3/C) 4/0 ALUMINUM ARMORED POWER CABLE BETWEEN ESB-15 AND POWER TRAILER, PROTECT IN PLACE. CONTRACTOR TO USE 15kV, 200 AMP RATED CONNECTIONS AND TERMINATIONS.

SCALE (A) 100 0 100 200 SCALE OF FEET 1" = 100'

SCALE (B) 10 0 10 20 SCALE OF FEET 1" = 10'

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: JUNE 2026	091 E102	POWER DISTRIBUTION REPLACEMENT (ESB-15, SR7, T-16) NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752
				PMIS/PKG NO. 318719
				SHEET 11 OF 51

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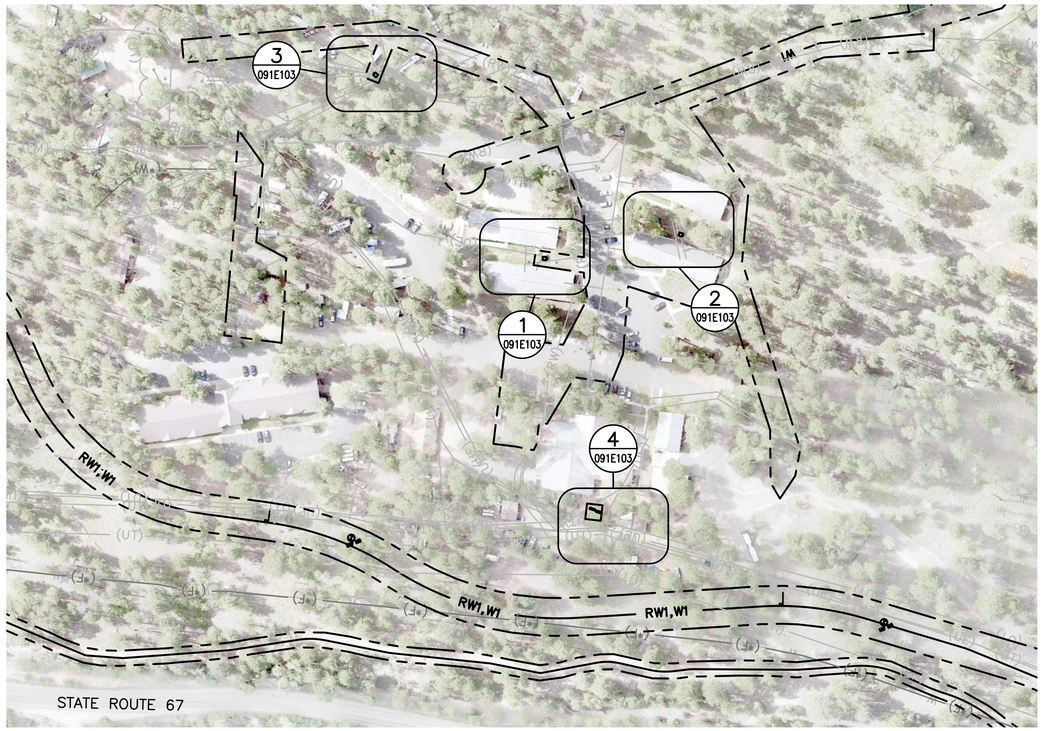
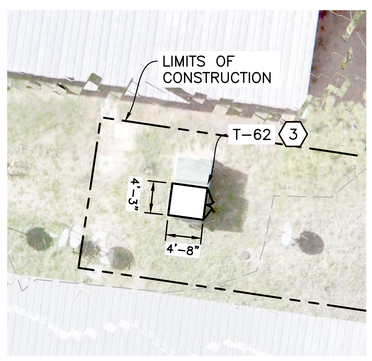


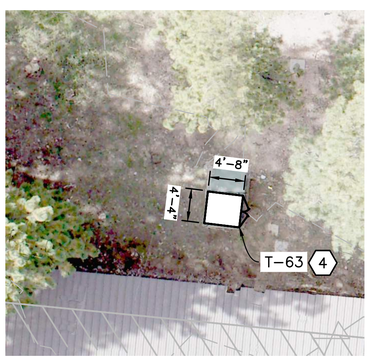
NORTH RIM CAMPGROUND SERVICE ROAD PLAN

SCALE (B)



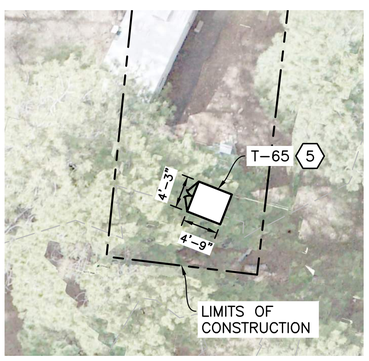
1 090E103 T-62

SCALE (A)



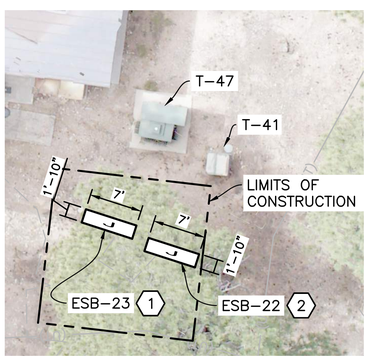
2 090E103 T-63

SCALE (A)



3 090E103 T-65

SCALE (A)



4 090E103 ESB-22 & ESB-23

SCALE (A)

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. SEE SHEET 091E101 FOR ADDITIONAL GENERAL NOTES.

C. REUSE EXIST ELBOW TERMINATIONS UNLESS OTHERWISE NOTATED. IF EXIST MEDIUM VOLTAGE POWER CABLE AND ELBOW TERMINATIONS CANNOT REACH REPLACED EQUIPMENT TERMINATION BUSHINGS, ADDITIONAL CABLE MAY BE SPLICED AT REPLACED EQUIPMENT AS NEEDED TO REACH EQUIPMENT CABLE TERMINATIONS. IN THESE INSTANCES, INSTALL ADDITIONAL 5 FT OF CABLE SLACK AT EQUIPMENT BASEMENT. RETERMINATE CABLE WITH APPROPRIATE ELBOW TERMINATIONS. CONTRACTOR TO MATCH ADDITIONAL MEDIUM VOLTAGE POWER CABLE VOLTAGE, SIZE, AND TYPE WITH EXIST MEDIUM VOLTAGE POWER CABLE. CONTRACTOR TO RECEIVE APPROVAL FROM CONTRACTING OFFICER PRIOR TO SPLICING AND TERMINATING MEDIUM VOLTAGE POWER CABLES AT REPLACED EQUIPMENT.

D. ACCESS FOR ALL VOLUME 9 AREAS SHALL BE THROUGH PREVIOUSLY DISTURBED AREAS. CONTRACTOR SHALL NOT DAMAGE ANY ROCK CURBING, EDGING, SIDEWALKS, ASPHALT. AREAS SHALL BE PROTECTED IF NEAR A CONSTRUCTION LOCATION.

E. INSTALL 200 AMP SURGE ARRESTER ELBOWS ON UNUSED TAP POSITIONS AT REPLACED TRANSFORMERS.

KEYNOTES: (X)

1. REPLACE EXIST JUNCTION ENCLOSURE ESB-23 WITH 4-POSITION PADMOUNT JUNCTION ENCLOSURE. INSTALL JUNCTION ENCLOSURE SLEEVE. RECONNECT EXIST MEDIUM VOLTAGE POWER CABLES AND ELBOWS FROM ESB-21, ESB-23A, T-41, AND T-14.

2. REPLACE EXIST JUNCTION ENCLOSURE ESB-22 WITH 4-POSITION PADMOUNT JUNCTION ENCLOSURE. INSTALL JUNCTION ENCLOSURE SLEEVE. RECONNECT EXIST MEDIUM VOLTAGE POWER CABLES AND ELBOWS FROM ESB-20, ESB-22A, ESB-26, AND T-47.

3. REPLACE EXIST TRANSFORMER T-62 WITH 250KVA, 14.4kV-240/120V (1-PH) PADMOUNT TRANSFORMER. INSTALL TRANSFORMER VAULT AND PAD. RECONNECT EXIST MEDIUM VOLTAGE POWER CABLES FROM ESB-24 WITH 200 AMP LOADBREAK ELBOW. RECONNECT EXIST SECONDARY SERVICE CONDUCTOR TO SECONDARY SIDE OF TRANSFORMER.

4. REPLACE EXIST TRANSFORMER T-63 WITH 250KVA, 14.4kV-240/120V (1-PH) PADMOUNT TRANSFORMER. INSTALL TRANSFORMER VAULT AND PAD. RECONNECT EXIST MEDIUM VOLTAGE POWER CABLES FROM ESB-25 WITH 200 AMP LOADBREAK ELBOW. RECONNECT EXIST SECONDARY SERVICE CONDUCTOR TO SECONDARY SIDE OF TRANSFORMER.

5. REPLACE EXIST TRANSFORMER T-65 WITH 250KVA, 14.4kV-240/120V (1-PH) PADMOUNT TRANSFORMER. INSTALL TRANSFORMER VAULT AND PAD. RECONNECT EXIST MEDIUM VOLTAGE POWER CABLES FROM ESB-24 WITH 200 AMP LOADBREAK ELBOW. RECONNECT EXIST SECONDARY SERVICE CONDUCTOR TO SECONDARY SIDE OF TRANSFORMER.

SCALE (A) 10 0 10 20 SCALE OF FEET 1" = 10'

SCALE (B) 100 0 100 200 SCALE OF FEET 1" = 100'

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 091 E103	TITLE OF SHEET POWER DISTRIBUTION REPLACEMENT (T-62, T-63, ESB-22, ESB-23) NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 12 OF 51

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GAS STATION AREA PLAN

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. SEE SHEET 091E101 FOR ADDITIONAL GENERAL NOTES.

C. REUSE EXIST ELBOW TERMINATIONS UNLESS OTHERWISE NOTATED. IF EXIST MEDIUM VOLTAGE POWER CABLE AND ELBOW TERMINATIONS CANNOT REACH REPLACED EQUIPMENT TERMINATION BUSHINGS, ADDITIONAL CABLE MAY BE SPLICED AT REPLACED EQUIPMENT AS NEEDED TO REACH EQUIPMENT CABLE TERMINATIONS. IN THESE INSTANCES, INSTALL ADDITIONAL 5 FT OF CABLE SLACK AT EQUIPMENT BASEMENT. RETERMINATE CABLE WITH APPROPRIATE ELBOW TERMINATIONS. CONTRACTOR TO MATCH ADDITIONAL MEDIUM VOLTAGE POWER CABLE VOLTAGE, SIZE, AND TYPE WITH EXIST MEDIUM VOLTAGE POWER CABLE. CONTRACTOR TO RECEIVE APPROVAL FROM CONTRACTING OFFICER PRIOR TO SPLICING AND TERMINATING MEDIUM VOLTAGE POWER CABLES AT REPLACED EQUIPMENT.

D. ACCESS FOR ALL VOLUME 9 AREAS SHALL BE THROUGH PREVIOUSLY DISTURBED AREAS. CONTRACTOR SHALL NOT DAMAGE ANY ROCK CURBING, EDGING, SIDEWALKS, ASPHALT. AREAS SHALL BE PROTECTED IF NEAR A CONSTRUCTION LOCATION.

KEYNOTES: (X)

1. REPLACE EXIST JUNCTION ENCLOSURE ESB-18 WITH 25kV 4-POSITION PADMOUNT JUNCTION ENCLOSURE. INSTALL JUNCTION ENCLOSURE SLEEVE. RECONNECT EXIST MEDIUM VOLTAGE POWER CABLES AND ELBOWS FROM ESB-17, ESB-18A, ESB-21, AND T-42 IN EXIST CONFIGURATION.

2. REPLACE EXIST JUNCTION ENCLOSURE ESB-19 WITH 25kV 4-POSITION PADMOUNT JUNCTION ENCLOSURE. INSTALL JUNCTION ENCLOSURE SLEEVE. RECONNECT EXIST MEDIUM VOLTAGE POWER CABLES AND ELBOWS FROM PTS-1, ESB-20, T-24, T-25, AND T-26 IN EXIST CONFIGURATION.

10 0 10 20
SCALE OF FEET
1" = 10'

REGISTERED PROFESSIONAL ENGINEER COLORADO
CERTIFICATE NO. 75014
TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: MAY 2026	SUB SHEET NO. 091 E104	TITLE OF SHEET POWER DISTRIBUTION REPLACEMENT (ESB-17, ESB-18) NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 13 OF 51

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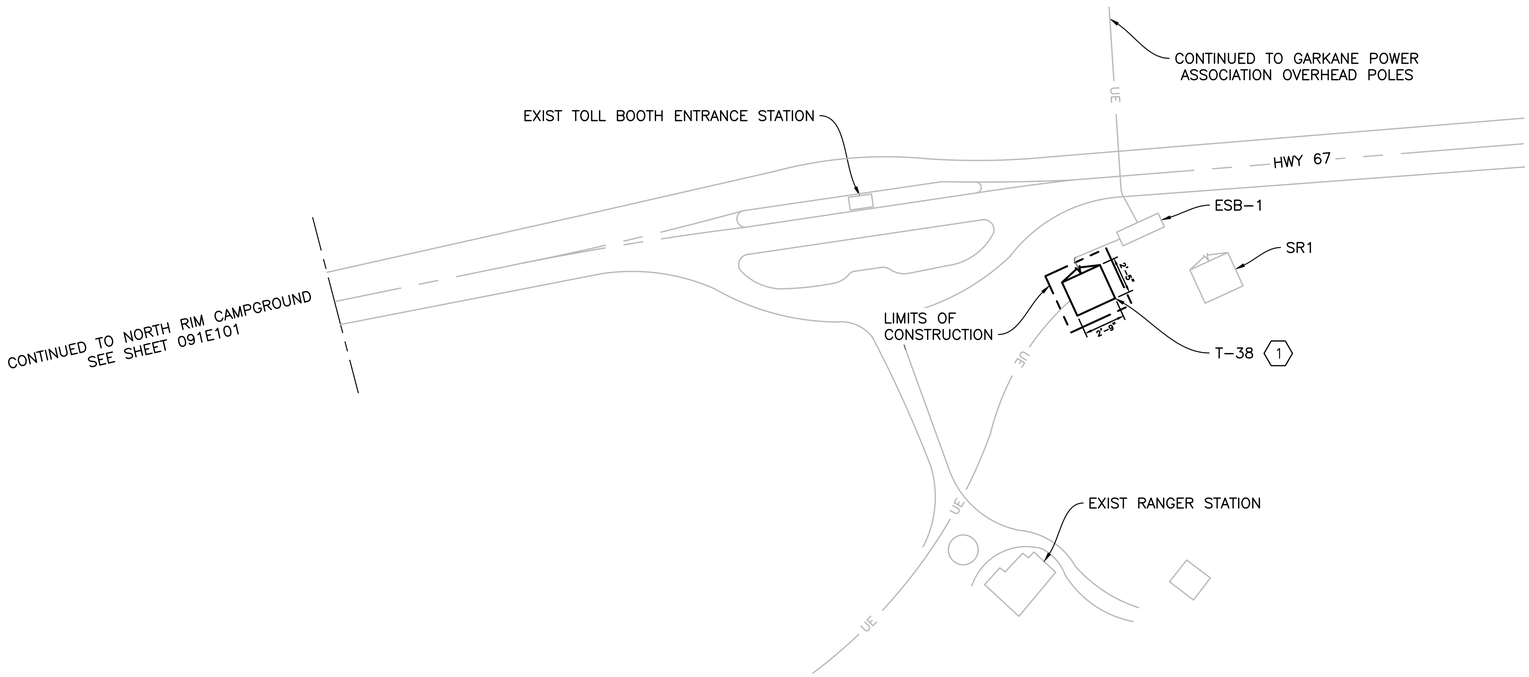

ENTRANCE STATION AREA PLAN

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. SEE SHEET 091E101 AND 091E106 FOR ADDITIONAL GENERAL NOTES.

C. SEE VOLUME 0, SHEET 008E509, DETAIL 3 FOR TYPICAL SINGLE–PHASE PADMOUNT TRANSFORMER DETAILS.

KEYNOTES: (X)

1. REPLACE EXIST TRANSFORMER T–38 WITH 25KVA, 14.4kV – 240/120V PADMOUNT TRANSFORMER AT EXIST LOCATION. REUSE TRANSFORMER PAD AND SLEEVE. RECONNECT EXIST MEDIUM VOLTAGE POWER CABLES AND ELBOWS FROM ESB–1. RECONNECT EXIST SECONDARY SERVICE CONDUCTOR TO SECONDARY SIDE OF TRANSFORMER.

60 0 60 120 1" = 60'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER OF THE STATE OF COLORADO
CERTIFICATE NO. 75014
TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: MAY 2026	SUB SHEET NO. 091 E105	TITLE OF SHEET POWER DISTRIBUTION REPLACEMENT (T–38) NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 14 OF 51

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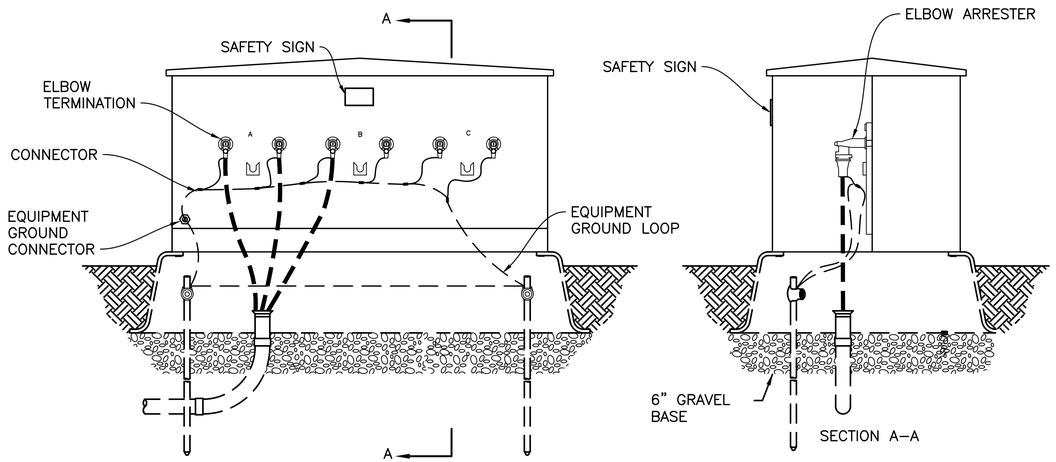

1 091E501 THREE–PHASE, LOOP FEED, PADMOUNTED SHUNT REACTOR

SCALE N.T.S.

NOTES:

1. PROVIDE SUFFICIENT PRIMARY NEUTRAL PIGTAIL AND CABLE SLACK TO PERMIT READY DISCONNECTION OF ELBOW AND MOUNTING ON PARKING STAND. TRAIN CABLES AS SHOWN.
2. NON–METALLIC SWEEPS SHOULD BE STABILIZED WITH CONCRETE ENCASEMENTS IF PULLING TENSIONS DICTATE.
3. LARGER RADIUS SWEEPS, WHICH MAY BE REQUIRED BY PULLING TENSIONS, MAY INCREASE BURIAL DEPTHS.

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: MAY 2026	SUB SHEET NO. 091 E501	TITLE OF SHEET POWER DISTRIBUTION SHUNT REACTOR DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 15 OF 51	

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GENERAL NOTES:

A. EQUIPMENT LATITUDE AND LONGITUDE SHOWN IN TABLES ARE FOR GENERAL LOCATING PURPOSES ONLY. CONTRACTOR TO CONFIRM ACTUAL INFIELD LOCATION AS NEEDED.

SWITCHGEAR TO BE INSTALLED DURING CONSTRUCTION								
JUNCATION ENCLOSURE NAME	NUMBER OF POSITIONS	VOLTAGE RATING	SOURCE	TAPS			FUSE PROTECTION	LOCATION
				MEDIUM VOLTAGE EQUIPMENT NAME	PHASE	CONFIGURATION		LATITUDE / LONGITUDE
ESB–12	6	24.9KV	PTS–2	ESB–12A	3–PH	NORMAL CLOSED	NONE	36°12'58.49"N 112°3'44.12"W
				ESB–16	3–PH	NORMAL CLOSED	24.9kV, 12A, K–SPEED*	
				T–35	3–PH	NORMAL CLOSED	24.9kV, 6A, K–SPEED*	
				T–49	3–PH	NORMAL CLOSED	24.9kV, 12A, K–SPEED*	
				WWTP	3–PH	NORMAL OPEN	24.9kV, 15A, K–SPEED*	
ESB–12A	4	24.9KV	ESB–12	ESB–13	3–PH	NORMAL CLOSED	24.9kV, 15A, K–SPEED*	36°12'58.79"N 112°3'43.78"W
				ESB–14	3–PH	NORMAL CLOSED	24.9kV, 12A, K–SPEED*	
				SPARE	–	–	NONE	
ESB–15	4	24.9KV	T–16	NORTH RIM POWER POLES	3–PH	NORMAL CLOSED	13.2kV, 40A, K–SPEED*	36°11'59.40"N 112°3'5.81"W
				SPARE	–	–	NONE	
				SPARE	–	–	NONE	
ESB–18A	4	24.9KV	ESB–18	T–49	3–PH	NORMAL OPEN	NONE	36°12'52.67"N 112°3'29.71"W
				T–48	3–PH	NORMAL CLOSED	NONE	
				T–27	3–PH	NORMAL CLOSED	NONE	

*ELECTRONIC PROTECTION IS ANTICIPATED. FUSE SIZES ARE PROVIDED FOR CONTRACTOR TO PROGRAM EQUIVALENT FUSE PROTECTION WHEN SWITCH IS INSTALLED.

25kV MEDIUM VOLTAGE POWER CABLE SCHEDULE			
CABLE ID	FROM	TO	CABLE TYPE
A	ESB–12	PTS–2	EXIST. 25KV (3/C + G) 3/0 AWG ALUMINUM ARMORED POWER CABLE
B	ESB–12	T–49	25KV (3/C + G) 4/0 AWG ALUMINUM ARMORED POWER CABLE
C	ESB–12	ESB–12A	25KV (3/C + G) 4/0 AWG ALUMINUM ARMORED POWER CABLE
D	ESB–12	T–35	25KV (3/C + G) 4/0 AWG ALUMINUM ARMORED POWER CABLE
E	ESB–12	POINT OF INTERCEPT "B" (ESB–16)	25KV (3/C + G) 4/0 AWG ALUMINUM ARMORED POWER CABLE
F	ESB–12	POINT OF INTERCEPT "D" (WWTP AREA)	25KV (3/C + G) 4/0 AWG ALUMINUM ARMORED POWER CABLE
G	ESB–12A	POINT OF INTERCEPT "A" (ESB–14)	25KV (3/C + G) 4/0 AWG ALUMINUM ARMORED POWER CABLE
H	ESB–12A	ESB–13B	EXIST. 25KV (3/C + G) 3/0 AWG ALUMINUM ARMORED POWER CABLE
I	ESB–13A	T–37	25KV (3/C + G) 4/0 AWG ALUMINUM ARMORED POWER CABLE
J	ESB–13A	T–49	25KV (3/C + G) 4/0 AWG ALUMINUM ARMORED POWER CABLE
K	ESB–13A	T–48	25KV (3/C + G) 4/0 AWG ALUMINUM ARMORED POWER CABLE
L	ESB–18A	T–27	25KV (3/C + G) 4/0 AWG ALUMINUM ARMORED POWER CABLE
M	ESB–18A	T–48	25KV (3/C + G) 4/0 AWG ALUMINUM ARMORED POWER CABLE
N	ESB–18A	T–27	25KV (3/C + G) 4/0 AWG ALUMINUM ARMORED POWER CABLE
O	ESB–18A	ESB–18	EXIST. 25KV (3/C + G) 3/0 AWG ALUMINUM ARMORED POWER CABLE

NOTE: INTERNAL PHASE CABLE IN ARMORED POWER CABLE TO HAVE CONCENTRIC NEUTRAL.

JUNCTION ENCLOSURES TO BE INSTALLED DURING CONSTRUCTION							
JUNCATION ENCLOSURE NAME	NUMBER OF POSITIONS	VOLTAGE RATING	SOURCE	TAPS		LOCATION	
				DOWNSTREAM ELECTRICAL EQUIPMENT	PHASE	LATITUDE	LONGITUDE
ESB–18	4	24.9KV	ESB–17	ESB–18A	3–PH	36°12'46.07"N	112°3'34.45"W
				ESB–21	3–PH		
				T–42	3–PH		
ESB–19	4	24.9KV	PTS–1	ESB–20	3–PH	36°12'46.18"N	112°3'34.52"W
				T–24	1–PH		
				T–25	1–PH		
				T–26	3–PH		
ESB–22	4	24.9KV	ESB–20	ESB–26	3–PH	36°12'28.83"N	112°3'22.16"W
				T–47	3–PH		
				ESB–22A	3–PH		
ESB–23	4	24.9KV	ESB–21	ESB–23A	3–PH	36°12'28.79"N	112°3'22.13"W
				T–14	3–PH		
				T–41	1–PH		
ESB–30	4	24.9KV	PS–1	SR3	3–PH	36°12'58.64"N	112°3'43.92"W
				MC–1	3–PH		
				MC–2	3–PH		

TRANSFORMERS & SHUNT REACTORS TO BE INSTALLED DURING CONSTRUCTION							
NAME	SIZE	VOLTAGE RATING	PHASE	Current Sensing Fuse (Amp Rating)	Current Limiting Fuse (Amp Rating)	LOCATION	
						LATITUDE	LONGITUDE
T–16	750 KVA	24.9GY/14.4KV – 12.47GY/7.2V	3–PH	40A	80A	36°12'3.80"N	112°3'12.65"W
T–27	45 KVA	24.9GY/14.4KV – 208GY/120V	3–PH	6A	30A	36°12'53.96"N	112°3'29.21"W
T–35	112.5 KVA	24.9GY/14.4KV – 480GY/277V	3–PH	6A	30A	36°12'58.40"N	112°3'44.24"W
T–36	1500 KVA	24.9GY/14.4KV – 480V	3–PH	65A	150A	36°12'58.46"N	112°3'44.42"W
T–38	25 KVA	14.4KV – 240/120V	1–PH	6A	30A	36°20'7.42"N	112°6'58.34"W
T–49	750 KVA	24.9KV DELTA – 480GY/277V	3–PH	30A	100A	36°12'52.72"N	112°3'29.65"W
T–62	250 KVA	14.4KV – 240/120V	1–PH	40A	100A	36°12'26.63"N	112°3'25.14"W
T–63	250 KVA	14.4KV – 240/120V	1–PH	40A	100A	36°12'28.00"N	112°3'26.61"W
T–65	250 KVA	14.4KV – 240/120V	1–PH	40A	100A	36°12'23.60"N	112°3'26.45"W
T–67	750 KVA	12.47KV DELTA – 480GY/277V	3–PH	65A	150A	36°11'36.61"N	112°2'3.61"W
SR7	75 KVA	24.9KV	3–PH	6A	30A	36°11'58.37"N	112°3'9.96"W

REGISTERED PROFESSIONAL ENGINEER – ELECTRICAL – CERTIFICATE NO. 58037 – TRAVIS T. MOORE – Date Signed 1/4/2026 – ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: AUG 2026	SUB SHEET NO. 091 E502	TITLE OF SHEET POWER DISTRIBUTION MEDIUM VOLTAGE EQUIPMENT INSTALLATION TABLES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 16 OF 51

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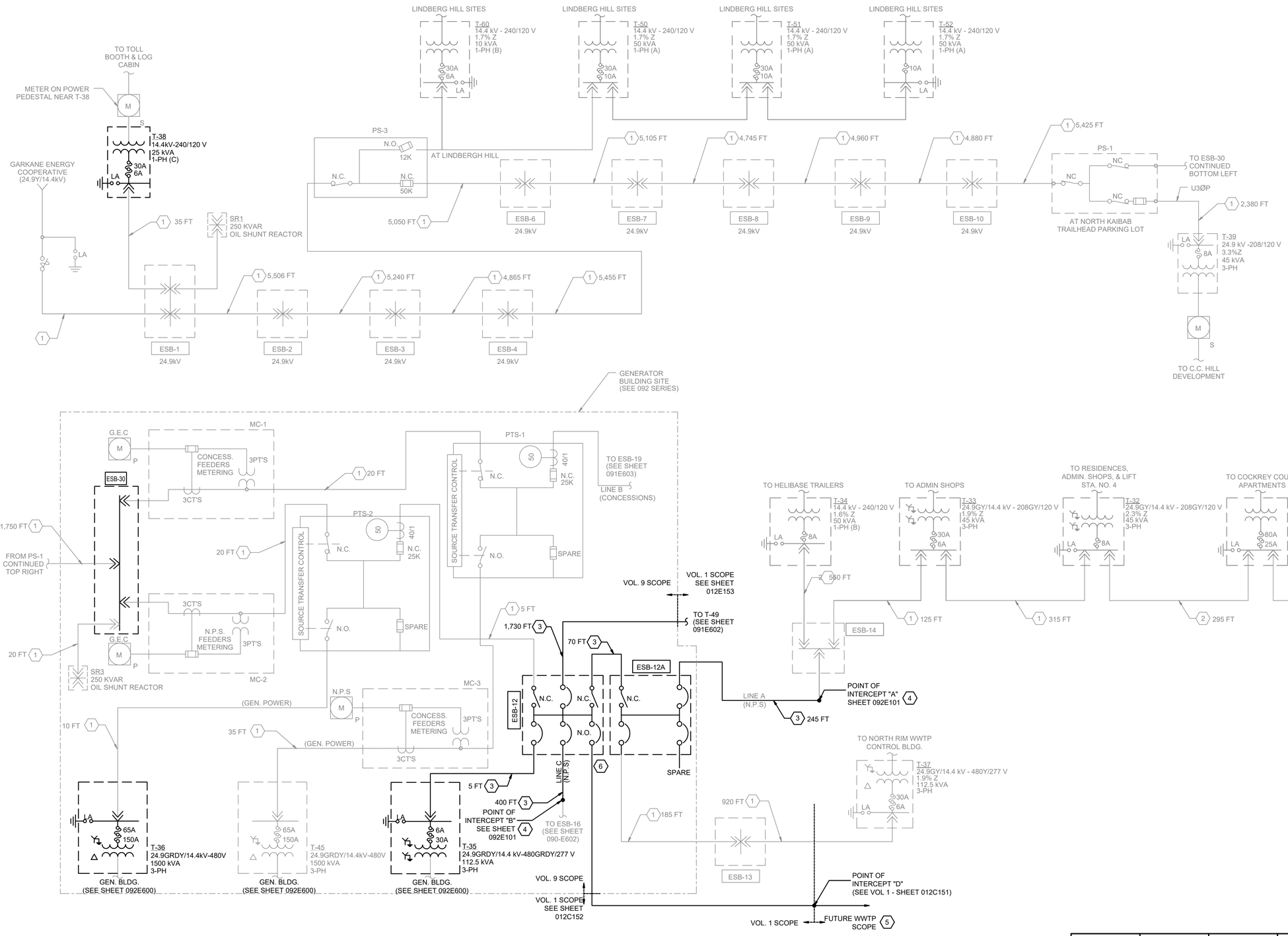

GENERAL NOTES:

A. REFER TO OVERALL NORTH RIM POWER DISTRIBUTION SYSTEM IMPROVEMENT PLANS IN THE 091E1 DRAWING SERIES.

B. LINES A AND C ARE OPERATED BY THE NATIONAL PARK SERVICES. LINES B AND D ARE OPERATED BY CONCESSIONS.

C. SEE VOLUME 0 FOR ONE–LINE LEGEND.

D. ALL MATERIALS AND INSTALLATION PRACTICES SHALL CONFORM TO IEEE, ANSI, AND NEMA STANDARDS.

E. CONTRACTOR SHALL VERIFY ALL DIMENSIONS AND CONDITIONS AT THE SITE BEFORE COMMENCING WORK.

F. CONTRACTOR SHALL COORDINATE WITH CONTRACTING OFFICER FOR ANY NECESSARY OUTTAGES AND TIE–INS.

G. CONTRACTOR SHALL PROVIDE ALL NECESSARY SAFETY MEASURES AND FOLLOW ALL OSHA REGULATIONS DURING CONSTRUCTION.

H. ALL WORK SHALL COMPLY WITH THE LATEST EDITION OF THE NATIONAL ELECTRIC CODE (NEC), NATIONAL ELECTRIC SAFETY CODE (NESC), DSC, NPS, AND APPLICATION FEDERAL, STATE, AND LOCAL ELECTRICAL CODES, REGULATIONS, ORDINANCES AND PROJECT STANDARDS.

I. TRANSFORMER FUSE SIZES AND PHASES BASED ON EXIST NPS DOCUMENTATION. CONTRACTOR TO FIELD VERIFY EQUIPMENT PHASING AND CONNECTIONS PRIOR TO WORK.

KEYNOTES ⟨X⟩

1. EXIST 25kV (3/C + G) 3/0 AWG ALUMINUM ARMORED POWER CABLE.
2. EXIST 25kV (1/C + G) 3/0 AWG ALUMINUM ARMORED POWER CABLE.
3. 25kV (3/C + G) 4/0 AWG ALUMINUM ARMORED POWER CABLE. EACH INTERNAL CABLE TO HAVE CONCENTRIC NEUTRAL.
4. 25kV MEDIUM VOLTAGE SPLICE POINT.
5. FUTURE WWTP PROJECT TO CONTINUE LINE INTO WWTP AREA.
6. LOCK AND LEAVE SWITCH POSITION FOR CONNECTION TO WWTP AREA IN THE OPEN POSITION. INSTALL SWITCH TAG AS DO NOT CLOSE. CONFIRM SWITCH IS LOCKED IN THE OPEN POSITION PRIOR TO ENERGIZING OF SWITCH ESB–12.

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75014 — TOU VANG — STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG DRAWN: D. VANG TECH. REVIEW: S. EVERETT DATE: JUNE 2026	SUB SHEET NO. 091 E601	TITLE OF SHEET POWER DISTRIBUTION ONE–LINE DIAGRAM MODIFIED NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 17 OF 51

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GENERAL NOTES:

A. REFER TO OVERALL NORTH RIM POWER DISTRIBUTION SYSTEM IMPROVEMENT PLANS IN THE 091E1 DRAWING SERIES.

B. LINES A AND C ARE OPERATED BY THE NATIONAL PARK SERVICES. LINES B AND D ARE OPERATED BY CONCESSIONS.

C. SEE VOLUME 0 FOR ONE-LINE LEGEND.

D. ALL MATERIALS AND INSTALLATION PRACTICES SHALL CONFORM TO IEEE, ANSI, AND NEMA STANDARDS.

E. CONTRACTOR SHALL VERIFY ALL DIMENSIONS AND CONDITIONS AT THE SITE BEFORE COMMENCING WORK.

F. CONTRACTOR SHALL COORDINATE WITH CONTRACTING OFFICER FOR ANY NECESSARY OUTTAGES AND TIE-INS.

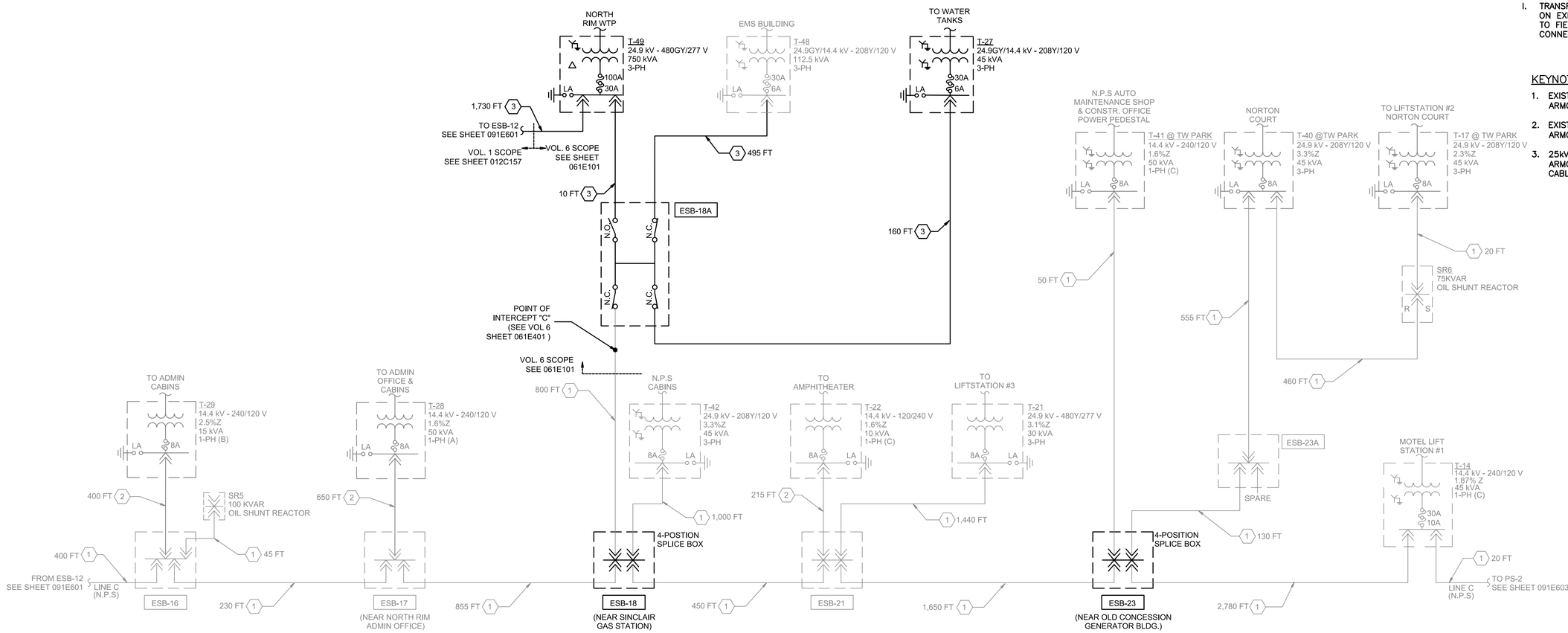
G. CONTRACTOR SHALL PROVIDE ALL NECESSARY SAFETY MEASURES AND FOLLOW ALL OSHA REGULATIONS DURING CONSTRUCTION.

H. ALL WORK SHALL COMPLY WITH THE LATEST EDITION OF THE NATIONAL ELECTRIC CODE (NEC), NATIONAL ELECTRIC SAFETY CODE (NESC), DSC, NPS, AND APPLICATION FEDERAL, STATE, AND LOCAL ELECTRICAL CODES, REGULATIONS, ORDINANCES AND PROJECT STANDARDS.

I. TRANSFORMER FUSE SIZES AND PHASES BASED ON EXIST NPS DOCUMENTATION. CONTRACTOR TO FIELD VERIFY EQUIPMENT PHASING AND CONNECTIONS PRIOR TO WORK.

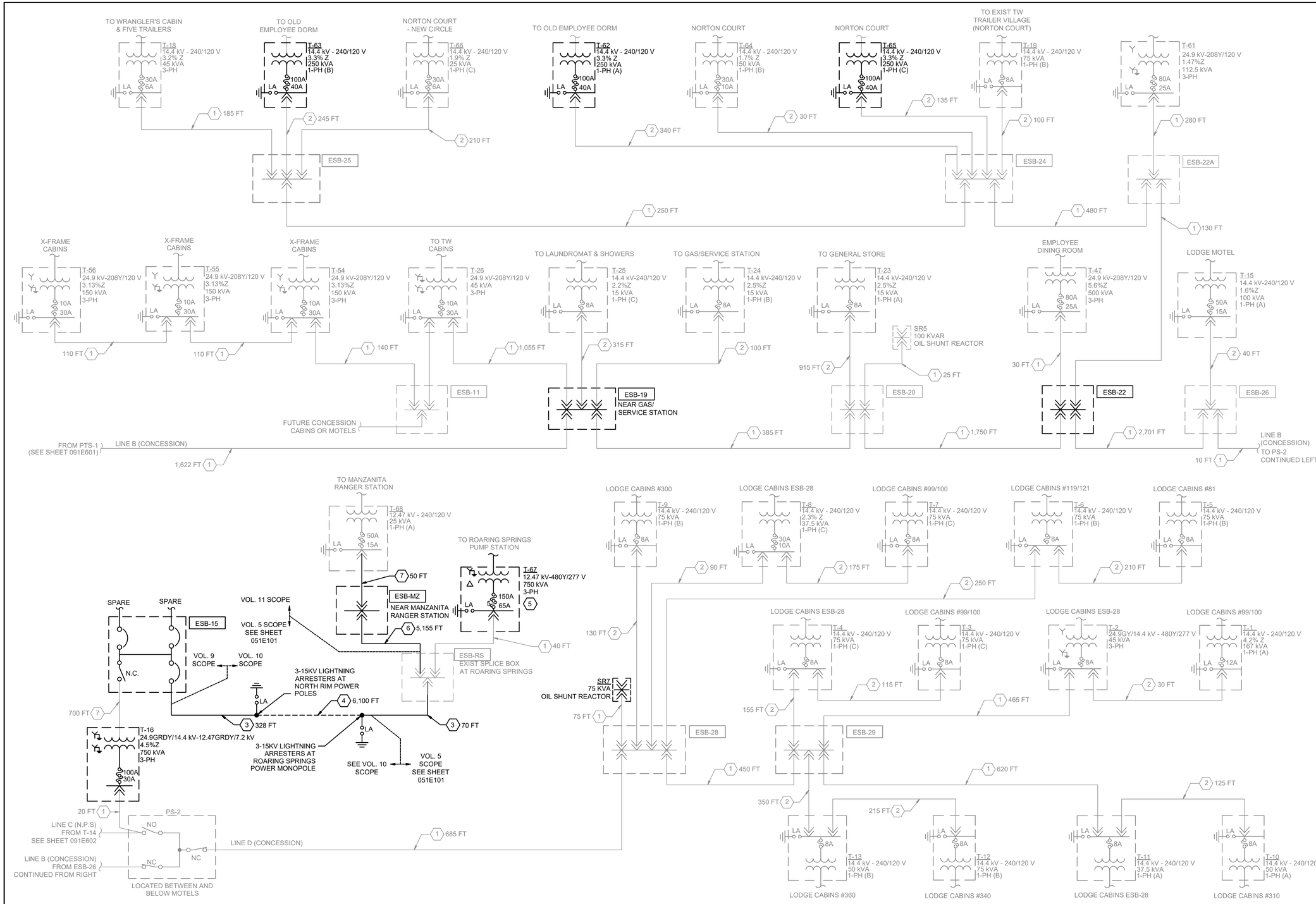
KEYNOTES (X)

1. EXIST 25kV (3/C + G) 3/0 AWG ALUMINUM ARMORED POWER CABLE.
2. EXIST 25kV (1/C + G) 3/0 AWG ALUMINUM ARMORED POWER CABLE.
3. 25kV (3/C + G) 4/0 AWG ALUMINUM ARMORED POWER CABLE. EACH INTERNAL CABLE TO HAVE CONCENTRIC NEUTRAL.



REGISTERED PROFESSIONAL ENGINEER OF ARIZONA
CERTIFICATE NO. 58037
TRAVIS T. MOORE
Date signed 04/30/2026
APPROVAL U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: AUG 2026	SUB SHEET NO. 091 E602	TITLE OF SHEET POWER DISTRIBUTION ONE-LINE DIAGRAM MODIFIED NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 18 OF 51



GENERAL NOTES:

A. REFER TO OVERALL NORTH RIM POWER DISTRIBUTION SYSTEM IMPROVEMENT PLANS IN THE 091E1 DRAWING SERIES.

B. LINES A AND C ARE OPERATED BY THE NATIONAL PARK SERVICES. LINES B AND D ARE OPERATED BY CONCESSIONS.

C. SEE VOLUME 0 FOR ONE–LINE LEGEND.

D. ALL MATERIALS AND INSTALLATION PRACTICES SHALL CONFORM TO IEEE, ANSI, AND NEMA STANDARDS.

E. CONTRACTOR SHALL VERIFY ALL DIMENSIONS AND CONDITIONS AT THE SITE BEFORE COMMENCING WORK.

F. CONTRACTOR SHALL COORDINATE WITH CONTRACTING OFFICER FOR ANY NECESSARY OUTTAGES AND TIE–INS.

G. CONTRACTOR SHALL PROVIDE ALL NECESSARY SAFETY MEASURES AND FOLLOW ALL OSHA REGULATIONS DURING CONSTRUCTION.

H. ALL WORK SHALL COMPLY WITH THE LATEST EDITION OF THE NATIONAL ELECTRIC CODE (NEC), NATIONAL ELECTRIC SAFETY CODE (NESC), DSC, NPS, AND APPLICATION FEDERAL, STATE, AND LOCAL ELECTRICAL CODES, REGULATIONS, ORDINANCES AND PROJECT STANDARDS.

I. TRANSFORMER FUSE SIZES AND PHASES BASED ON EXIST NPS DOCUMENTATION. CONTRACTOR TO FIELD VERIFY EQUIPMENT PHASING AND CONNECTIONS PRIOR TO WORK.

KEYNOTES ⟨X⟩

1. EXIST 25kV (3/C + G) 3/0 AWG ALUMINUM ARMORED POWER CABLE.
2. EXIST 25kV (1/C + G) 3/0 AWG ALUMINUM ARMORED POWER CABLE.
3. 15kV (3–1/C + G) 4/0 AWG ALUMINUM URD CABLE WITH CONCENTRIC NEUTRAL IN (3) 3" CONDUIT. SEE VOLUME 10 FOR INSTALLATION DETAILS BETWEEN ESB–15 AND NORTH RIM POWER POLES. SEE VOLUME 5 FOR INSTALLATION BETWEEN ROARING SPRINGS MONOPOLE AND ESB–RS.
4. SEE VOLUME 10 FOR OVERHEAD INSTALLATION DETAILS.
5. SEE VOLUME 5 FOR TRANSFORMER T–67 REPLACEMENT DETAILS.
6. 15kV (3/C) 1/0 AWG ALUMINUM ARMORED POWER CABLE. EACH INTERNAL CABLE TO HAVE CONCENTRIC NEUTRAL. SEE VOLUME 5 AND 11 FOR INSTALLATION DETAILS.
7. EXIST 15kV (3–1/C + G) #2 AWG ALUMINUM ARMORED POWER CABLE.
8. 15KV (1/C) 1/0 AWG ALUMINUM URD CABLE WITH FULL CONCENTRIC NEUTRA IN 3" PVC CONDUIT.

REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL) CERTIFICATE NO. 75014 TOU VANG Date Signed

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: JUNE 2026	SUB SHEET NO. 091 E603	TITLE OF SHEET POWER DISTRIBUTION ONE–LINE DIAGRAM MODIFIED NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 19 OF 51

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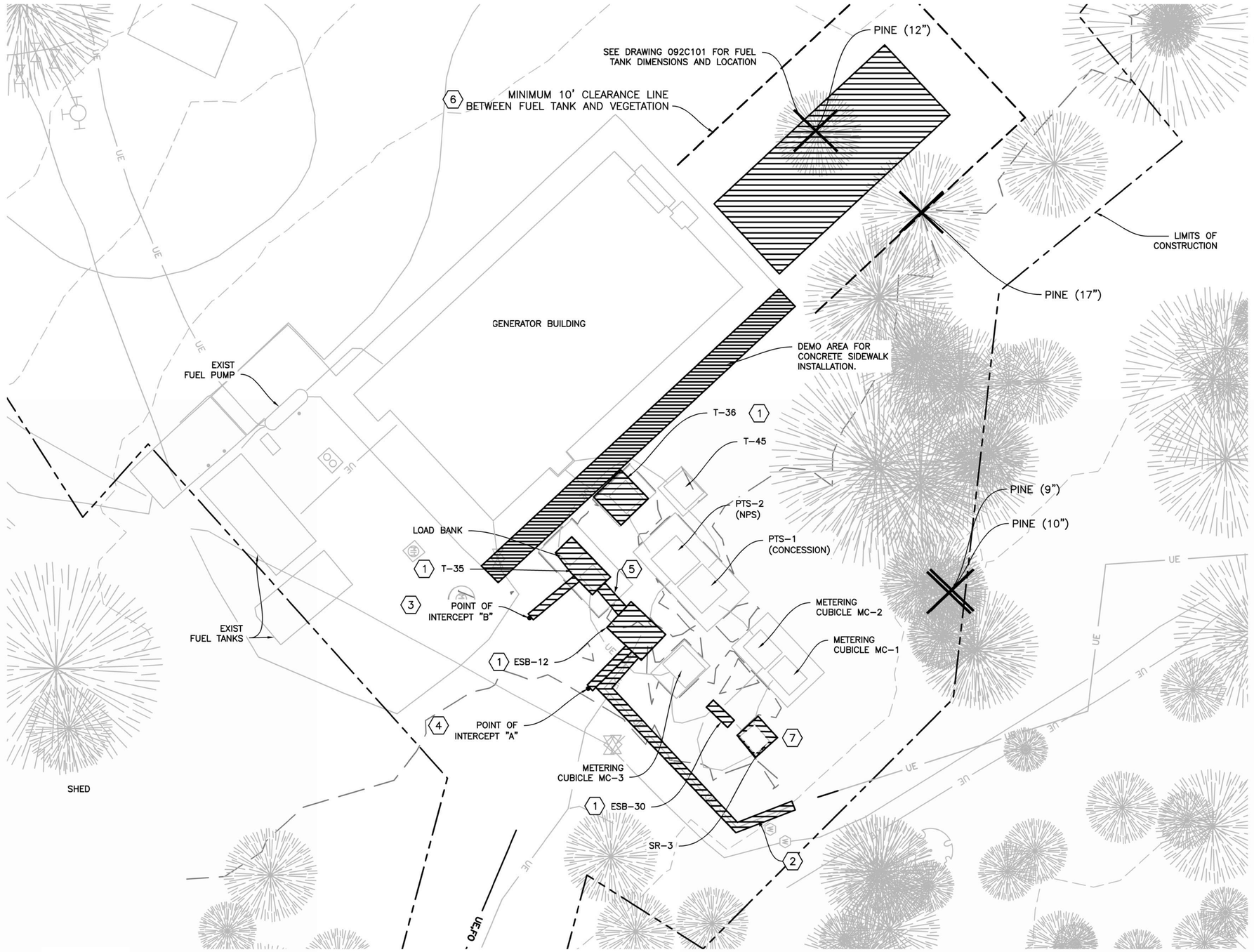

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. REFER TO OVERALL NORTH RIM POWER DISTRIBUTION SYSTEM ONE-LINE DRAWINGS 091E601 TO 091E603 FOR ADDITIONAL INFORMATION ON POWER SYSTEM IMPROVEMENTS.

C. LOCATION OF ALL UNDERGROUND UTILITIES SHOWN ARE APPROXIMATE AND TO BE VERIFIED IN FIELD BY CONTRACTOR.

D. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BE THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.

E. PROVIDE ROOT PRESERVATION AND TREE/PLANT PROTECTION. CONTRACTOR SHALL PROTECT TREES IN PLACE IF NOT NEEDED FOR REMOVAL.

F. CABLE LENGTHS, ENCLOSURE LOCATIONS, AND NORTHING/EASTING COORDINATES ARE APPROXIMATE. FINAL EQUIPMENT LOCATIONS AND CABLE TRENCH LENGTHS TO BE VERIFIED IN FIELD BY CONTRACTOR AND APPROVED BY NPS CONTRACTING OFFICER.

G. PROVIDE TEMPORARY PROTECTION FOR REUSED EQUIPMENT PRIOR TO NEW EQUIPMENT INSTALLATIONS.

TREE TYPE

XXX (X")
- PINE = PONDEROSA PINE
- ASP = ASPEN

TREE TRUNK DIAMETER

INDICATES TREE TO BE DEMOLISHED

KEY NOTES: (X)

1. REMOVE EXIST PADMOUNT TRANSFORMER, JUNCTION ENCLOSURES, LOAD BANK, AND SWITCHGEAR. REFER TO SHEET 092E101 FOR ELECTRICAL DISTRIBUTION IMPROVEMENT PLANS AT THE NORTH RIM POWER DELIVERY CENTER.
2. LOCATE EXIST UNDERGROUND MEDIUM VOLTAGE POWER CABLE BETWEEN EXIST JUNCTION ENCLOSURE ESB-12 AND ESB-13. EXCAVATE AND EXPOSE CABLE FROM ESB-12 TO SWITCHGEAR ESB-12A LOCATION. CABLE TO BE REROUTED AND REUSED AS SHOWN ON SHEET 091E101.
3. LOCATE EXIST UNDERGROUND MEDIUM VOLTAGE POWER CABLE BETWEEN EXIST JUNCTION ENCLOSURE ESB-16 AND TRANSFORMER T-35. EXCAVATE AND EXPOSE CABLE FROM T-35 TO POINT OF INTERCEPT "B".
4. LOCATE EXIST UNDERGROUND MEDIUM VOLTAGE POWER CABLE BETWEEN EXIST JUNCTION ENCLOSURES ESB-12 AND ESB-14. EXCAVATE AND EXPOSE CABLE FROM ESB-12 TO POINT OF INTERCEPT "A". CABLE TO BE SPLICED AND REROUTED AS SHOWN ON SHEET 092E101.
5. LOCATE AND REMOVE EXIST UNDERGROUND MEDIUM VOLTAGE POWER CABLE AND TERMINATIONS BETWEEN EXIST JUNCTION ENCLOSURE ESB-12 AND TRANSFORMER T-35. RECONNECT AS SHOWN ON SHEET 092E101.
6. MAINTAIN MINIMUM 10 FEET SEPARATION BETWEEN FUEL TANK AREA AND ADJACENT VEGETATION.
7. REMOVE AND REPLACE DAMAGED FOUNDATION PAD UNDERNEATH SR-3.

10 0 10 20
SCALE OF FEET 1" = 10'

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO. 75014 TOU VANG
State of Arizona

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	T. VANG CADD D. VANG TECH. REVIEW: S. EVERETT DATE: MAY 2026	092 X101	POWER DISTRIBUTION CENTER DEMOLITION PLAN POWER DISTRIBUTION CENTER AREA NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 20 OF 51

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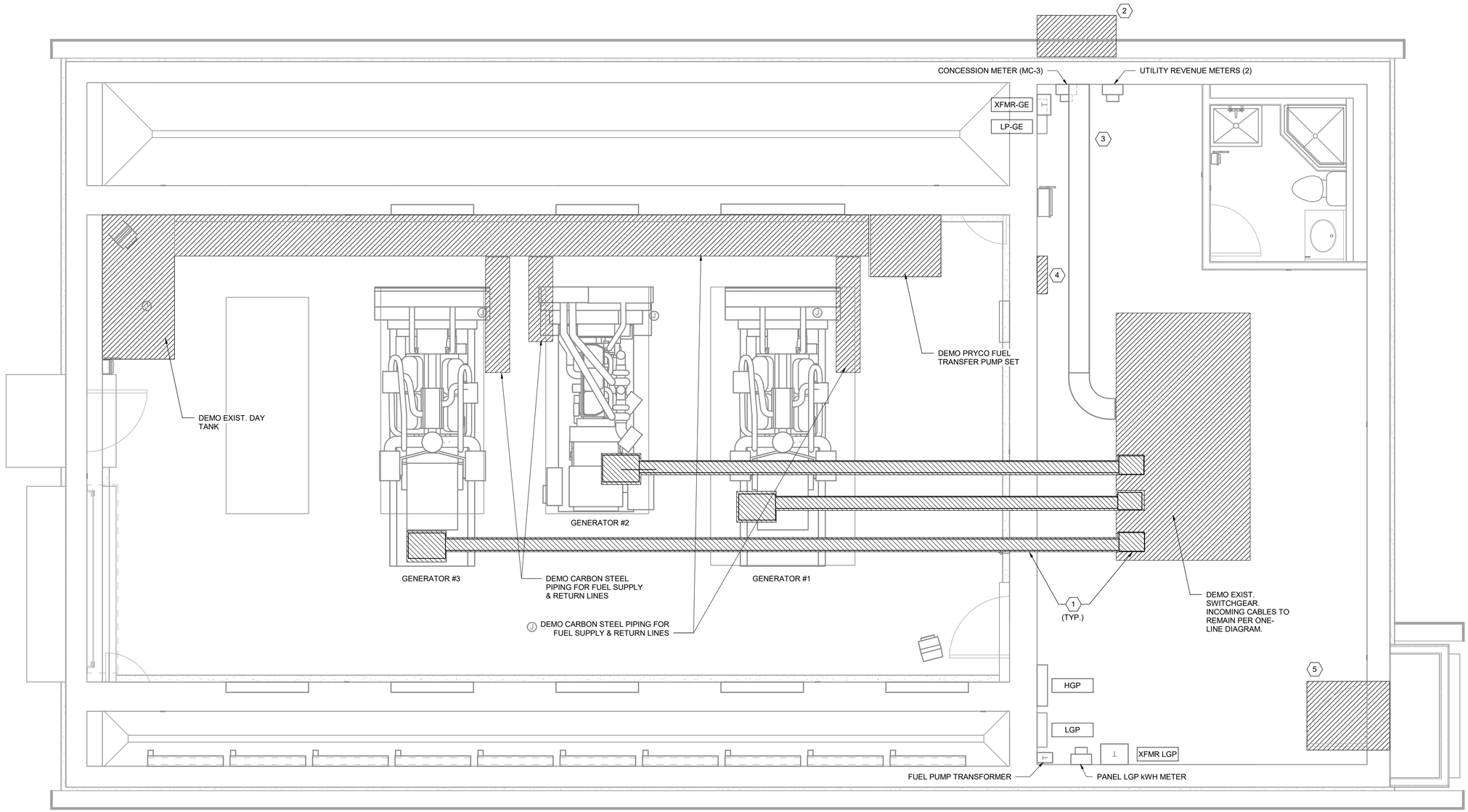

GENERAL NOTES:

A. REMOVE EXISTING SWITCHGEAR, AND ALL ASSOCIATED RACEWAYS, CIRCUITS, HARDWARE, AND ACCESSORIES.

B. REMOVE ALL RACEWAYS, CIRCUITS, HARDWARE, AND ACCESSORIES ASSOCIATED WITH DEMOLISHED EQUIPMENT AND SYSTEMS.

KEY NOTES: (X)

1. REMOVE EXISTING BUSWAY SYTEMS BETWEEN GENERATORS AND SWITCHGEAR, INCLUDING TAP BOXES, WALL PENETRATIONS, ANCILLARY CONDUCTORS, AND ACCESSORY ITEMS.
2. REMOVE EXISTING EXTERIOR WIREWAY.
3. MODIFY EXISTING CABLE TRAY AS REQUIRED FOR INSTALLATION OF NEW EQUIPMENT. PROTECT EXISTING CABLES FROM SWITCHGEAR TO TRANSFORMER T-45 FOR REUSE AND RECONNECTION TO NEW SWITCHGEAR.
4. REMOVE LOAD BANK CONTROL PANEL.
5. REMOVE DOOR, DOOR FRAME AND TRANSOM. DOOR WILL BE REPLACED WITH A NEW DOOR THAT SWINGS OUTWARD.

SCALE (A) 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

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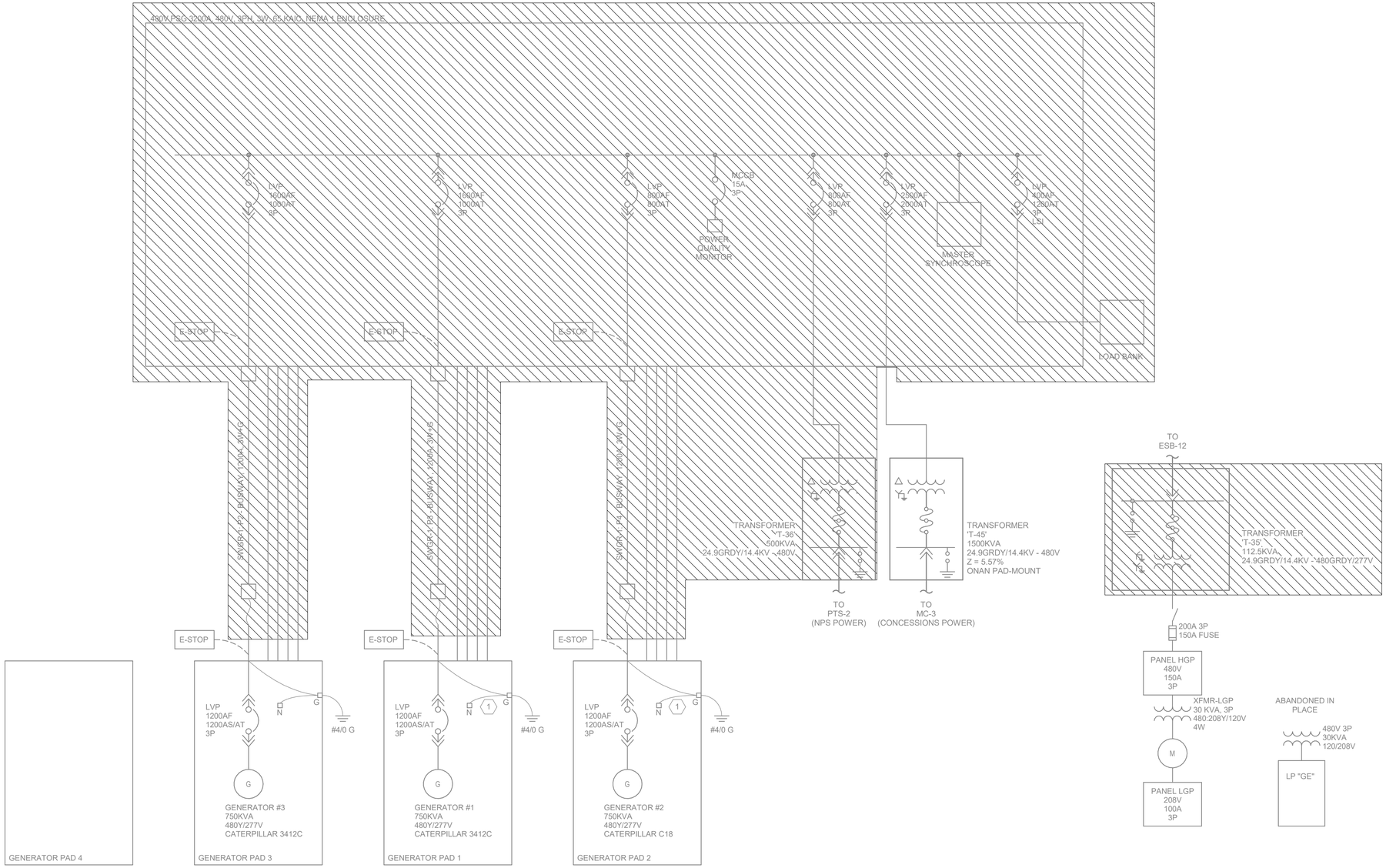
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. MCDONALD CADD J. STUJENSKE TECH. REVIEW: J. FERGUSON DATE: MAY 2026	SUB SHEET NO. 092 X102	TITLE OF SHEET POWER DISTRIBUTION CENTER DEMOLITION PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 21 OF 51

GENERAL NOTES:

A. REFER TO SHEET 092E603 FOR ADDITIONAL SYSTEM CONTROL, INSTRUMENT, AND NETWORK WIRING.

B. REFER TO SHEET 092Y601 FOR SCADA NETWORK BLOCK DIAGRAM.



REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL) CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 8/13/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO DRAWN: A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 092 X601	TITLE OF SHEET POWER DISTRIBUTION CENTER GENERATOR BUILDING DEMOLITION ONE-LINE DIAGRAM NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 22 OF 51

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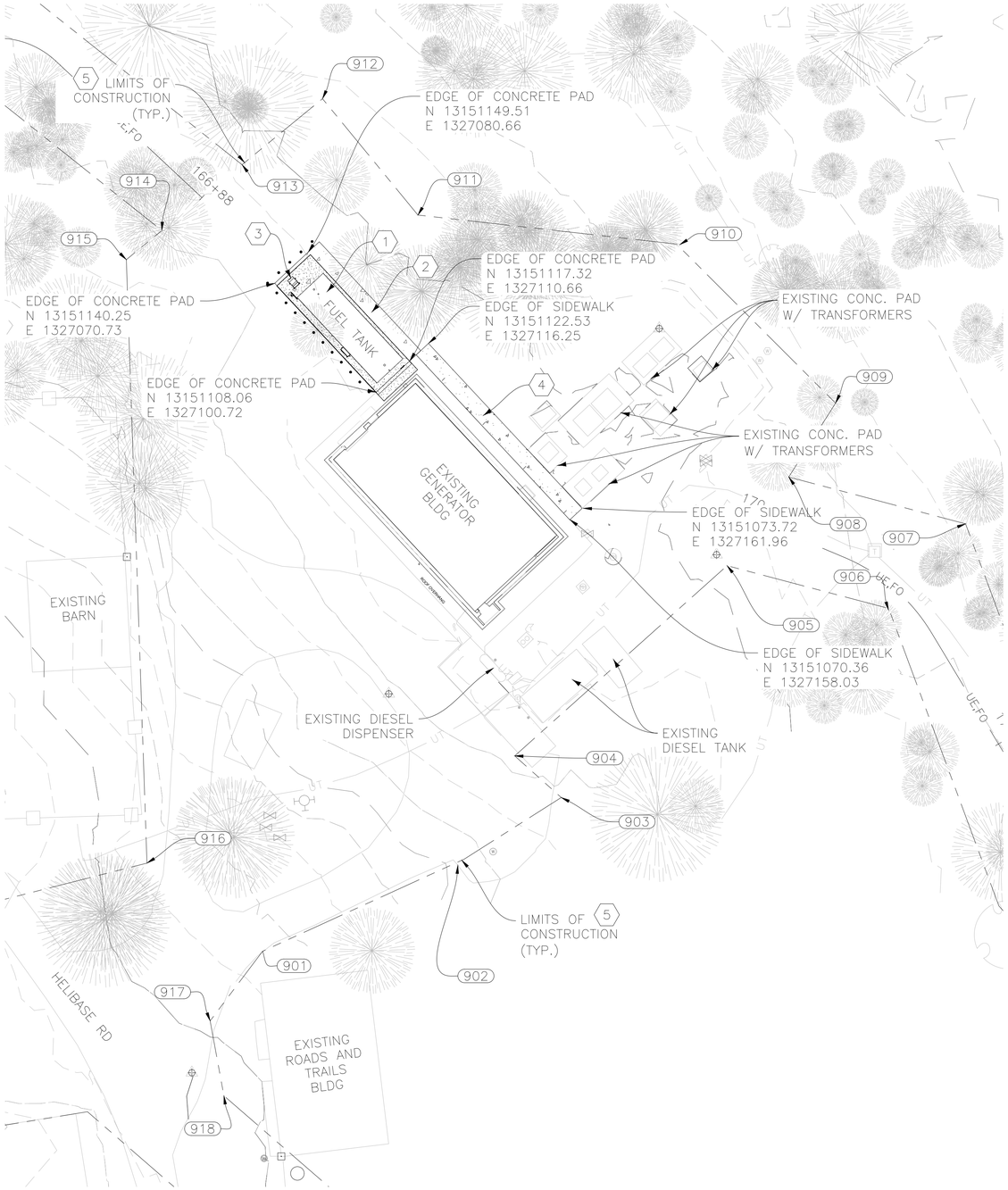

LIMITS OF CONSTRUCTION		
POINT NO.	NORTHING	EASTING
901	13150942.75	1327067.13
902	13150968.21	1327124.87
903	13150988.21	1327155.43
904	13151000.51	1327141.68
905	13151056.76	1327204.48
906	13151043.61	1327252.20
907	13151069.12	1327274.98
908	13151082.97	1327222.70
909	13151104.82	1327236.54
910	13151151.66	1327190.48
911	13151160.42	1327113.06
912	13151194.82	1327084.85
913	13151175.31	1327061.06
914	13151155.85	1327037.34
915	13151147.24	1327026.85
916	13150968.82	1327032.95
917	13150922.17	1327051.63
918	13150900.03	1327055.70

GENERAL NOTES:

A. CONTRACTOR TO RE-GRAVEL DISTURBED AREAS WITH A COMPACTED ADOT CLASS 2 AGGREAGATE BASE COURSE.

KEY NOTES: (X)

1. NEW 18' WIDE X 44' LONG 20,000 GALLON DIESEL FUEL TANK.
2. CONCRETE PAD AND FOUNDATION PER STRUCTURAL DRAWINGS.
3. DIESEL FUEL FILL BOX.
4. 5'X6" THICK CONCRETE SIDEWALK. TIE NEW SIDEWALK INTO EXISTING SIDEWALK USING LONGITUDINAL JOINTING BAR #4@18" O.C.
   - 4.1. QUANTITY = 333 SF
5. AVAILABLE CONSTRUCTION LIMITS ARE SHOWN IN THE LIMITS OF CONSTRUCTION TABLE.

20 0 20 40 1" = 20'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
Date Signed
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: P. OXBORROW TECH. REVIEW: J. GLOVER DATE: MAY 2026	SUB SHEET NO. 092 C101	TITLE OF SHEET POWER DISTRIBUTION CENTER SITE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 23 OF 51

4/15/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\092-C101.dwg

GENERAL

G1. SCOPE
THE NOTES ON THIS SHEET AND THE STANDARD STRUCTURAL DETAILS ARE GENERAL AND APPLY TO THE ENTIRE PROJECT WHETHER SPECIFICALLY CALLED OUT OR NOT, EXCEPT WHERE THERE ARE SPECIFIC INDICATIONS TO THE CONTRARY ON STRUCTURAL SHEETS. IF THERE ARE QUESTIONS, THEY SHALL BE SUBMITTED TO THE STRUCTURAL ENGINEER AND ANSWERED IN WRITING PRIOR TO CONSTRUCTION.

G2. APPLICABLE SPECIFICATIONS AND CODES
- A. INTERNATIONAL BUILDING CODE, IBC 2024, WITH APPLICABLE EDITIONS OF THE CODE REFERENCED STANDARDS.
- B. AUTHORITY HAVING JURISDICTION: NATIONAL PARK SERVICE (WIND, SNOW, AND SEISMIC DESIGN CRITERIA IN ACCORDANCE WITH THE MORE CONSERVATIVE OF COCONINO COUNTY, AZ AND NATIONAL PARK SERVICE.)
- C. BUILDING CODE REQUIREMENTS FOR REINFORCED CONCRETE, ACI 318-19.
- D. MINIMUM DESIGN LOADS AND ASSOCIATED CRITERIA FOR BUILDINGS AND OTHER STRUCTURES, ASCE 7-22.
- E. RCSC'S SPECIFICATION FOR STRUCTURAL JOINTS USING HIGH-STRENGTH BOLTS.
- F. AMERICAN WELDING SOCIETY, AWS D1.1.
- G. ASTM MATERIAL STANDARDS AS NOTED.

G3. DESIGN CRITERIA
1. APPLIES TO ALL STRUCTURES (UNO)
- A. DEAD LOAD:
  - 1. ACTUAL TRIBUTARY STRUCTURE WEIGHT
  - 2. ROOF SUPERIMPOSED DEAD LOAD: 20 PSF
- B. LIVE LOAD:
  - 1. WALKWAYS, STAIRS, GRATING: 100 PSF
  - 2. SLAB ON GRADE: 250 PSF
  - 3. STORAGE ABOVE CEILING: 20 PSF
  - 4. ROOF: 20 PSF (NOT REDUCIBLE)
- C. WIND:
  - 1. ULTIMATE WIND SPEED (V_{ult}): 111 MPH
  - 2. TORNADO SPEED (V_T): NOT APPLICABLE PER ASCE 7 FIGURES 32.1-1 AND 32.1-2
  - 3. NOMINAL WIND SPEED (V_{ASD}): 78 MPH
  - 4. RISK CATEGORY: IV
  - 6. EXPOSURE: C
  - 7. INTERNAL PRESSURE COEFFICIENT (GC_{pi}): ±0.18
  - 8. DESIGN WIND PRESSURE: 16 PSF
- D. SEISMIC:
  - 1. DIESEL GENERATOR FUEL PAD:
    - a. RISK CATEGORY: III
    - b. IMPORTANCE FACTOR (I_e): 1.5
    - c. SPECTRAL RESPONSE ACCELERATION, S_S = 0.41g
    - d. SPECTRAL RESPONSE ACCELERATION, S_1 = 0.11g
    - e. SITE CLASS: C
    - f. SEISMIC DESIGN CATEGORY: B
    - g. SPECTRAL RESPONSE ACCELERATION, S_{DS} = 0.3g
    - h. SPECTRAL RESPONSE ACCELERATION, S_{D1} = 0.107g
    - i. BASIC SEISMIC FORCE RESISTING SYSTEM: ASCE 7 CHAPTER 15 HORIZONTAL, SADDLE-SUPPORTED WELDED STEEL VESSELS
    - j. ANALYSIS PROCEDURE: EQUIVALENT LATERAL FORCE
    - k. DESIGN BASE SHEAR: 28.5 KIP
    - l. SEISMIC RESPONSE COEFFICIENT (C_s) = 0.15
    - m. RESPONSE MODIFICATION FACTOR (R) = 3.0
    - n. OVERSTRENGTH FACTOR = 2.0
- E. SNOW LOAD:
  - 1. GROUND SNOW LOAD (p_g) = 256 PSF
  - 2. EXPOSURE FACTOR (C_e) = 1.0
  - 3. SLOPED ROOF FACTOR (C_s) = 1.0
  - 4. THERMAL FACTOR (C_t) = 1.1
  - 3. RISK CATEGORY = IV
  - 4. FROST DEPTH = 36 INCHES
- F. ICE LOADS:
  - 1. ICE THICKNESS: 0.02 IN
  - 2. ICE THICKNESS HEIGHT FACTOR: 1.00
  - 3. DESIGN ICE THICKNESS: 0.02 IN
  - 4. DESIGN WEIGHT OF ICE: 0.09 PSF
- G. FLOOD CRITERIA: NOT APPLICABLE
- H. ROOF RAIN LOAD
  - 1. 15-MINUTE PRECIPITATION INTENSITY: 8.6 IN/HR
  - 2. 60-MINUTE PRECIPITATION INTENSITY: 3.58 IN/HR

G4. THE FOLLOWING NON-CONTRACTUAL GEOTECHNICAL REPORT WAS DEVELOPED FOR THIS PROJECT AND IS THE BASIS OF THIS STRUCTURAL DESIGN:

GEOTECHNICAL FIRM NAME:	LITHOS ENGINEERING
ADDRESS:	2750 S. WADSWOTH BLVD., SUITE D-200 DENVER, CO 80227
REPORT NUMBER:	21186
REPORT DATE:	JANUARY 16, 2024
MAXIMUM ALLOWABLE SOIL BEARING =	3,000 PSF
COEFFICIENT OF FRICTION SLIDING =	0.35
ACTIVE LATERAL EARTH PRESSURE =	42 PCF
AT-REST LATERAL EARTH PRESSURE =	63 PCF
PASSIVE LATERAL EARTH PRESSURE =	200 PCF

G5. SAFETY
SAFETY AND STRUCTURE STABILITY DURING CONSTRUCTION ARE THE SOLE RESPONSIBILITY OF THE CONTRACTOR. STRUCTURES HAVE BEEN DESIGNED TO RESIST THE DESIGN LIVE LOADS ONLY AS A COMPLETED STRUCTURE.

G6. OPENINGS
OPENINGS FOR PIPES, DUCTS, CONDUITS, ETC. ARE NOT ALL SHOWN ON THE STRUCTURAL DRAWINGS. COORDINATE AND PROVIDE OPENINGS AS REQUIRED TO ACCOMMODATE ALL WORK SHOWN OR SPECIFIED IN THE CONTRACT DOCUMENTS AND OTHERWISE REQUIRED FOR THE FURNISHING OF A FUNCTIONALLY COMPLETE PROJECT. REINFORCE AROUND OPENINGS PER STANDARD STRUCTURAL DETAILS UNLESS OTHERWISE SHOWN.

G7. SPECIAL INSPECTIONS
SPECIAL INSPECTIONS ARE REQUIRED IN ACCORDANCE WITH CHAPTER 1 AND CHAPTER 17 OF THE IBC. PAYMENT FOR THESE INSPECTIONS IS NOT THE RESPONSIBILITY OF THE CONTRACTOR IN ACCORDANCE WITH THE REQUIREMENTS OF SPECIFICATION SECTION 01 40 00. THE CONTRACTOR SHALL PROVIDE FOR FULL ACCESS TO THE WORK BY THE SPECIAL INSPECTOR AND SHALL PROVIDE FOR THESE INSPECTIONS IN THE OVERALL CONSTRUCTION SCHEDULE. THE SPECIAL INSPECTION PROGRAM SHALL BE IN ACCORDANCE WITH SPECIFICATION SECTION 01 40 00 AND "STATMENT OF STRUCTURAL TESTS AND SPECIAL INSPECTIONS" DOCUMENTS INCLUDED AS ATTACHMENTS TO SPECIFICATION SECTION 01 40 00.

GENERAL CONTINUED:

G8. STANDARD DETAILS
THE STANDARD DETAILS IN VOLUME 00 DEPICT TYPICAL DETAILING TO BE USED ON THIS PROJECT. IF CONDITIONS ARE NOT EXPLICITLY SHOWN ON THE DRAWINGS THEY SHALL BE MADE SIMILAR TO THE STANDARD DETAILS. OBTAIN APPROVAL OF CONTRACTING OFFICER IN WRITING FOR SIMILAR CONDITIONS PRIOR TO CONSTRUCTION.

G9. THE CONTACTOR SHALL FIELD VERIFY ALL DIMENSIONS AND ELEVATIONS OF EXISTING CONSTRUCTION AS REQUIRED TO COORDINATE NEW CONSTRUCTION. SUBMIT REQUIRED CHANGES FOR APPROVAL.

G10. CONTRACTOR TO SUBMIT FOR REVIEW ALL EQUIPMENT SIZES, OPERATING WEIGHTS, VIBRATION FORCES, SUPPORT LOCATIONS, ALONG WITH ANY FLOOR OPENINGS, NOTCHES, AND RECESSES REQUIRED BY SUCH EQUIPMENT. CONCRETE SUPPORT PADS AND/OR FRAMING REQUIRED TO SUPPORT SAID EQUIPMENT SHALL NOT BE FABRICATED AND PLACED UNTIL THE CONCRETE SUPPORT PADS AND/OR FRAMING IS APPROVED TO SUPPORT THE EQUIPMENT.

G11. THE CONTRACTOR SHALL FIELD VERIFY ALL DIMENSIONS AND ELEVATIONS OF EXISTING CONSTRUCTION AS REQUIRED TO COORDINATE NEW CONSTRUCTION. SUBMIT REQUIRED CHANGES FOR APPROVAL. TEH PLANS AND DETAILS FOR THE EXISTING STRUCTURES PRESENTED IN THE DRAWINGS ARE FOR REFERENCE ONLY AND BASED ON THE FOLLOWING AS-BUILT DRAWINGS:

1. NPS GENERATOR BUILDING, DATED JAN 1987.

CONCRETE

C1. DESIGN STRENGTHS:
F'c = 4,500 PSI (UNO)
Fy = 60,000 PSI REINFORCEMENT BARS - ASTM A615, GR 60

C2. CONCRETE COVER
UNLESS OTHERWISE NOTED, PROVIDE CONCRETE COVER FOR REINFORCING AS FOLLOWS:

CONCRETE DEPOSITED AGAINST EARTH:	3"
ALL OTHER:	2"
WATERSTOPS:	3 1/2"

SEE DRAWINGS FOR EXCEPTIONS

C3. SEE SPECIFICATIONS FOR REINFORCING PLACEMENT REQUIREMENTS.

C4. REFER TO OTHER DISCIPLINE DRAWINGS PRIOR TO CONSTRUCTION FOR EMBEDDED ITEMS AND PENETRATIONS NOT SHOWN ON STRUCTURAL DRAWINGS. AS REQUIRED TO ACCOMMODATE ALL WORK SHOWN OR SPECIFIED IN THE CONTRACT DOCUMENTS AND OTHERWISE REQUIRED FOR THE FURNISHING OF A FUNCTIONALLY COMPLETE PROJECT. REINFORCE AROUND OPENINGS PER STANDARD STRUCTURAL DETAILS UNLESS OTHERWISE SHOWN.

C5. PROVIDE 3/4" CHAMFERS AT ALL EXPOSED EDGES. NOT ALL CHAMFERS MAY BE SHOWN ON DRAWINGS.

C6. FIELD ADJUST REINFORCING AT OPENINGS AND EMBEDDED ITEMS AS INDICATED.

C7. ANCHOR BOLTS NOT SPECIFIED BY ENGINEER SHALL BE DESIGNED AND CERTIFIED BY A REGISTERED PROFESSIONAL ENGINEER LICENSED IN THE STATE OF ARIZONA, RETAINED BY THE CONTRACTOR, IN ACCORDANCE WITH APPLICABLE PROJECT AND CODE REQUIREMENTS. SUBMIT AS A SHOP DRAWING FOR REVIEW AND APPROVAL BY THE ENGINEER. COORDINATE LOCATION, SIZE AND EMBEDMENT PRIOR TO CASTING CONCRETE.

C8. CONTINUOUS WATERSTOP SHALL BE INSTALLED IN JOINTS SUBJECT TO STATIC WATER PRESSURE.

C9. ABSOLUTELY NO WELDING OF REINFORCING BARS OR TORCHING TO BEND REINFORCING BARS SHALL BE ALLOWED.

C10. CONTRACTOR SHALL SUBMIT A CONCRETE PLACEMENT PLAN{PER SPECIFICATIONS IDENTIFYING JOINT TYPES, JOINT LOCATIONS AND CONCRETE PLACEMENT SEQUENCE.

C11. ALL CAST IN PLACE AND POST-INSTALLED ANCHORS INDICATED IN THE STRUCTURAL DOCUMENTS SHALL COMPLY WITH CHAPTER 17 OF ACI 318, APPENDIX E OF ACI 350, AND CHAPTER 19 OF THE IBC. ALL EXPANSION AND ADHESIVE ANCHORS SHALL HAVE THE ICC REPORT SHOWING EQUIVALENT LOAD CAPACITY. SUBMIT AND INSTALL PER THE ICC EVALUATION REPORT.

EARTHWORK:

E1. ALL EARTHWORK AND SITE PREPERATION SHALL BE PERFORMED IN STRICT ACCORDANCE WITH THE SPECIFICATIONS. ALL FOUNDATION EXCAVATIONS SHALL BE OBSERVED AND APPROVED BY THE LICENSED GEOTECHNICAL ENGINEER.

E2. ANY EXISTING FILLS OR UNSUITABLE SOILS AS DETERMINED BY THE GEOTECHNICAL ENGINEER SHALL BE EXCAVATED AND REPALCED WITH PROPERLY COMPACTED GRANULAR FILL PER SPECIFICATIONS.

E3. CONTRACTOR SHALL EXERCISE EXTREME CARE WHEN EXCAVATING OR GRADING ADJACENT TO EXISTING STRUCTURES SO AS TO NOT DAMAGE OR UNDERMINE FOUNDATIONS, WALLS, SLABS, UTILITIES, ETC.

E4. DO NOT EXCAVATE BELOW THE BEARING ELEVATION OF ANY COMPLETED FOOTING NOR ANY CLOSER TO THE FOOTING THAN A SLOPE OF TWO HORIZONTAL (MEASURED FROM EDGE OF FOOTING TO NEAREST POINT IN EXCAVATION) TO ONE VERTICAL.

DELEGATED DESIGN SUBMITTALS:

D1. PORTIONS OF THE STRUCTURE HAVE ELEMENTS OF PROPRIETARY DESIGN AND FABRICATION, WHICH SHALL BE SUBMITTED BY THE SUPPLIER FOR APPROVAL.

D2. DELEGATED DESIGN SUBMITTAL WILL BE REVIEWED AND APPROVED BY THE CONTRACTING OFFICER FOR COMPLIANCE WITH THE SPECIFIED DESIGN REQUIREMENTS.

D3. DELEGATED DESIGN SUBMITTAL ITEMS SHALL NOT BE INSTALLED UNTIL THEIR DESIGN AND SUBMITTED DOCUMENTS HAVE BEEN APPROVED BY THE AUTHORITY HAVING JURISDICTION.

D4. DELEGATED DESIGN SUBMITTALS INCLUDE:
1. STEEL FUEL STORAGE TANK AND TANK ANCHORAGE

Registered Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK STATE OF ARIZONA Date Signed 05/4/2026

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. PEREZ CADD R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 092 S001	TITLE OF SHEET POWER DISTRIBUTION CENTER GENERAL NOTES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 24 OF 51

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GENERAL NOTES:

A. REFER TO SHEET 092S001 FOR GENERAL NOTES.

B. COORDINATE THE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR PIPING EQUIPMENT, ANCHORS, CONDUITS, HOUSEKEEPING PADS, AND ANY OTHER ITEMS ATTACHED TO OR EMBEDDED IN STRUCTRUAL ELEMENTS THAT MAY NOT BE SHOWN.

C. SEE CIVIL DRAWINGS FOR PAD COORDINATES.

KEYNOTES

1 18" THICK CONCRETE SLAB. REINFORCE WITH #6@10", EACH WAY, TOP AND BOTTOM. PLACE SHORTER BARS IN OUTER MAT. PROVIDE STANDARD HOOKS EACH END OF TOP BARS.

2 ABOVE GROUND FUEL STORAGE TANK. REFER TO MECHANICAL DRAWINGS FOR PLACEMENT.

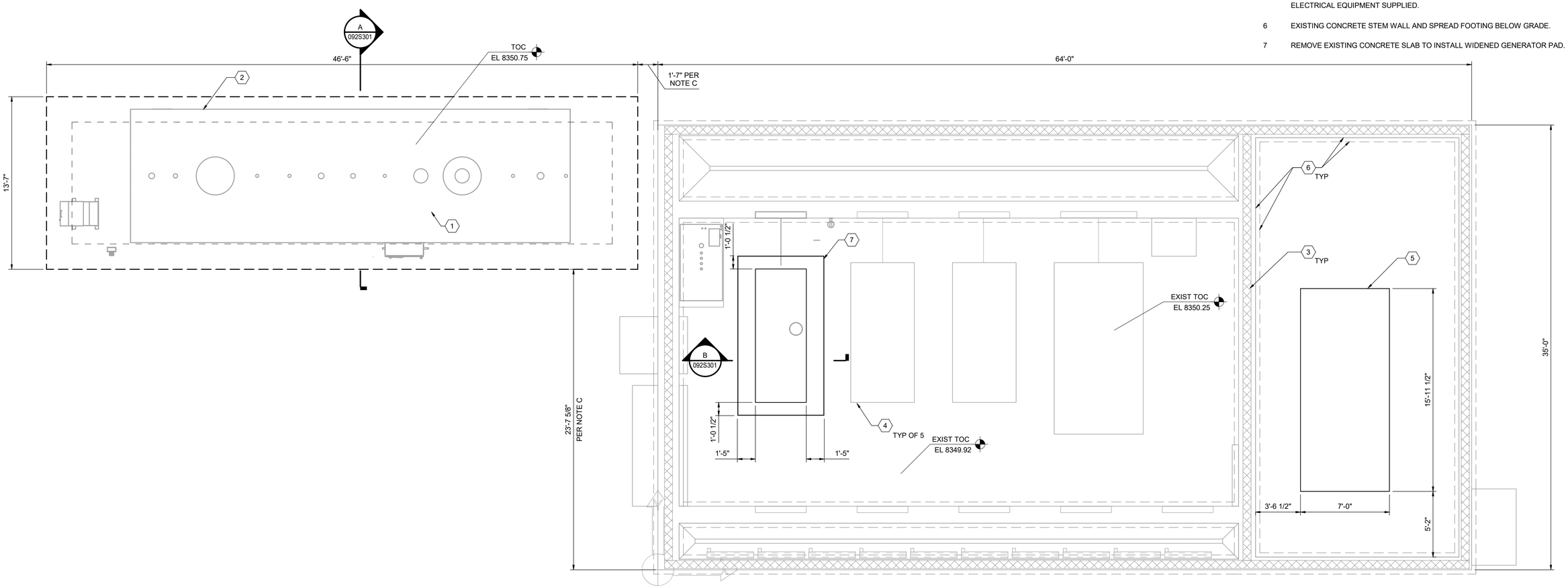
3 EXISTING CMU PERIMERTER WALLS.

4 EXSITING CONCRETE HOUSEKEEPING PADS.

5 NEW SHALLOW EQUIPMENT SUPPORT PAD. REFER TO VOLUME 00 FOR TYPICAL SUPPORT PAD DETAILS. COORDINATE SUPPORT PAD DIMENSIONS WITH ELECTRICAL EQUIPMENT SUPPLIED.

6 EXISTING CONCRETE STEM WALL AND SPREAD FOOTING BELOW GRADE.

7 REMOVE EXISTING CONCRETE SLAB TO INSTALL WIDENED GENERATOR PAD.



1 092S101 GROUND LEVEL PLAN
SCALE A

SCALE A 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil)
CERTIFICATE NO.
72480
COLIN MULDERICK
State Registered
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. PEREZ DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 092 S101	TITLE OF SHEET POWER DISTRIBUTION CENTER GROUND LEVEL PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 25 OF 51

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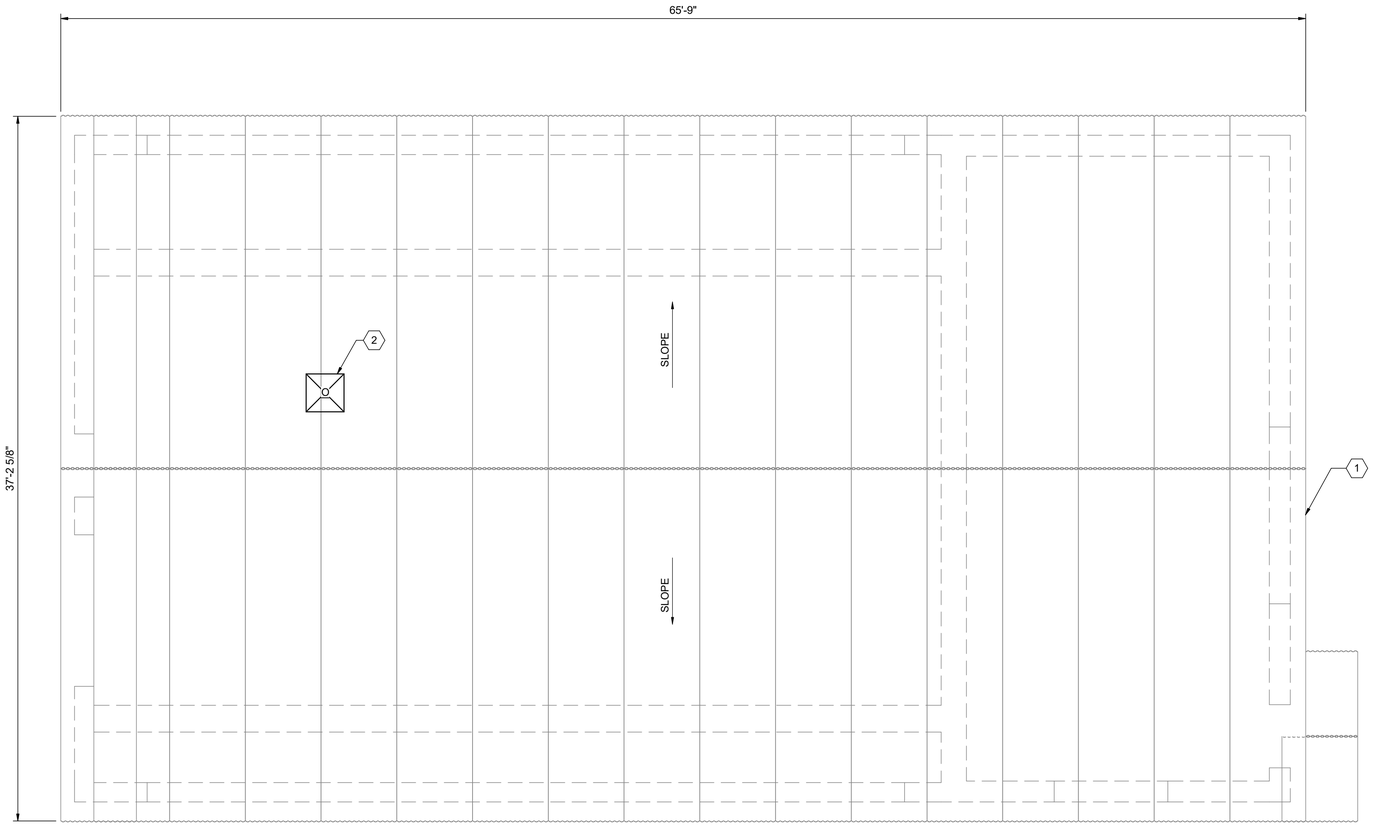

1 092S102 ROOF FRAMING PLAN
SCALE A

GENERAL NOTES:

A. REFER TO SHEET 092S001 FOR GENERAL NOTES.

B. COORDINATE THE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR PIPING EQUIPMENT, PENETRATIONS, AND ANY OTHER ITEMS ATTACHED TO OR EMBEDDED IN STRUCTRUAL ELEMENTS THAT MAY NOT BE SHOWN.

KEYNOTES

1 CONTRACTOR TO FIELD VERIFY SPACING OF EXISTING WOOD TRUSSES PRIOR TO CUTTTING EXISTING PLYWOOD DECKING FOR NEW OPENINGS.

2 NEW ROOF OPENING. COORDINATE SIZE AND LOCATION WITH MECHANICAL DRAWINGS. SEE DETAIL 2 ON SHEET 09S301 FOR OPENING SUPPORT FRAMING.

SCALE A 1 0 1 2 1" = 1'-0"
SCALE OF FEET

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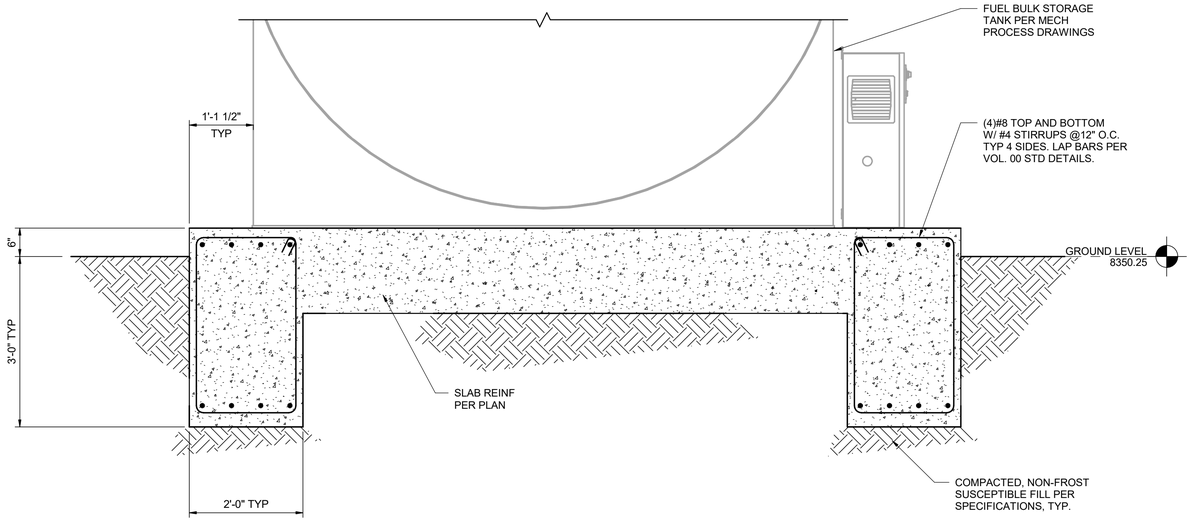
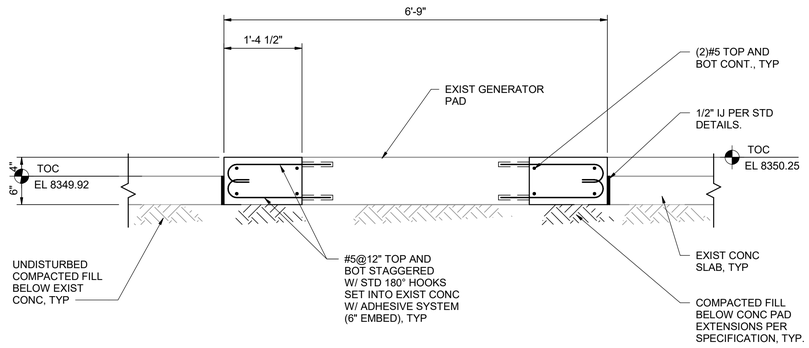
Registered Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK STATE OF ARIZONA Digitally signed: 04/17/2026

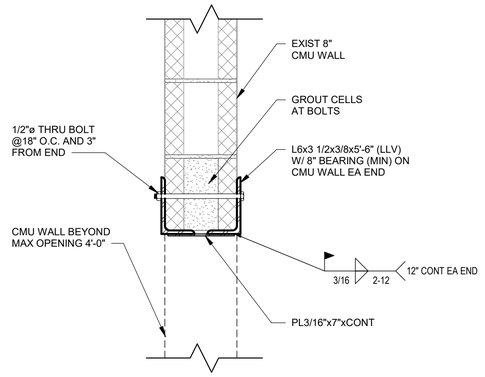
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. PEREZ DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 092 S102	TITLE OF SHEET POWER DISTRIBUTION CENTER ROOF FRAMING PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 26 OF 51



A 092S301 SECTION THRU BULK FUEL TANK FOUNDATION
SCALE (A)



B 092S301 SECTION THRU WIDENED GENERATOR PAD
SCALE (A)



NOTES:
1. REFER TO MECHANICAL DRAWINGS FOR NEW OPENING SIZE AND LOCATION.
2. MINIMUM SEPARATION BETWEEN BOTTOM OF DUCT OPENING AND TOP OF DOOR JAMB TO BE 18".

1 092S301 NEW OPENING SUPPORT AT EXISTING CMU WALL
SCALE (B)

SCALE (B) 12 6 0 12 1-1/2" = 1'-0"
SCALE OF FEET
SCALE (A) 1 0 1 2 3 3/4" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Utah) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed 04/17/2026

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. PEREZ CADD R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 092 S301	TITLE OF SHEET POWER DISTRIBUTION CENTER SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 27 OF 51

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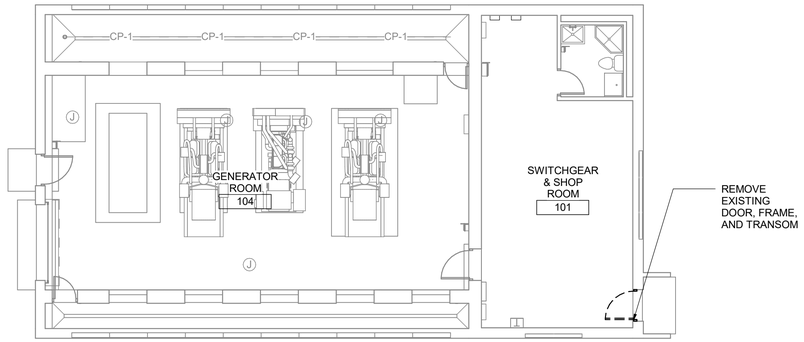
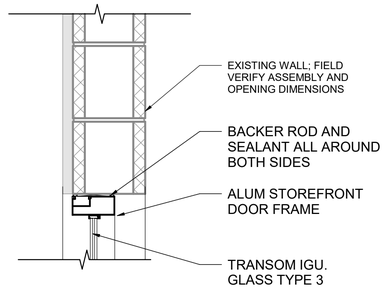
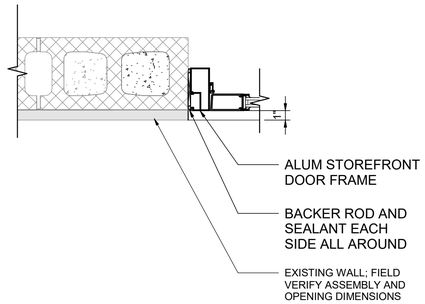


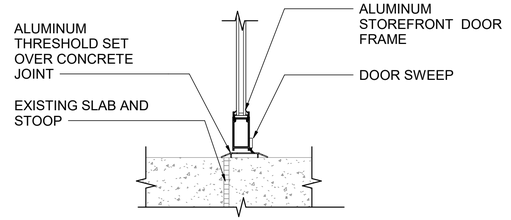
1 092A101 GROUND LEVEL FLOOR PLAN - DEMO
SCALE (A)



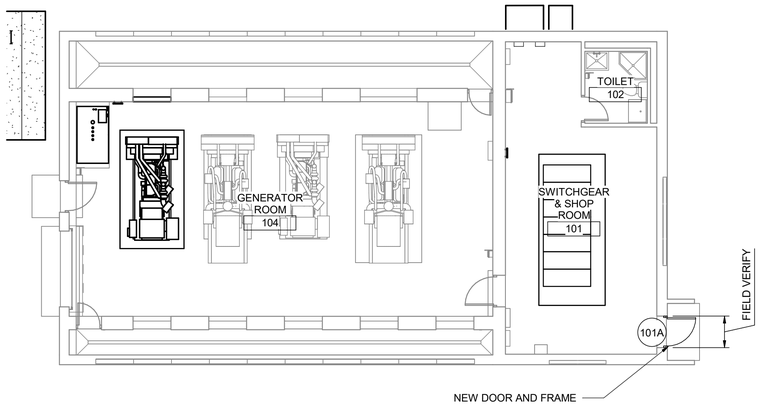
2 092A101 DOOR HEAD DETAIL
SCALE (B)



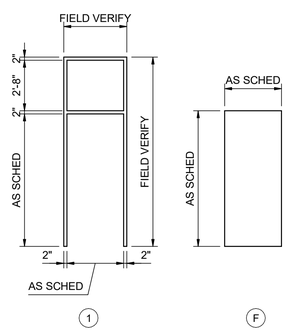
3 092A101 DOOR JAMB DETAIL
SCALE (B)



4 092A101 DOOR SILL DETAIL
SCALE (B)



5 092A101 GROUND LEVEL FLOOR PLAN
SCALE (A)



6 092A101 DOOR AND FRAME ELEVATIONS
NO SCALE

DOOR SCHEDULE

DOOR NUMBER	PR	WIDTH	HEIGHT	DOOR TYPE	DOOR MATERIAL	DOOR FINISH	FRAME TYPE	FRAME MATERIAL	FRAME FINISH	GLASS	RATING	HARDWARE SET	DETAILS HEAD	DETAILS JAMB	DETAILS SILL	*REMARKS
101A		3' - 0"	7' - 0"	F	AL	FAP/FRP	1	AL	FAP	3	-	1	2/092A101	3/092A101	4/092A101	A

MATERIAL AND FINISH LEGEND

MATERIAL		FINISH	
AL	ALUMINUM/FIBERGLASS REINFORCED POLYMER HYBRID	FAP	FACTORY APPLIED PAINT
		FRP	FIBERGLASS REINFORCED POLYMER

NOTES:
1. SEE DOOR AND FRAME TYPE DETAILS ON THIS SHEET FOR DOOR AND FRAME ELEVATIONS.
2. SEE SECTION 09 06 00 FOR COLOR SCHEDULE.

REMARKS:
A. TRANSOM TO BE REMOVABLE.

SCALE (B) 12 6 0 12 — 1-1/2" = 1'-0"
SCALE OF FEET

SCALE (A) 8 0 8 16 — 1/8" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 2026.04.01 ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. HEANEY	092 A101	POWER DISTRIBUTION CENTER ARCHITECTURAL PLANS AND DETAILS
	R. MILBRATH		
	TECH. REVIEW: R. MCKINLEY		
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO.	113 / 182752
PMIS/PKG NO.	318719
SHEET	28 OF 51

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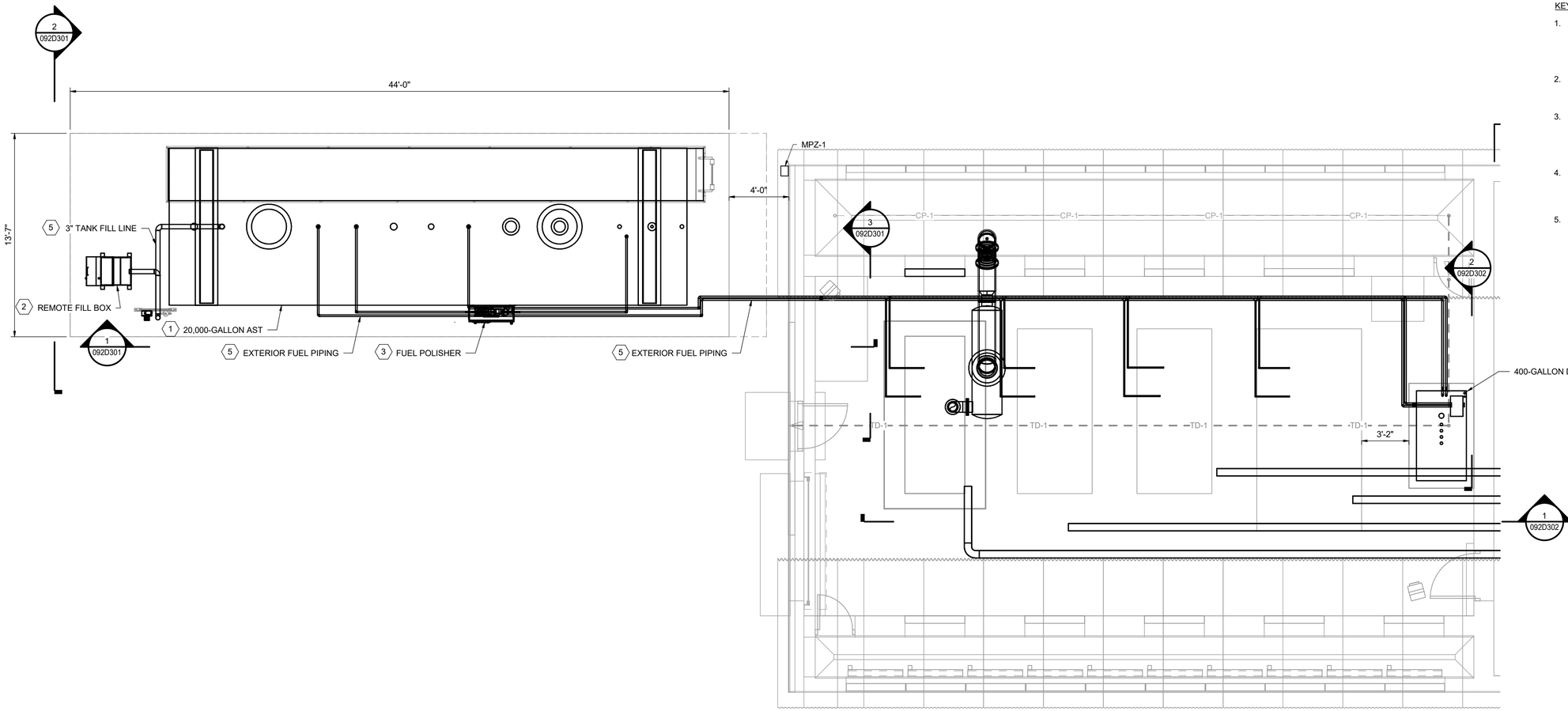

1 092D101 DIESEL GENERATOR FUEL SYSTEM PLAN
SCALE A

GENERAL NOTES:

A. SEE SHEETS 092-D601 AND 092-D602 FOR PROCESS AND INSTRUMENTATION DIAGRAMS OF THE GENERATOR FUELING AND STORAGE SYSTEMS. SHEETS INCLUDE TANK NOZZLE SCHEDULE, PUMP SCHEDULE AND VALVE LOCATIONS.

B. PIPING, VALVES, AND FITTINGS SHALL BE FURNISHED PER SPECIFICATION 33 52 10.

C. CONTRACTOR SHALL DESIGN, FURNISH, AND INSTALL STEEL PIPE SUPPORTS IN ACCORDANCE WITH SPECIFICATION 33 52 10.

D. ELECTRICALLY GROUND ALL PIPING AND COMPONENTS AS PER SPECIFICATIONS.

E. TANK CATWALK AND STAIRS TO BE SUPPLIED BY TANK MANUFACTURER. CONTRACTOR WILL BE RESPONSIBLE FOR INSTALLATION. PLATFORM AND STAIRS MATERIAL SHALL BE GALVANIZED STEEL AND/OR ALUMINUM AS PER SPECIFICATION 33 56 13.

KEY NOTES: (X)

1. 20,000-GALLON ABOVEGROUND STORAGE TANK LISTED PER UL-2085, DESIGNATED FOR DIESEL SERVICE. PROVIDE LABELING IN ACCORDANCE WITH SHEET 092-D502. MOUNT TANK SADDLE SUPPORTS TO CONCRETE EQUIPMENT PAD WITH ANCHOR BOLTS. REFER TO SPECIFICATION 33 56 13.
2. REMOTE SPILL CONTAINER FOR FUEL DELIVERY TRUCK FILL CONNECTION. INSIDE THE CONTAINER, PROVIDE POPPETED DRY-BREAK DISCONNECT, COUPLER, AND DUST CAP.
3. PACKAGED FUEL POLISHING SYSTEM WITH HEATED WEATHER-PROOF ENCLOSURE. MOUNT ENCLOSURE TO CONCRETE EQUIPMENT PAD WITH ANCHOR BOLTS. CONNECT 1-INCH STEEL PIPING TO TANK-TOP NOZZLES FOR SUPPLY AND RETURN LINES FROM POLISHER.
4. 400-GALLON DAY TANK LISTED PER UL-142, DESIGNATED FOR DIESEL SERVICE. FURNISH WITH PACKAGED PUMPSET ASSEMBLY CONFIGURED WITH DUPLEX SUPPLY PUMPS AND RETURN PUMP.
5. EXTERIOR FUEL PIPING SHALL BE HEAT TRACED AND INSULATED. REFER TO SPECIFICATION 33 52 10 FOR MORE INFORMATION.

SCALE A 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

Registered Professional Engineer CERTIFICATE NICHOLAS S. MCDONALD 0917 Date Signed ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. MCDONALD DRAWN: J. STUJENSKE TECH. REVIEW: J. FERGUSON DATE: MAY 2026	SUB SHEET NO. 092 D101	TITLE OF SHEET POWER DISTRIBUTION CENTER FUEL BULK TANK PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 29 OF 51

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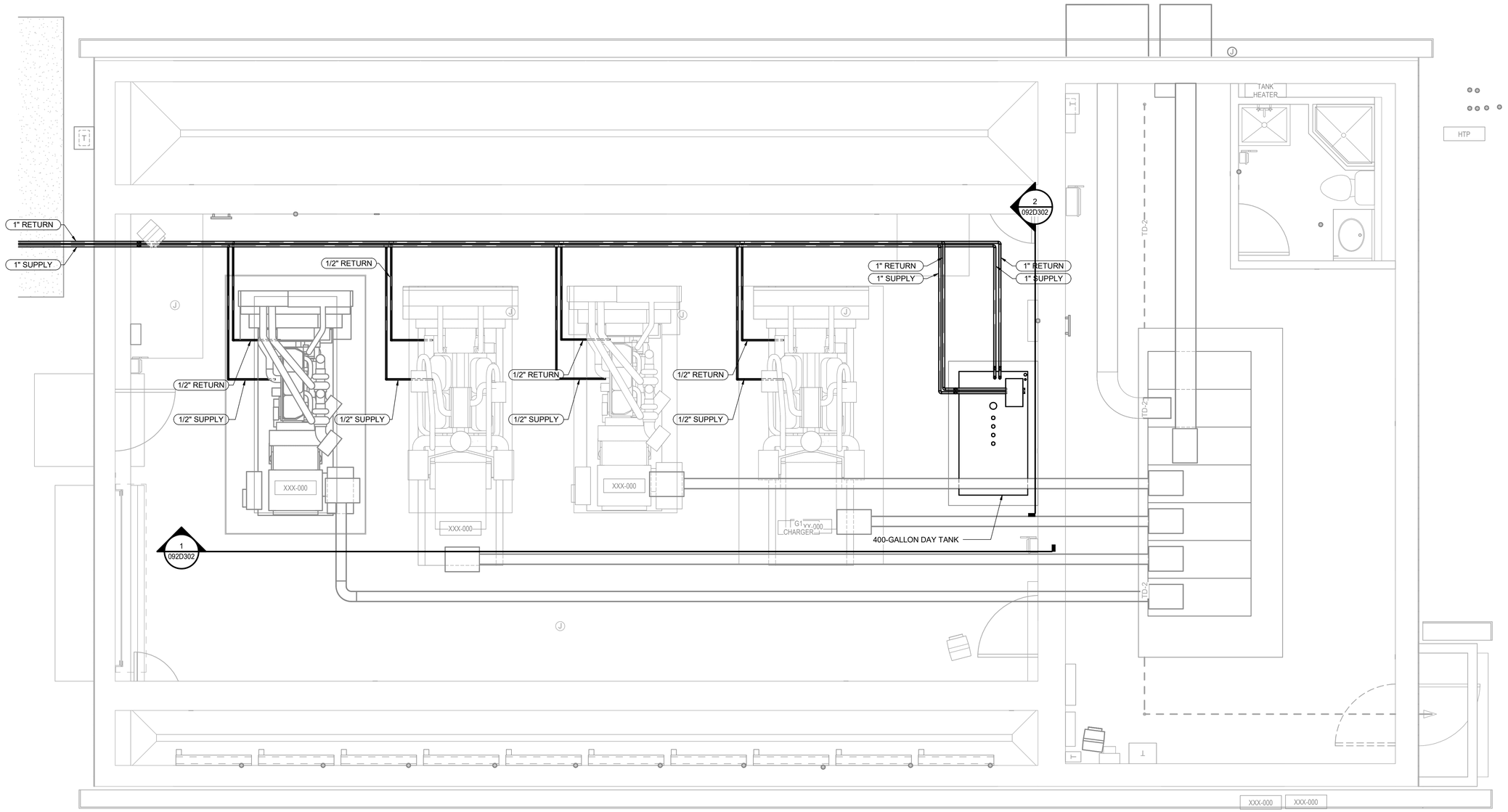

1 092D102 GENERATOR ROOM PIPING PLAN

SCALE A 2 0 2 4 6 3/8" = 1'-0"

SCALE OF FEET

Registered Professional Engineer Certificate No. 60401 NICHOLAS S. MCDONALD Date Signed 08/18/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. MCDONALD CADD J. STUJENSKE TECH. REVIEW: J. FERGUSON DATE: MAY 2026	SUB SHEET NO. 092 D102	TITLE OF SHEET POWER DISTRIBUTION CENTER GENERATOR ROOM PIPING PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 30 OF 51

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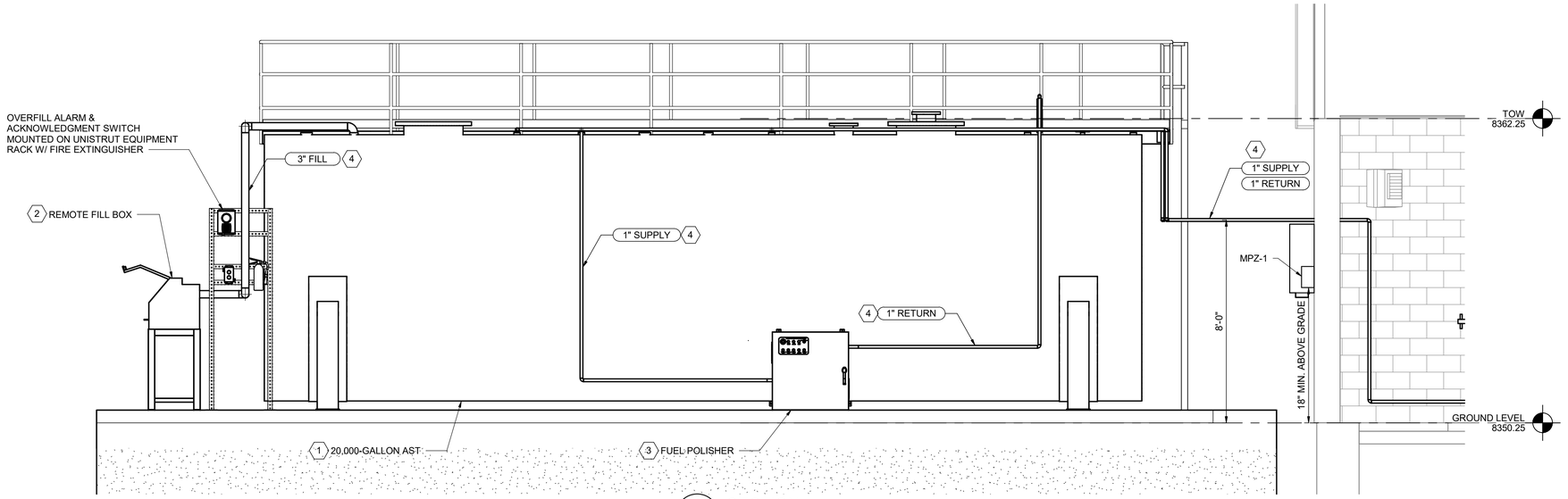
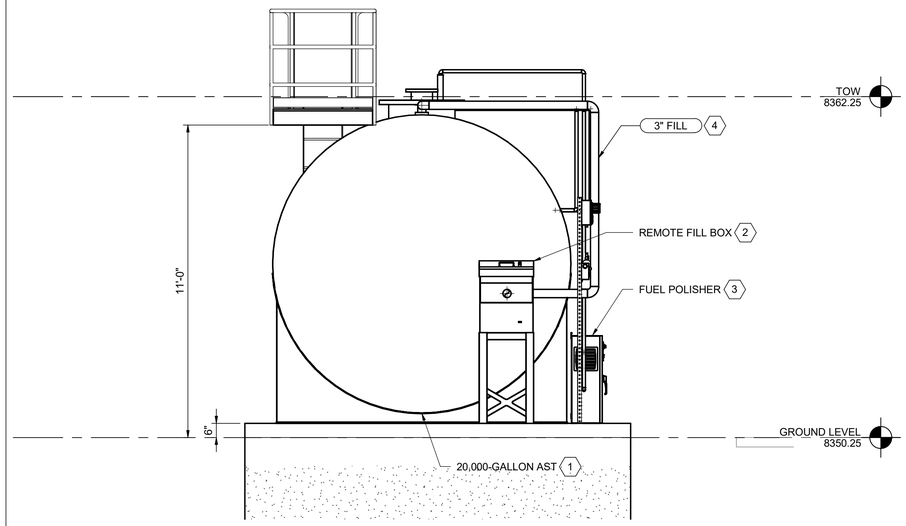
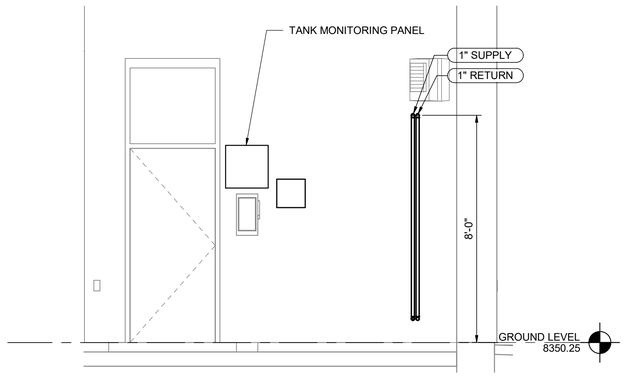


1 092D301 SECTION



2 092D301 SECTION



3 092D301 SECTION

GENERAL NOTES:

A. SEE SHEETS 092-D601 AND 092-D602 FOR PROCESS AND INSTRUMENTATION DIAGRAMS OF THE GENERATOR FUELING AND STORAGE SYSTEMS. SHEETS INCLUDE TANK NOZZLE SCHEDULE, PUMP SCHEDULE AND VALVE LOCATIONS.

B. PIPING, VALVES, AND FITTINGS SHALL BE FURNISHED PER SPECIFICATION 33 52 10.

C. CONTRACTOR SHALL DESIGN, FURNISH, AND INSTALL STEEL PIPE SUPPORTS IN ACCORDANCE WITH SPECIFICATION 33 52 10.

D. ELECTRICALLY GROUND ALL PIPING AND COMPONENTS AS PER SPECIFICATIONS.

E. TANK CATWALK AND STAIRS TO BE SUPPLIED BY TANK MANUFACTURER. CONTRACTOR WILL BE RESPONSIBLE FOR INSTALLATION. PLATFORM AND STAIRS MATERIAL SHALL BE GALVANIZED STEEL AND/OR ALUMINUM AS PER SPECIFICATION 33 56 13.

KEY NOTES: (X)

1. 20,000-GALLON ABOVEGROUND STORAGE TANK LISTED PER UL-2085, DESIGNATED FOR DIESEL SERVICE. PROVIDE LABELING IN ACCORDANCE WITH SHEET 092-D502. MOUNT TANK SADDLE SUPPORTS TO CONCRETE EQUIPMENT PAD WITH ANCHOR BOLTS. REFER TO SPECIFICATION 33 56 13.
2. REMOTE SPILL CONTAINER FOR FUEL DELIVERY TRUCK FILL CONNECTION. INSIDE THE CONTAINER, PROVIDE POPPETED DRY-BREAK DISCONNECT, COUPLER, AND DUST CAP.
3. PACKAGED FUEL POLISHING SYSTEM WITH HEATED WEATHER-PROOF ENCLOSURE. MOUNT ENCLOSURE TO CONCRETE EQUIPMENT PAD WITH ANCHOR BOLTS. CONNECT 1-INCH STEEL PIPING TO TANK-TOP NOZZLES FOR SUPPLY AND RETURN LINES FROM POLISHER.
4. EXTERIOR FUEL PIPING SHALL BE HEAT TRACED AND INSULATED. REFER TO SPECIFICATION 33 52 10 FOR MORE INFORMATION.

SCALE (A) 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

Registered Professional Engineer — CERTIFICATE NO. — NICHOLAS S. MCDONALD — Date Signed — 08/18/2026 — ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	N. MCDONALD	092 D301	POWER DISTRIBUTION CENTER FUEL BULK TANK SECTION	113 182752
	CADD J. STUJENSKE			PMIS/PKG NO. 318719
	TECH. REVIEW: J. FERGUSON			SHEET 31 OF 51
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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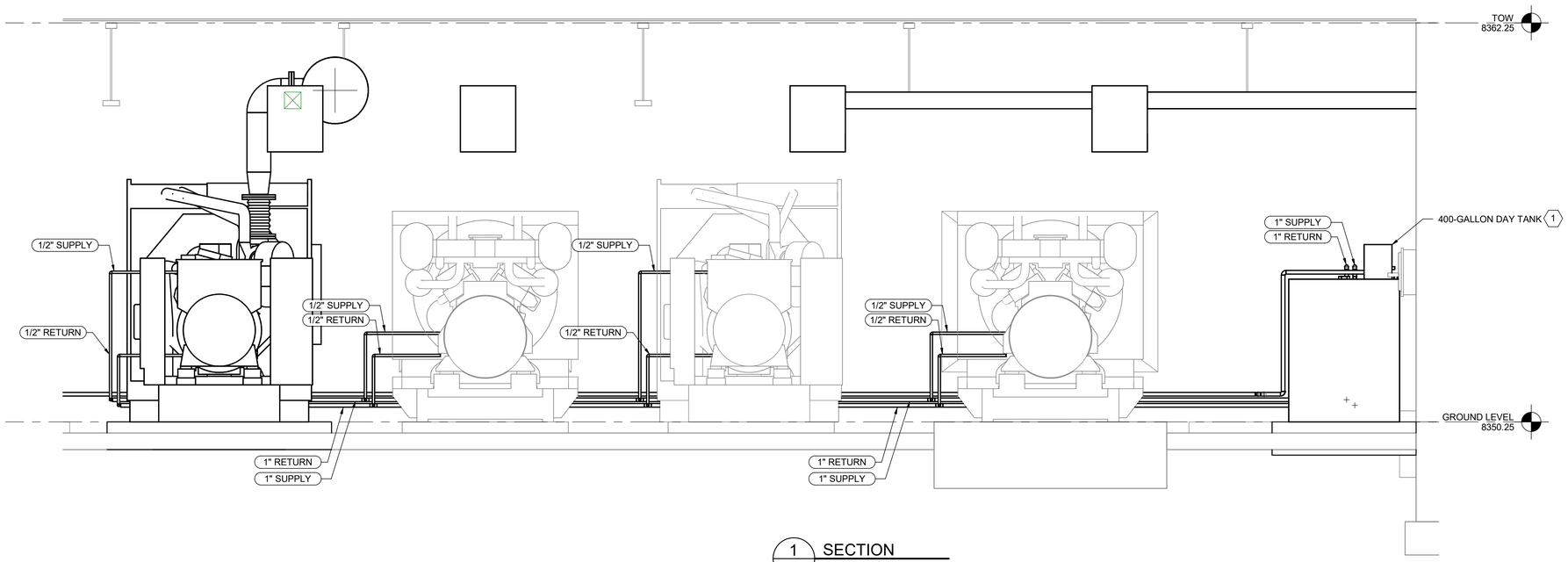
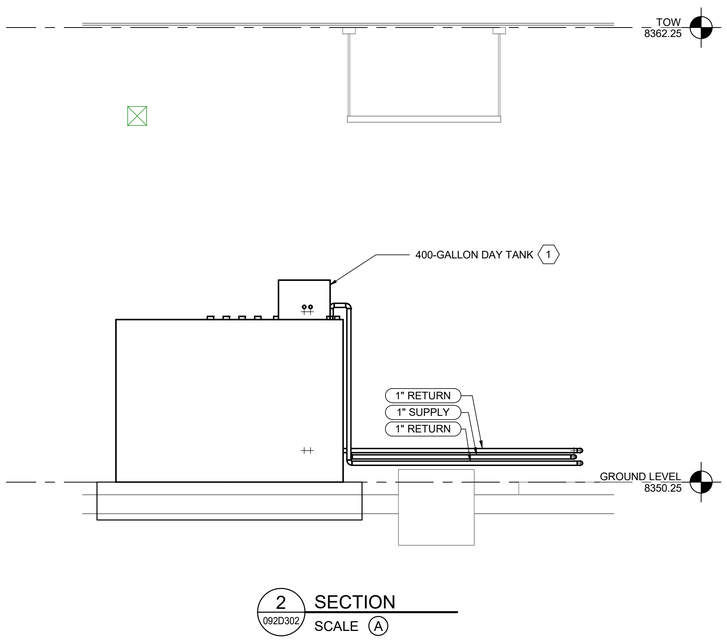


1 092D302 SECTION
SCALE A



2 092D302 SECTION
SCALE A

GENERAL NOTES:

A. SEE SHEETS 092-D601 AND 092-D602 FOR PROCESS AND INSTRUMENTATION DIAGRAMS OF THE GENERATOR FUELING AND STORAGE SYSTEMS. SHEETS INCLUDE TANK NOZZLE SCHEDULE, PUMP SCHEDULE AND VALVE LOCATIONS.

B. PIPING, VALVES, AND FITTINGS SHALL BE FURNISHED PER SPECIFICATION 33 52 10.

C. CONTRACTOR SHALL DESIGN, FURNISH, AND INSTALL STEEL PIPE SUPPORTS IN ACCORDANCE WITH SPECIFICATION 33 52 10.

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KEY NOTES: X

1. 400-GALLON DAY TANK LISTED PER UL-142, DESIGNATED FOR DIESEL SERVICE. FURNISH WITH PACKAGED PUMPSET ASSEMBLY CONFIGURED WITH DUPLEX SUPPLY PUMPS AND RETURN PUMP.

SCALE A 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

Registered Professional Engineer
CERTIFICATE NO.
NICHOLAS S. MCDONALD
Date Signed 08/18/2026
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. MCDONALD DRAFTED: J. STUJENSKE TECH. REVIEW: J. FERGUSON DATE: MAY 2026	SUB SHEET NO. 092 D302	TITLE OF SHEET POWER DISTRIBUTION CENTER FUEL BULK TANK SECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 32 OF 51

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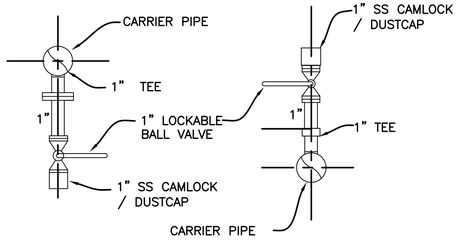
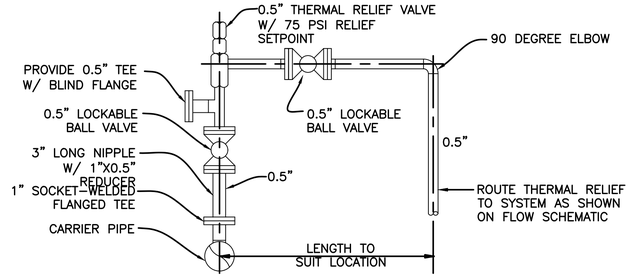


NOTES:

1. SOCKET–WELDED FLANGED TEE IS SHOWN. ALL PIPING JOINTS SHALL BE THREADED OR SOCKET–WELDED.
2. INSTALL LOW POINT DRAINS AT LOWEST ELEVATION OF EACH PIPING SEGMENT AND AS OTHERWISE SHOWN ON FLOW SCHEMATICS TO ALLOW FOR COMPLETE DRAINAGE OF SYSTEM.
3. INSTALL AIR VENTS AT HIGHEST ELEVATION OF EACH PIPING SEGMENT AND AS OTHERWISE SHOWN ON FLOW SCHEMATICS TO ALLOW FOR COMPLETE REMOVAL OF TRAPPED AIR FROM THE SYSTEM.
4. LPD AND HPV PIPING AND COMPONENT MATERIAL SHALL MATCH CARRIER PIPE MATERIAL.

1 092D501 DETAIL – LOW POINT DRAIN & HIGH POINT VENT

SCALE NOT TO SCALE

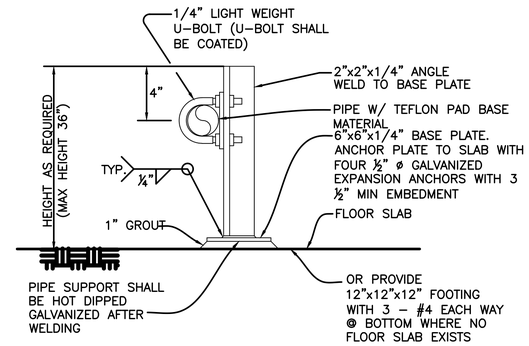


NOTES:

1. ALL PIPING JOINTS SHALL BE THREADED OR SOCKET–WELDED.
2. VALVES SHALL BE SECURED IN THE OPEN POSITION.
3. INSTALL THERMAL RELIEF VALVE IN VERTICAL POSITION.

2 092D501 DETAIL – THERMAL RELIEF VALVE (TRV) ASSEMBLY

SCALE NOT TO SCALE

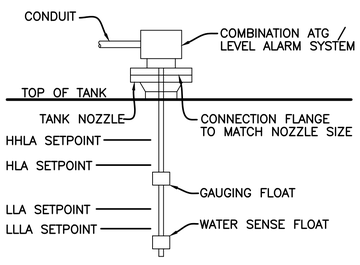


NOTES:

1. MAXIMUM PIPE SUPPORT SPACING:
   1.1. 1" PIPE – 7';
   1.2. 2" PIPE – 10'

3 092D501 DETAIL – PIPE SUPPORT (3/4" TO 2" PIPE)

SCALE NOT TO SCALE



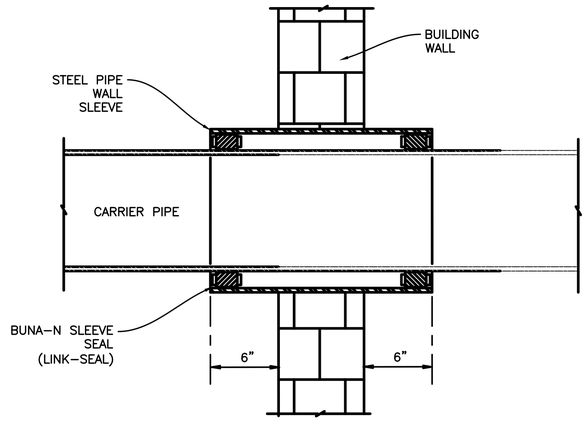
LEVEL ALARM HEIGHT				
	RISING LEVEL ACTUATOR POINT		FALLING LEVEL ACTUATOR POINT	
	HIGH–HIGH	HIGH	LOW	LOW–LOW
PRODUCT LEVEL	95% VOLUME	90% VOLUME	15% VOLUME	5% VOLUME
SYSTEM RESPONSE	OVERFILL PREVENTION VALVE CLOSES	ACTIVATES AUDIBLE AND VISUAL ALARM	ACTIVATES AUDIBLE AND VISUAL ALARM	ACTIVATES AUDIBLE AND VISUAL ALARM

NOTES:

1. WATER SENSE FLOAT SHALL TRIGGER ALERT AT MONITORING PANEL.

4 092D501 DETAIL – TANK LEVEL ALARM

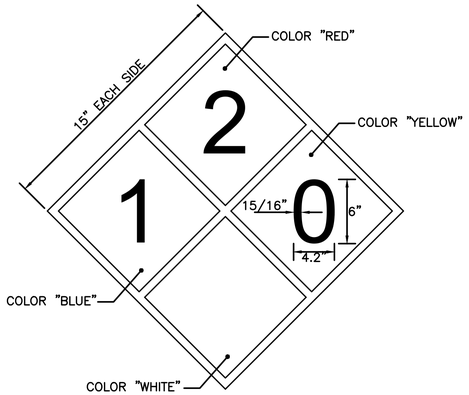
SCALE NOT TO SCALE



5 092D501 DETAIL – FUEL PIPE PENETRATION

SCALE NOT TO SCALE

A/E FIRM: PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. MCDONALD / C. IHEKWEAZU	SUB SHEET NO. 092 D501	TITLE OF SHEET: POWER DISTRIBUTION CENTER FUEL SYSTEM MECHANICAL DETAILS	DRAWING NO. 113 182752
	TECH. REVIEW: J. FERGUSON			PMIS/PKG NO. 318719
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	SHEET 33 OF 51



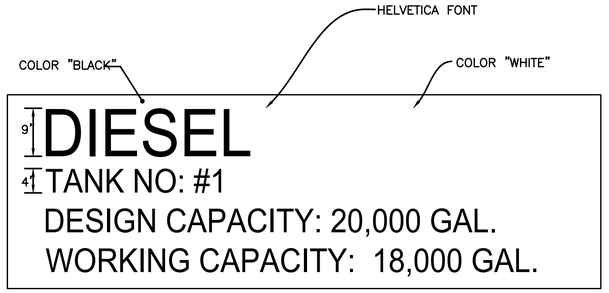
FUEL RATINGS			
	FIRE (TOP)	HEALTH (LEFT)	REACTIVITY (RIGHT)
DIESEL	2	1	0

NOTES:

1. PLACE ONE LABEL ON EACH SIDE AND END OF EACH TANK.
2. PLACARD SHALL COMPLY WITH REQUIREMENTS OF NFPA 704. DIMENSIONS SHOWN ARE MINIMUM FOR VISIBILITY AT 300 FT.

1 092D502 DETAIL – TYPICAL NFPA 704 PLACARD (TANKS)

NO SCALE

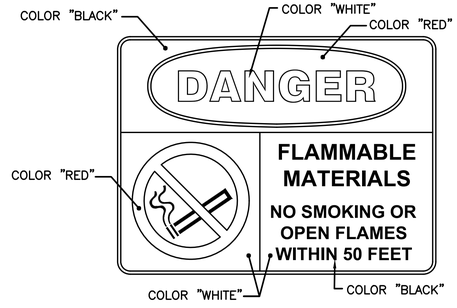


NOTES:

1. PLACE ONE LABEL ON EACH SIDE AND END OF EACH TANK.
2. ADJUST MARKINGS FOR OTHER TANKS ACCORDINGLY. REFER TO FLOW SCHEMATICS ON SHEETS F301 AND F302 FOR INFORMATION ON EACH TANK.

2 092D502 DETAIL – TYPICAL TANK LABEL

NO SCALE



NOTES:

1. PLACE ONE LABEL ON EACH SIDE AND END OF EACH TANK.

3 092D502 DETAIL – NO SMOKING LABEL

NO SCALE

IN CASE OF FIRE, SPILL OR RELEASE

1) USE EMERGENCY PUMP SHUT OFF

2) REPORT THE ACCIDENT!

FIRE DEPARTMENT TELEPHONE NO. _____

FACILITY ADDRESS _____

_____

_____

NOTES:

1. POST EMERGENCY PROCEDURE SIGN IN A CONSPICUOUS LOCATION.
2. SIGN SHALL BE RED LETTERING ON A WHITE BACKGROUND

4 092D502 DETAIL – EMERGENCY PROCEDURE SIGN

NO SCALE

STORAGE TANK IS EQUIPPED WITH:

HIGH LEVEL ALARM, DESIGNED TO ACTIVATE AT 90% PRODUCT LEVEL.

MECHANICAL OVERFILL PROTECTION VALVE, DESIGNED TO SHUT–OFF FLOW AT 95% PRODUCT LEVEL.

NOTES:

1. POST DRIVER WARNING SIGN ADJACENT TO REMOTE FILL CONTAINER TO ENSURE ATTENTION AND CLARITY PRIOR TO FILLING OPERATION.
2. SIGN SHALL BE RED LETTERING ON A WHITE BACKGROUND

5 092D502 DETAIL – DRIVER WARNING SIGN

NO SCALE

GENERAL NOTES:

A. STATE AND LOCAL CODE REQUIREMENTS FOR LABELING SHALL TAKE PRECEDENCE OVER ALL INDICATED DETAILS ON THIS DRAWING.

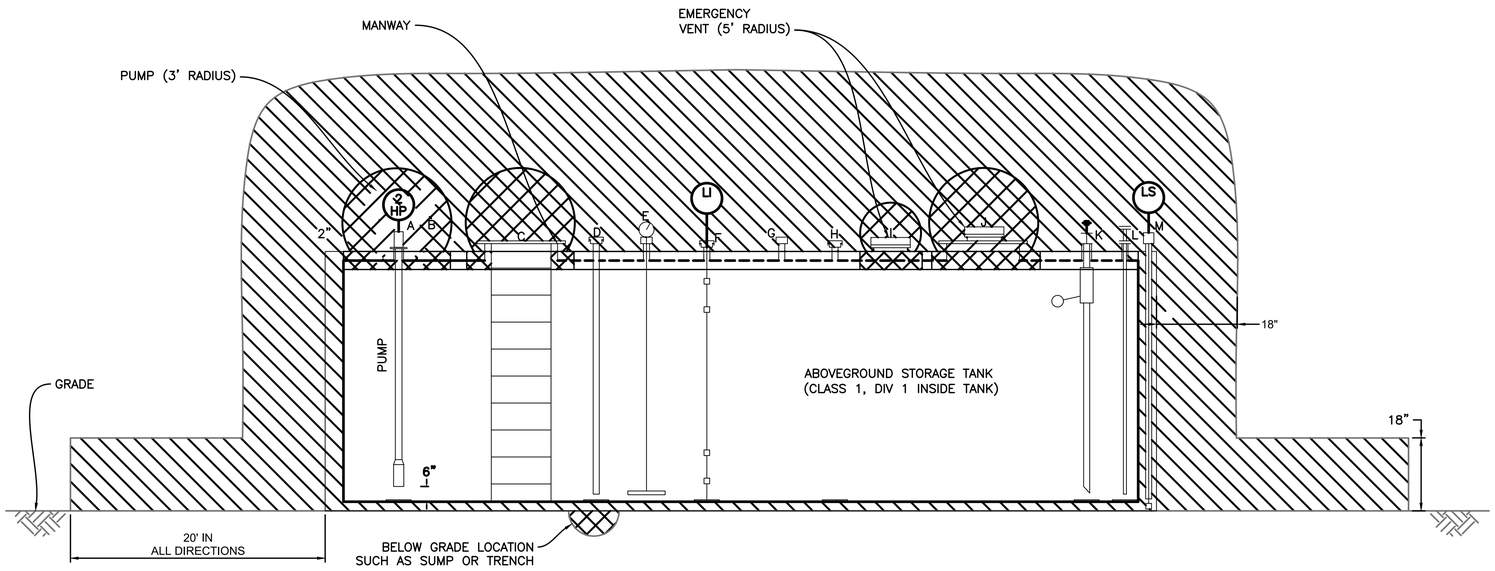
B. LABELS STANDARD TO THE INDUSTRY AND APPROVED BY THE OWNER WILL BE ACCPETABLE.

C. ALL LABELS MUST BE ABLE TO WITHSTAND EXTREME WEATHER CONDITIONS FOR AT LEAST ONE YEAR, AND MUST BE RESISTANT TO FUEL, OIL, GREASE, SOLVENTS, DETERGENTS, AND WATER.

D. ALL SIGNS AND LABELS SHALL BE POSTED/APPLIED IN A CONSPICUOUS LOCATION.

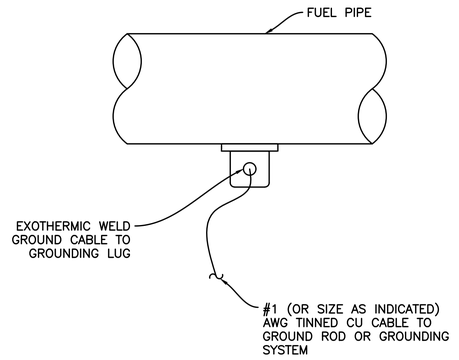
4/29/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\ELEC\092D501-504 & 092D601-602.dwg

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. MCDONALD CADD C. IHEKWEAZU TECH. REVIEW: J. FERGUSON DATE: MAY 2026	SUB SHEET NO. 092 D502	TITLE OF SHEET POWER DISTRIBUTION CENTER FUEL SYSTEM TANK AREA LABELS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 34 OF 51



1 092D503 DETAIL – ABOVE GROUND FUEL STORAGE TANK HAZARDOUS AREA

NO SCALE

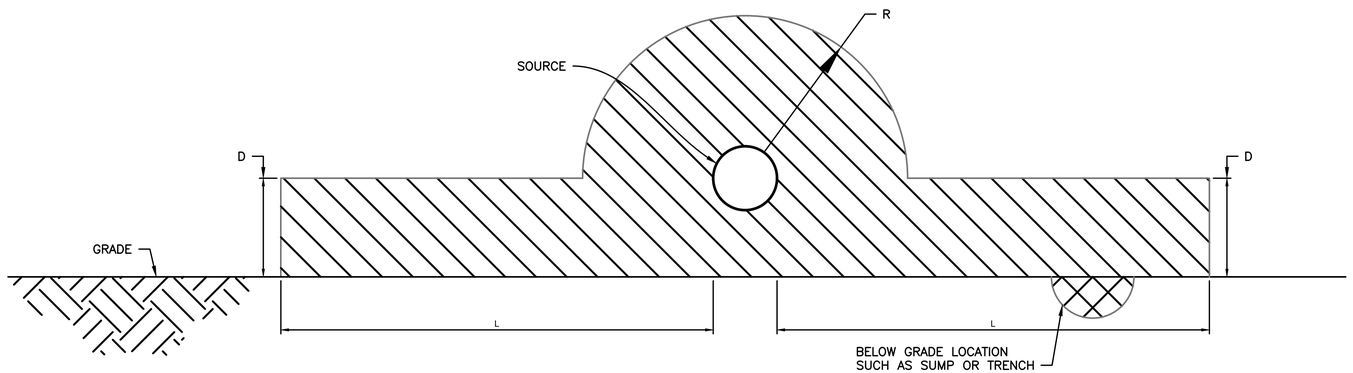


3 092D503 DETAIL – FUEL PIPE GROUNDING

NO SCALE

NOTES:

1. IN LOCATIONS WHERE IT IS IMPRACTICAL TO LOCATE THE GROUNDING TAB ON THE BOTTOM SIDE OF THE PIPE, LOCATING THE TAB ON THE TOP SIDE OF THE PIPE SHALL BE ACCEPTABLE.



2 092D503 DETAIL – ABOVE GROUND FUEL PIPING HAZARDOUS AREA

NO SCALE

LEVEL	DISTANCE IN FEET L	R	D
LIQUID 275 PISG OR LESS	10	3	2

CLASS 1, DIVISION 1 HAZARDOUS AREA

CLASS 1, DIVISION 2 HAZARDOUS AREA

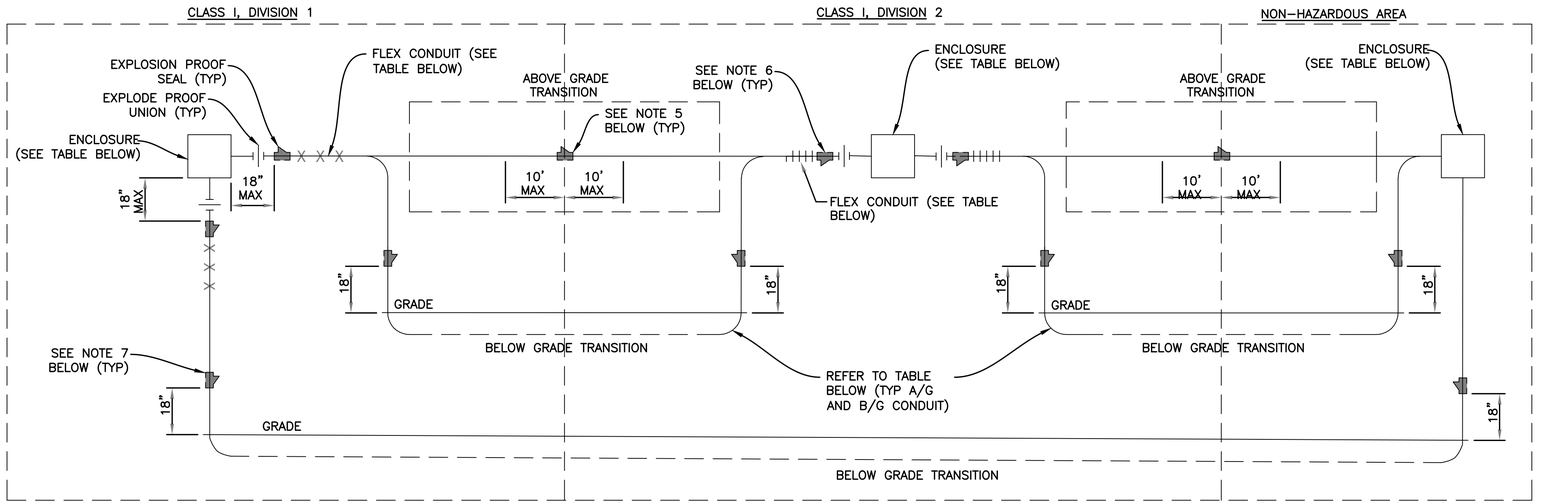
NON–CLASSIFIED, NON–HAZARDOUS AREA

NOTE:

1. SHOWN FOR PIPING INSTALLED ABOVE GRADE. ALSO APPLIED TO BURIED PIPING WITH ABOVE GRADE VENTS, SUCH AS MOTOR OPERATED VALVE BODY BURIED AND THE MOTOR OPERATOR ABOVE GRADE.

4/29/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\ELEC\092D501-504 & 092D601-602.dwg

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	N. MCDONALD	092 D503	POWER DISTRIBUTION CENTER FUEL SYSTEM ELECTRICAL DETAILS	113 182752
	CADD C. IHEKWEAZU			PMIS/PKG NO. 318719
	TECH. REVIEW: J. FERGUSON			SHEET 35 OF 51
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	



MATERIAL REQUIREMENTS (SEE NOTE 1)

CONDUIT					ENCLOSURES		
A/G	FLEXIBLE	DIRECT BURIED	CONCRETE-ENCASED	DUCTBANK ELBOWS	CLASS I, DIVISION 1	CLASS I, DIVISION 2	NON-HAZARDOUS AREA
RGS, 3/4" MIN. (SEE NOTE 2)	EXPLOSION PROOF LIQUID-TIGHT (CID1) LIQID-TIGHT (CID2 AND NON-HAZARDOUS)	PVC COATED RGS 1" MIN	RGS	PVC COATED RGS	NEMA 4/7 (SEE NOTE 3)	NEMA 4 (SEE NOTE 4)	NEMA 4 (OUTDOORS) NEMA 1 OR 12 (OUTDOORS)

NOTES:

1. MATERIAL REQUIREMENTS LISTED HERE SHALL BE USED UNLESS OTHERWISE NOTED.
2. EMT PERMITTED WHEN USED IN NON-HAZARDOUS INDOOR LOCATIONS.
3. NEMA 7 NOT REQUIRED IF ENCLOSURE IS LISTED AS SUITABLE CLASS I, DIVISION 1 LOCATIONS.
4. NEMA 4/7 REQUIRED WHEN ENCLOSURE CONTAINS ARCING OR SPARKING DEVICES.
5. NO UNIONS, COUPLINGS, BOX, OR OTHER FITTING EXCEPT FOR A LISTED EXPLOSION PROOF REDUCER AT THE SEAL IS PERMITTED BETWEEN THE SEAL AND BOUNDARY. IN PLACES WHERE CONDUIT IS TRANSITIONING ABOVE GRADE FROM CLASS I DIVISION 2 TO CLASS I DIVISION 1 AREAS, LOCATE SEAL ON DIVISION 2 SIDE OF BOUNDARY WITHIN 10'.
6. PROVIDE SEAL WITHIN 18" OF ENCLOSURE, SIMILAR TO CLASS 1, DIVISION 1 INSTALLATION, WHEN ENCLOSURE IS REQUIRED TO BE EXPLOSION PROOF IN A CLASS I, DIVISION 2 AREA, SEAL INSTALLED ON INSTRUMENTATION SHALL BE INSTALLED BETWEEN FLEXIBLE CONDUIT AND INSTRUMENTATION.
7. NO UNIONS, COUPLINGS, BOX, OR OTHER FITTING EXCEPT FOR A LISTED EXPLOSION PROOF REDUCER AT THE SEAL IS PERMITTED BETWEEN THE AND GRADE.
8. EXTERNAL BONDING JUMPER SECURED WITH CABLE TIES REQUIRED FOR LIQUID-TIGHT FLEXIBLE CONDUIT PER NFPA 70, ARTICLE 501.30.

1 / 092D504 DETAIL – TYPICAL CONDUIT AND EQUIPMENT REQUIREMENTS FOR HAZARDOUS AND NON-HAZARDOUS AREA

NO SCALE

GENERAL NOTES:

A. IT IS THE CONTRACTORS RESPONSIBILITY TO PROVIDE A FULLY FUNCTIONAL SYSTEM, ENSURE A COMPLETE WORKING SYSTEM WHICH SHALL COMPLY WITH MANUFACTURER'S INSTRUCTIONS, NFPA 30, NFPA 30A, AND NFPA 70, OSHA REQUIREMENTS, ARIZONA DEPARTMENT OF ENVIRONMENTAL QUALITY, AND ARIZONA STATE FIRE CODE.

B. ALL BRANCH CIRCUIT WIRING FROM PANELBOARD TO THE FUEL SYSTEM SHALL BE TYP XHHW-2 GASOLINE AND OIL RESISTANT SOLID WIRE A MINIMUM OF 12-WIRE IN 3/4" CONDUIT. ALL ABOVEGROUND CONDUIT SHALL BE RIGID GALVANIZED STEEL (RGS). ALL ABOVEGROUND CONDUIT BODIES AND PULL BOXES SHALL BE EXPLOSION PROOF WITHIN A CLASS 1, DIVISION 2 HAZARDOUS ZONE. ALL UNDERGROUND CONDUIT SHALL BE RIGID NON-METALLIC CONDUIT ENCASED IN RED CONCRETE 3" THICK MINIMUM. ALL LONG RADIUS 90-DEGREE ELBOWS EMERGING FROM THE EARTH OR CONCRETE SHALL BE PVC-COATED RGS CONDUIT. THE FIRST FITTING ABOVE GRADE SHALL BE AN NEC APPROVED EXPLOSION-PROOF SEAL.

C. THE ELECTRICAL INSTALLATION SHALL COMPLY WITH THE NATIONAL ELECTRICAL CODE, ARTICLES 500, 501, 502, 504, 514, AND ALL RELATED ARTICLES THEREOF.

D. PROVIDE AND INSTALL AN INTEGRATED AUTOMATIC TANK GAUGING (ATG) TANK LEVEL AND LEAK MONITORING SYSTEM WITH A HIGH LEVEL ALARM AND TEST/ACKNOWLEDGMENT SWITCH. CONTRACTOR SHALL FOLLOW THE MANUFACTURER'S INSTALLATION MANUAL AND WIRING DIAGRAMS TO PROVIDE A FULLY FUNCTIONAL SYSTEM. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO PROVIDE AND INSTALL THE APPROPRIATE EQUIPMENT, PROBES, SENSORS, CONDUCTOR WIRING, COMMUNICATION CABLING, AND CONDUITS FOR THE SYSTEM.

E. NO SPLICES SHALL BE MADE IN THE WIRE RUN BETWEEN A SENSOR OR PROBE JUNCTION BOX AND THE MONITORING PANEL INPUT/OUTPUT MODULES.

F. ALL ELECTRICAL POWER WITHIN THE BOUNDS OF THE CLASS 1 DIVISION 1 & 2 AREAS SHALL BE CONNECTED TO AN EMERGENCY FUEL SHUTOFF (EFSO / E-STOP) SWITCH. THE EFSO SWITCH SHALL DISCONNECT SIMULTANEOUSLY FROM THE SOURCE OF SUPPLY VIA SHUNT TRIP. ALL CONDUCTOR WIRING SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH NEC ARTICLE 514-11(A).

G. EFSO SWITCH SHALL BE A MOMENTARY MUSHROOM-STYLE PUSHBUTTON CLEARLY IDENTIFIED AND EASILY ACCESSIBLE IN A REMOTE LOCATION.

H. INTRINSICALLY SAFE INPUT WIRING SHALL BE INSTALLED PER NEC ARTICLE 504 AND LABELED PER NEC ARTICLE 504.80. NON-INTRINICALLY SAFE WIRING SHALL NOT SHARE SPACE WITH INTRINSICALLY SAFE WIRING IN RACEWAYS, CONDUIT BOXES, JUNCTION BOXES, OR GUTTERS.

I. CONTRACTOR SHALL SUBMIT TO THE CONTRACTING OFFICER A COPY OF THE TANK MONITORING SYSTEM WARRANTY AND CERTIFICATION LETTERS FROM THE MANUFACTURER TO ENSURE A FULLY-FUNCTIONAL INSTALLATION.

J. CONTRACTOR SHALL PROVIDE A START-UP, COMMISSIONING, AND TRAINING SESSION TO CITY OF CONCORD PERSONNEL DEMONSTRATING OPERATION OF THE FUELING SYSTEM.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. MCDONALD CADD C. IHEKWEAZU TECH. REVIEW: J. FERGUSON DATE: MAY 2026	SUB SHEET NO. 092 D504	TITLE OF SHEET POWER DISTRIBUTION CENTER FUEL SYSTEM HAZARDOUS CLASSIFIED AREA DETAIL NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 36 OF 51

4/29/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DWG\CAD_HDR\DRAWINGS\ELEC\092D501-504 & 092D601-602.dwg

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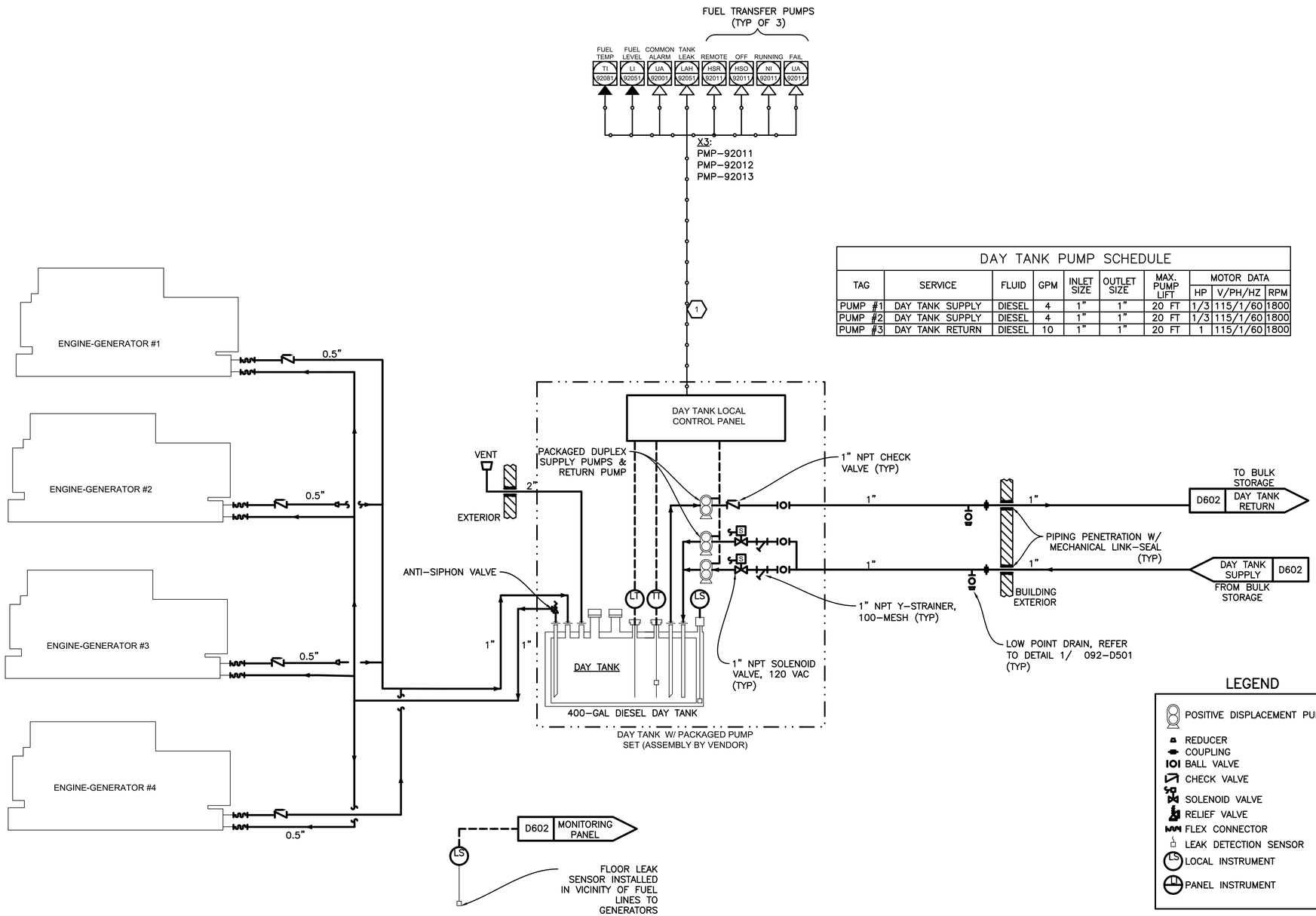


KEY NOTES: (X)

1. MODBUS TCP/IP COMMUNICATIONS TO SCADA NETWORK.

DAY TANK PUMP SCHEDULE

TAG	SERVICE	FLUID	GPM	INLET SIZE	OUTLET SIZE	MAX. PUMP LIFT	MOTOR DATA		
							HP	V/PH/HZ	RPM
PUMP #1	DAY TANK SUPPLY	DIESEL	4	1"	1"	20 FT	1/3	115/1/60	1800
PUMP #2	DAY TANK SUPPLY	DIESEL	4	1"	1"	20 FT	1/3	115/1/60	1800
PUMP #3	DAY TANK RETURN	DIESEL	10	1"	1"	20 FT	1	115/1/60	1800

LEGEND

- POSITIVE DISPLACEMENT PUMP
- REDUCER
- COUPLING
- BALL VALVE
- CHECK VALVE
- SOLENOID VALVE
- RELIEF VALVE
- FLEX CONNECTOR
- LEAK DETECTION SENSOR
- LS — LOCAL INSTRUMENT
- PANEL INSTRUMENT

ABBREVIATIONS

CS	CARBON STEEL
DIA	DIAMETER
DP	DIFFERENTIAL PRESSURE
DN	DOWN
DWG	DRAWING
EA	EACH
F	FILTER
FS	FLOW SENSOR
GAL	GALLON
GPM	GALLONS PER MINUTE
HP	HORSEPOWER
LA	LEVEL ALARM
LC	LOCK CLOSED
LI	LEVEL INDICATOR
LO	LOCK OPEN
LS	LEVEL SWITCH / LEAK SENSOR
LT	LEVEL TRANSMITTER
NC	NORMALLY CLOSED
NO	NORMALLY OPEN
NPT	NATIONAL PIPE THREAD
PS	PRESSURE SENSOR
QTY	QUANTITY
RPM	REVOLUTIONS PER MINUTE
SS	STAINLESS STEEL
TT	TEMPERATURE TRANSMITTER
TYP	TYPICAL
W/	WITH
W/O	WITHOUT
WS	WATER SENSOR

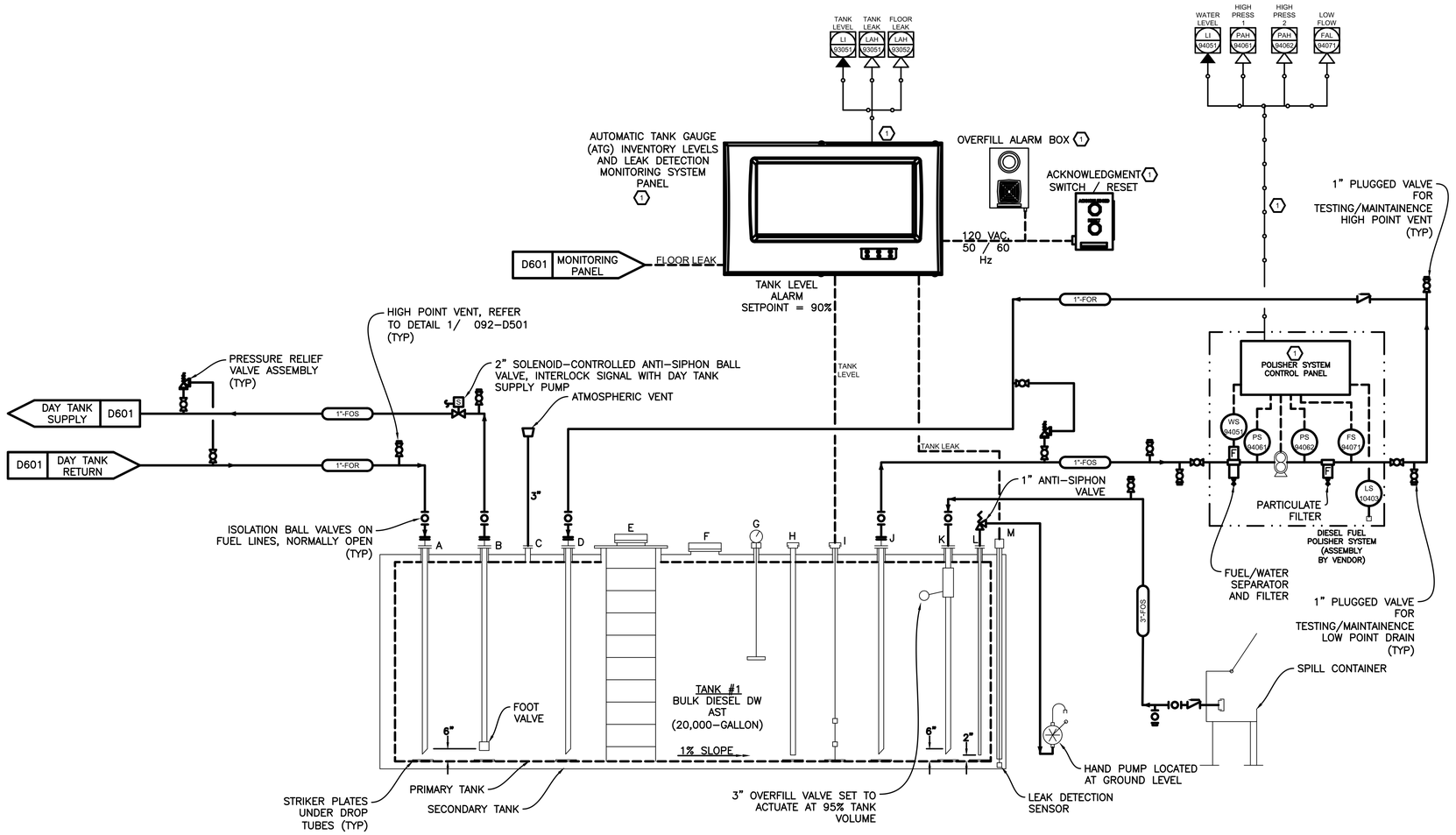
1 / 092D601 — DETAIL – NORTH RIM DAY TANK AND ENGINE-GENERATORS (DIESEL) SCHEMATIC

NO SCALE

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	N. MCDONALD	092 D601	POWER DISTRIBUTION CENTER FUEL SYSTEM DAY TANK & GENSET FLOW SCHEMATIC	113 182752
	C. IHEKWEAZU		NORTH RIM - GRAND CANYON NATIONAL PARK	PMIS/PKG NO. 318719
	TECH. REVIEW: J. FERGUSON			SHEET 37 OF 51
	DATE: MAY 2026			

KEY NOTES: (X)

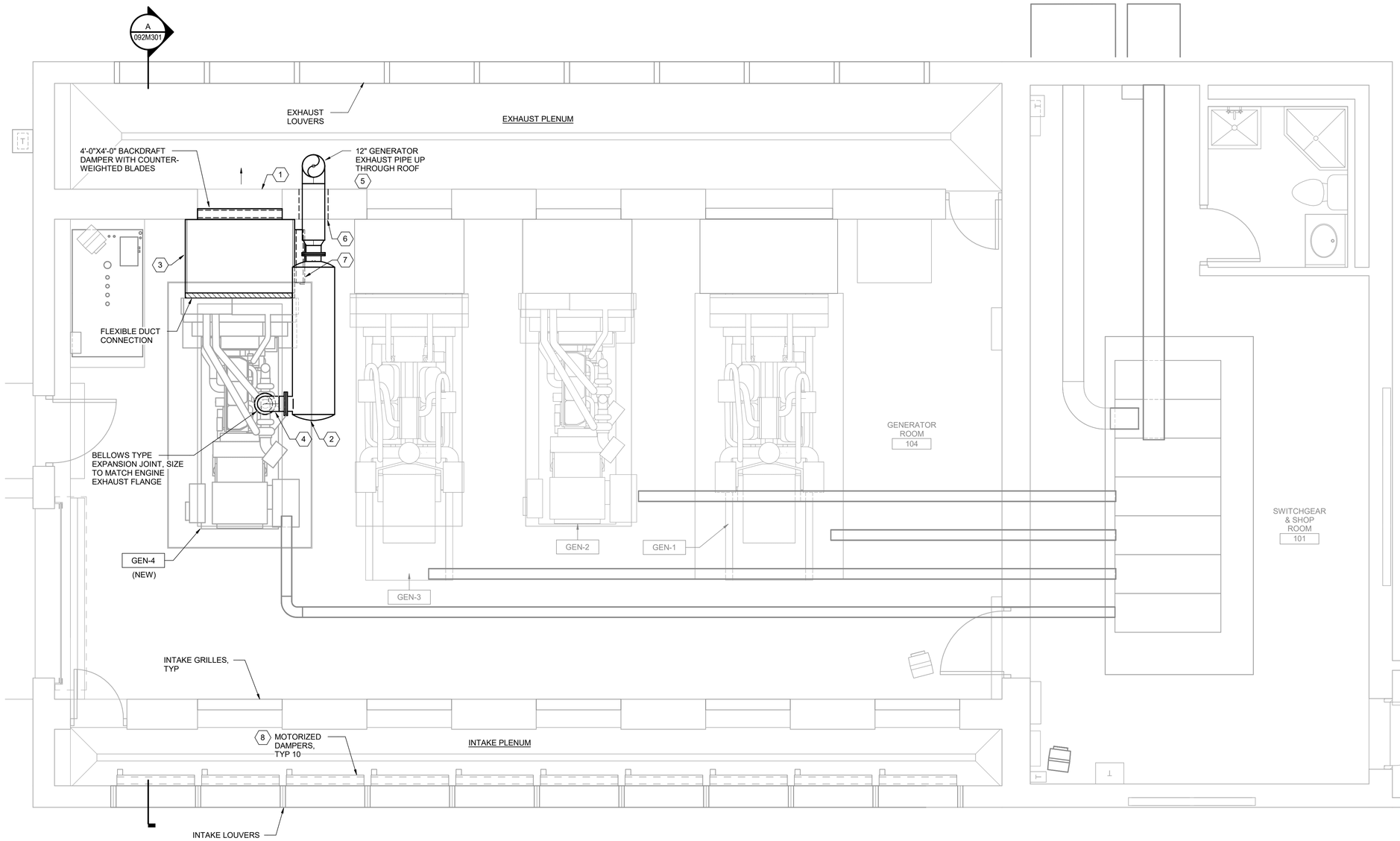
1. MODBUS TCP/IP COMMUNICATIONS TO SCADA NETWORK.



TANK #1 (DIESEL) NOZZLE SCHEDULE				
TANK NOZZLE	QTY	SIZE	TYPE	FUNCTION
A	1	2"	FLANGE	DAY TANK RETURN PIPING W/ 1" DROP TUBE
B	1	2"	FLANGE	DAY TANK SUPPLY PIPING W/ 1" DROP TUBE
C	1	3"	FLANGE	NORMAL/ATMOSPHERIC VENT
D	1	2"	FLANGE	FUEL POLISHER RETURN PIPING W/ 1" DROP TUBE
E	1	30"/8"	FLANGE	MANWAY W/ EMERGENCY VENT (PRIMARY TANK)
F	1	8"	FLANGE	EMERGENCY VENT (SECONDARY TANK)
G	1	2"	FNPT	MECHANICAL CLOCK GAUGE
H	1	2"	FNPT	MANUAL GAUGING & SAMPLING W/ HATCH
I	1	4"	FNPT	ATG LEVEL PROBE
J	1	2"	FNPT	FUEL POLISHER SUPPLY PIPING W/ 1" DROP
K	1	6"	MNPT	FILL PIPING W/ 3" OVERFILL PREVENTION VALVE
L	1	2"	FNPT	WATER DRAW-OFF W/ 1" DROP TUBE
M	1	2"	FNPT	INTERSTITIAL LEAK DETECTION

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	N. MCDONALD CADD C. IHEKWEAZU TECH. REVIEW: J. FERGUSON DATE: MAY 2026	092 D602	POWER DISTRIBUTION CENTER FUEL SYSTEM TANK LAYOUT FLOW SCHEMATIC NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 38 OF 51

4/29/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\D



1 092M101 HVAC PLAN
SCALE A

KEYNOTES: (X)

1. REMOVE 4'-0" X 4'-0" CMU WALL KNOCKOUT PANEL. INSTALL BACKDRAFT DAMPER IN WALL OPENING.
2. EXHAUST SILENCER SUSPENDED FROM ABOVE, TYPE 304 STAINLESS STEEL CONSTRUCTION. PROVIDE SEISMIC BRACING.
3. EXHAUST DUCT BETWEEN RADIATOR AND WALL, GALVANIZED STEEL, PROVIDE 1" INTERNAL ACOUSTIC FIBERGLASS LINING.
4. EXHAUST PIPE UPSTREAM OF SILENCER SHALL BE SCHEDULE 40 STAINLESS STEEL, TYPE 304, WITH BUTT WELDED FITTINGS.
5. EXHAUST PIPE DOWNSTREAM OF SILENCER SHALL BE EQUAL TO METAL-FAB INC. MODEL IPIC, UL LISTED INDUSTRIAL PRESSURE CHIMNEY SYSTEM. TYPE 304 STAINLESS STEEL CONSTRUCTION. PROVIDE 2" THICK INSULATION AND MAINTAIN MINIMUM REQUIRED CLEARANCE TO COMBUSTIBLES PER MANUFACTURER REQUIREMENTS. INSTALL ALL MATERIALS IN ACCORDANCE WITH MANUFACTURER INSTRUCTIONS.
6. WALL PENETRATION ASSEMBLY, EQUAL TO METAL-FAB IPICWP.
7. PROVIDE LOW-LEAKAGE OPPOSED BLADE RETURN AIR DAMPER, 30"WX36"H, WITH MANUAL QUADRANT ACTUATOR.
8. EXISTING MOTORIZED DAMPERS, INTERLOCK WITH NEW GENERATOR OPERATION.

SCALE A 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

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A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

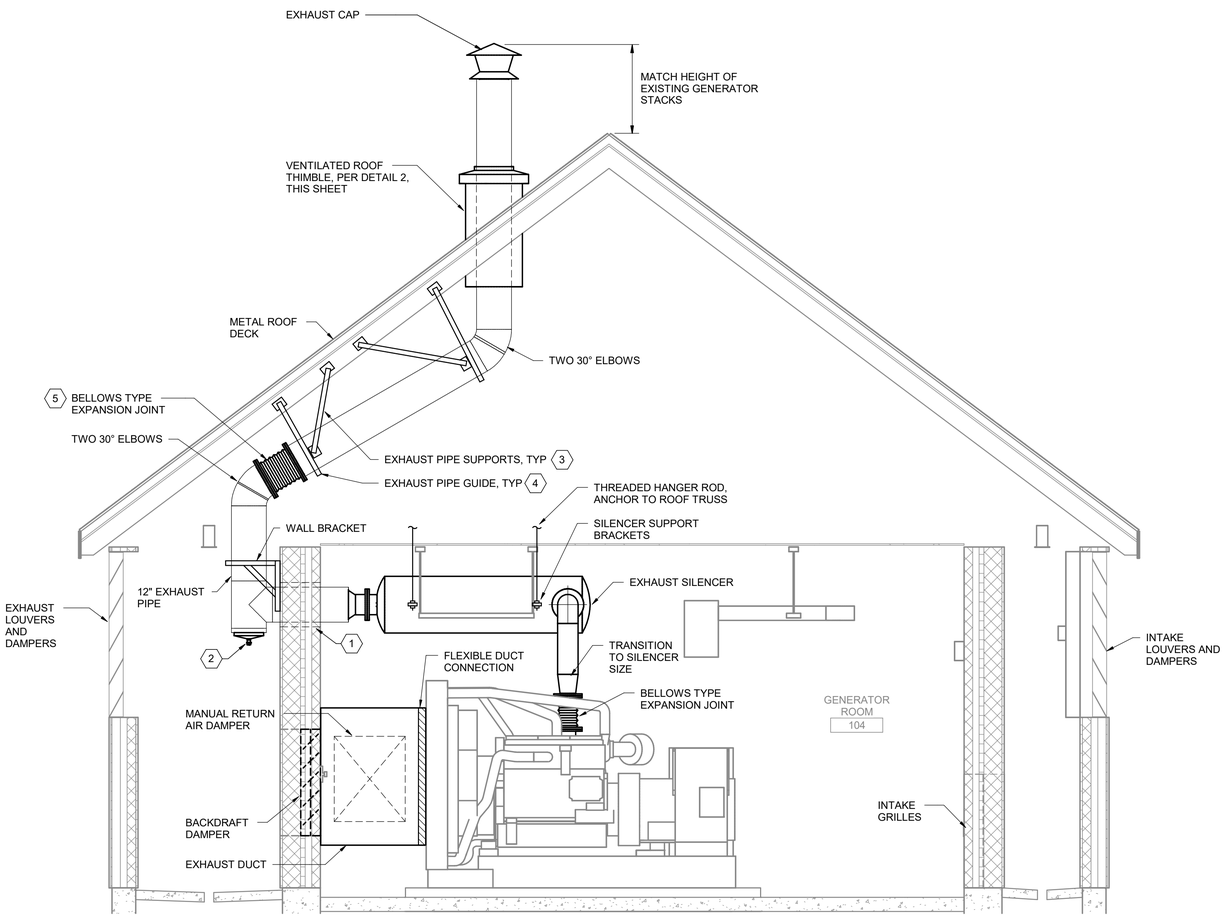
DESIGNED: A. PUTMAN
DRAWN: B. BAXTER
TECH. REVIEW: H. PEDRAZA
DATE: MAY 2026

SUB SHEET NO. 092 M101

TITLE OF SHEET
POWER DISTRIBUTION CENTER
HVAC PLAN

NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 39 OF 51

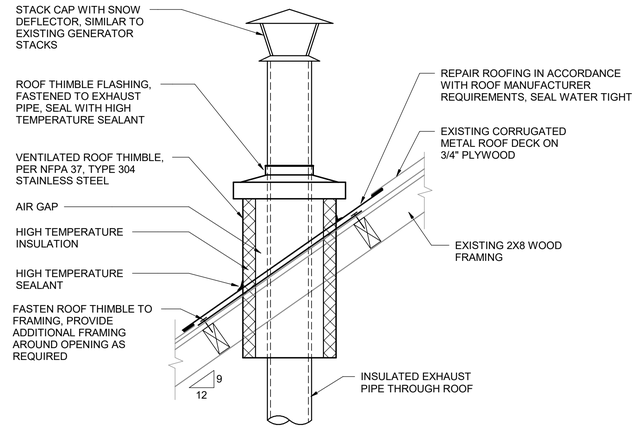


A 092M101 GENERATOR EXHAUST SECTION

SCALE A

KEYNOTES: X

1. WALL PENETRATION ASSEMBLY, EQUAL TO METAL-FAB IPICWP.
2. DRAIN TEE WITH THREADED AND CAPPED DRAIN CONNECTION.
3. EXHAUST PIPE SUPPORTS, EQUAL TO METAL-FAB IPICWSA.
4. EXHAUST PIPE GUIDES, EQUAL TO METAL-FAB IPICWG.
5. EXPANSION JOINT, EQUAL TO METAL-FAB PICBJ.



NOTES:

1. SEALANTS IN CONTACT WITH EXHAUST PIPE OR ROOF THIMBLE SHALL BE HIGH TEMPERATURE SEALANTS, PER MANUFACTURER REQUIREMENTS.
2. MAINTAIN MINIMUM REQUIRED CLEARANCE TO COMBUSTIBLE MATERIALS, PER ROOF THIMBLE AND EXHAUST PIPE MANUFACTURER REQUIREMENTS.

1 DETAIL - GENERATOR EXHAUST ROOF PENETRATION

SCALE A 2 0 2 4 6 3/8" = 1'-0"

SCALE OF FEET

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 092 M301	TITLE OF SHEET POWER DISTRIBUTION CENTER MECHANICAL SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 40 OF 51

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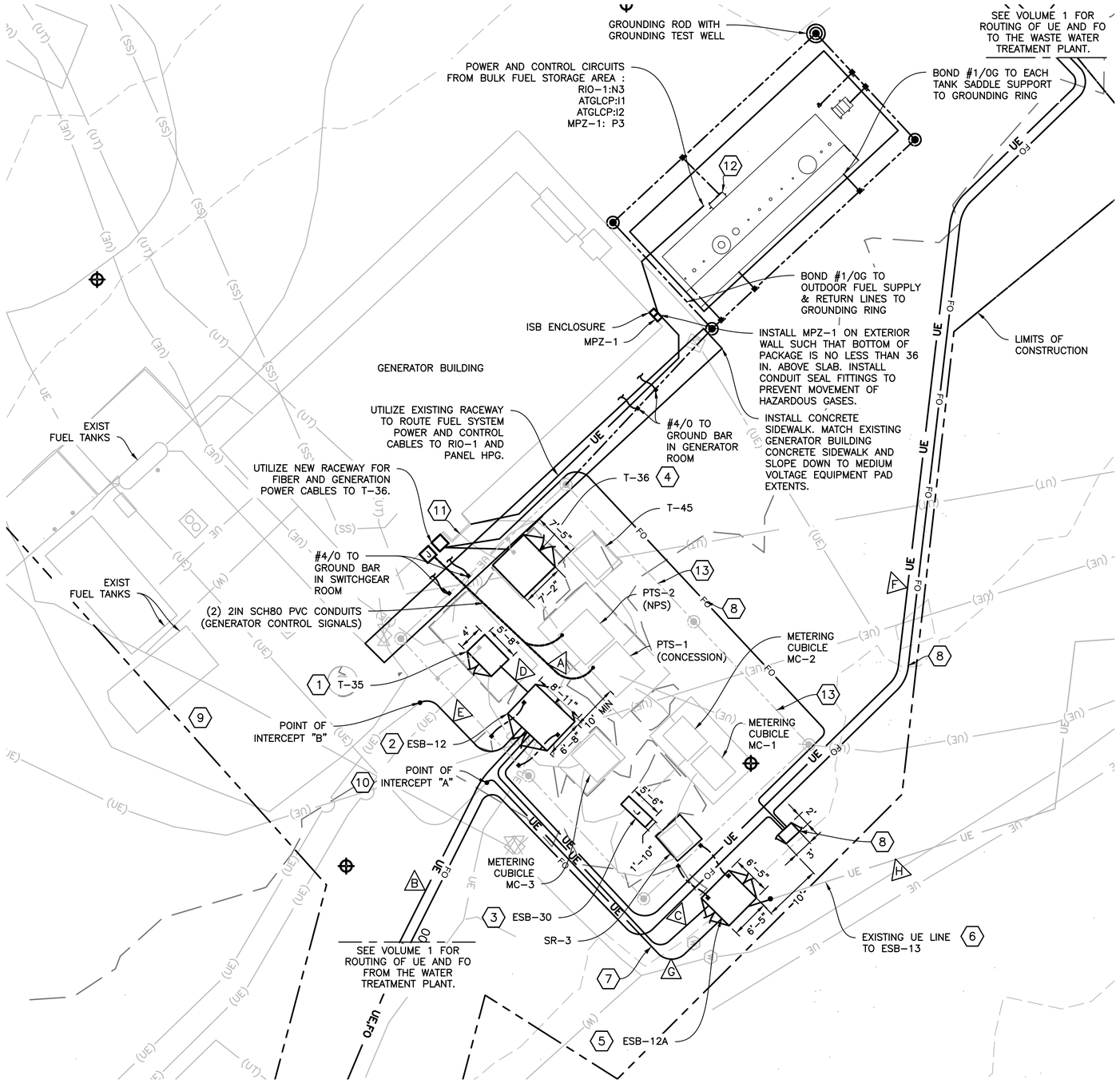

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. SEE SHEET 091E106 FOR GENERAL NOTES.

C. SEE SHEET 008E506 FOR TYPICAL THREE-PHASE PADMOUNT TRANSFORMER DETAILS.

D. SEE SHEET 008E506 DETAIL 1 FOR TYPICAL THREE-PHASE PADMOUNT JUNCTION ENCLOSURE DETAILS.

E. SEE SHEET 091E501 DETAIL 1 FOR TYPICAL THREE-PHASE PADMOUNT SWITCHGEAR ENCLOSURE DETAILS.

F. SEE SHEET 091E502 FOR MEDIUM VOLTAGE POWER CABLE SCHEDULE. (X)

KEY NOTES: (X)

1. REPLACE EXIST TRANSFORMER T-35 WITH 112.5KVA, 24.9GRDY/14.4kV -480GRDY/277V PADMOUNT TRANSFORMER. TERMINATE 25kV (3/C) 4/0 ALUMINUM ARMORED POWER CABLE FROM ESB-12. REUSE EXIST EQUIPMENT PAD AND RECONNECT TO PAD GROUND. TERMINATE 25kV (3/C) 4/0 ALUMINUM ARMORED POWER CABLE FROM ESB-12 WITH 200 AMP LOAD BREAK ELBOWS. INSTALL 200 AMP SURGE ARRESTER ELBOWS ON UNUSED BUSHINGS. RECONNECT EXIST SECONDARY CONDUCTORS TO RESTORE SERVICE.

2. REPLACE EXIST JUNCTION ENCLOSURE ESB-12 WITH 25kV 6-WAY (2 SOURCE - 4 PROTECTED TAP) PADMOUNT SWITCHGEAR AND EQUIPMENT VAULT AND PAD. TERMINATE EXIST MEDIUM VOLTAGE POWER CABLE FROM PTS-2 AND CONNECTED TO ESB-12A ON SOURCE POSITION. TERMINATE 25kV (3/C) 4/0 ALUMINUM ARMORED POWER CABLE TO T-35, T-49, POINT OF INTERCEPT "D" (WWTP AREA, SEE VOL 1), AND ESB-16 ON PROTECTED TAP POSITIONS WITH ELBOW TERMINATIONS. INSTALL 200 AMP PROTECTIVE CAPS ON UNUSED 200 AMP BUSHINGS.

3. REPLACE EXIST JUNCTION ENCLOSURE ESB-30 WITH 4-POSITION PADMOUNT JUNCTION ENCLOSURE AND JUNCTION ENCLOSURE SLEEVE. TERMINATE EXIST MEDIUM VOLTAGE POWER CABLES AND EXIST ELBOWS FROM PS-1, SR3, MC-1, AND MC-2.

4. REPLACE EXIST TRANSFORMER T-36 WITH 1500KVA, 24.9kV-480V PADMOUNT TRANSFORMER. TERMINATE EXIST MEDIUM VOLTAGE POWER CABLE AND ELBOWS FROM PTS-2. REUSE EXIST EQUIPMENT PAD AND RECONNECT TO PAD GROUND. INSTALL 200 AMP SURGE ARRESTER ELBOWS ON UNUSED BUSHINGS. INSTALL SECONDARY CONDUCTOR TO POWER GENERATION SWITCHGEAR IN GENERATION BUILDING. CONTRACTOR TO VERIFY AND UTILIZE EXIST CONDUIT AND BUILDING PENETRATION.

5. INSTALL ESB-12A, 4-WAY (1 SOURCE - 3 PROTECTED TAP) PADMOUNT SWITCHGEAR AND EQUIPMENT VAULT AND PAD. TERMINATE 25kV (3/C) 4/0 ALUMINUM ARMORED POWER CABLE FROM ESB-12 ON SOURCE POSITION WITH 200 AMP LOADBREAK ELBOWS. TERMINATE 25kV (3/C) 4/0 ALUMINUM ARMORED POWER CABLE TO ESB-14 AND ESB-13 ON PROTECTED TAP POSITIONS WITH 200 AMP LOAD BREAK ELBOWS. INSTALL 200 AMP PROTECTIVE CAPS ON UNUSED 200 AMP BUSHINGS.

6. LOCATE EXIST MEDIUM VOLTAGE POWER CABLE BETWEEN SWITCHGEAR ESB-12 AND JUNCTION ENCLOSURE ESB-13. PULL BACK EXIST MEDIUM VOLTAGE POWER CABLE FROM EXIST JUNCTION ENCLOSURE ESB-12 TO SWITCHGEAR ESB-12A FOUNDATION. EXIST MEDIUM VOLTAGE POWER CABLE LENGTH SHALL BE SUFFICIENT ENOUGH TO REACH EQUIPMENT CABLE TERMINATION BUSHINGS AND HAVE AT MINIMUM 5 FT OF CABLE SLACK AT EQUIPMENT BASEMENT.

7. INSTALL 25kV (3/C) 4/0 AWG ALUMINUM ARMORED POWER CABLE BETWEEN ESB-12 AND ESB-12A.

8. INSTALL MIN NEMA 4X RATED FOPP91 FIBER CABINET, FOR LAYOUT SEE DETAIL 1-3 009Y501. FOR FIBER NETWORK DIAGRAM, REFER TO 009Y101. INSTALL 24-STRAND SM FIBER IN ORANGE TUBE OF NEW 4-WAY MICRODUCT BETWEEN FIBER PEDESTAL ENCLOSURE AND GENERATOR BUILDING. INSTALL 144-STRAND SM FIBER IN ORANGE TUBE OF 7-WAY MICRODUCT TO WTP. INSTALL 7 WAY MICRODUCT TO FIBER PULLBOX NEAR WWTP, REFER TO VOLUME 1.

9. LOCATE EXIST MEDIUM VOLTAGE POWER CABLE BETWEEN EXIST JUNCTION ENCLOSURE ESB-16 AND TRANSFORMER T-35. REMOVE EXIST CABLE TERMINATIONS ON MEDIUM VOLTAGE POWER CABLE TO ESB-16 AT TRANSFORMER T-35. EXPOSE AND PULL BACK EXIST CABLE TO POINT OF INTERCEPT "B" FROM TRANSFORMER T-35. INSTALL BURIED CABLE SPLICE AT POINT OF INTERCEPTION BETWEEN EXIST MEDIUM VOLTAGE POWER CABLE AND 25kV (3/C) 4/0 ALUMINUM ARMORED POWER CABLE. INSTALL 25kV (3/C) 4/0 ALUMINUM ARMORED POWER CABLE FROM SPLICE POINT TO ESB-12 AND TERMINATE CABLE IN PROTECTED TAP POSITION WITH 200 AMP LOADBREAK ELBOWS RESTORING CONNECTION TO ESB-16.

10. LOCATE EXIST. MEDIUM VOLTAGE POWER CABLE BETWEEN EXIST JUNCTION ENCLOSURE ESB-14 AND ESB-12. REMOVE EXIST CABLE TERMINATIONS ON MEDIUM VOLTAGE POWER CABLE TO ESB-14 AT JUNCTION ENCLOSURE ESB-12. EXPOSE AND PULL BACK EXIST CABLE TO POINT OF INTERCEPT "A" FROM JUNCTION ENCLOSURE ESB-12. INSTALL BURIED CABLE SPLICE AT POINT OF INTERCEPTION BETWEEN EXIST MEDIUM VOLTAGE POWER CABLE AND 25kV (3/C) 4/0 ALUMINUM ARMORED POWER CABLE. INSTALL 25kV (3/C) 4/0 ALUMINUM ARMORED POWER CABLE FROM SPLICE POINT TO ESB-12A AND TERMINATE CABLE IN PROTECTED TAP POSITION WITH 200 AMP LOADBREAK ELBOWS RESTORING CONNECTION TO ESB-14.

11. CONTRACTOR TO DISCONNECT, PULL AND REMOVE EXIST. SERVICE POWER CABLE IN EXIST. RACEWAY BETWEEN TRANSFORMER T-36 AND EXIST. SWITCHGEAR IN GENERATOR BUILDING. CONTRACTOR TO IDENTIFY, CUT, AND REMOVE EXIST. ABOVE GROUND SERVICE POWER CONDUIT BETWEEN TRANSFORMER T-36 AND EXIST. BUILDING SERVICE ENTRANCE. INSTALL CONDUIT PLUGS AND ABANDON EXIST. BURIED CONDUIT IN PLACE.

12. CONNECT GROUNDING TABS OF FUEL PIPING TO GROUND RING.

13. APPROXIMATE LOCATION OF EXISTING GROUND RING. FIELD-VERIFY EXACT LOCATION.

10 0 10 20
SCALE OF FEET
1" = 10'

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG DRAWN: D. VANG TECH. REVIEW: S. EVERETT DATE: JUNE 2026	SUB SHEET NO. 092 E101	TITLE OF SHEET POWER DISTRIBUTION CENTER ELECTRICAL SITE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113
182752
PMIS/PKG NO. 318719
SHEET 41 OF 51

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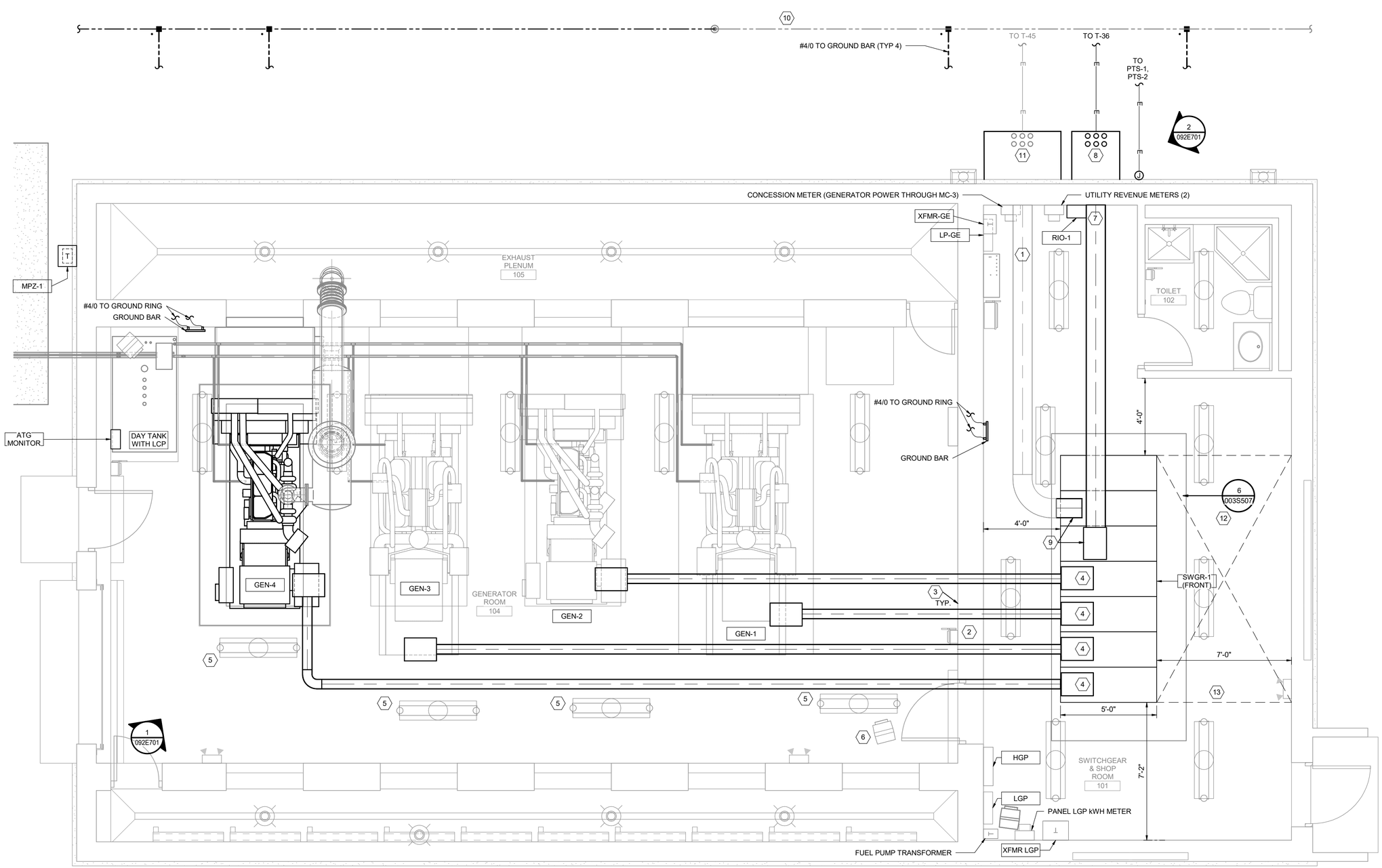

SCALE (A) 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

GENERAL NOTES

A. FIRESEAL ALL INTERNAL BUILDING PENETRATIONS BETWEEN GENERATOR ROOM AND OTHER ROOMS.

B. FIRESEAL AND WEATHER-SEAL ALL BUILDING PENETRATIONS BETWEEN BUILDING INTERIOR AND BUILDING EXTERIOR.

KEYNOTES

1 REUSE EXISTING CABLE TRAY AND CABLES TO T-45. MODIFY AND EXTENT CABLE TRAY AS REQUIRED TO ALIGN WITH NEW SWITCHGEAR.

2 CREATE PENETRATION FOR NEW GENERATOR 4 BUSWAY. ROUTE BUSWAY AS REQUIRED TO AVOID EXISTING GENERATOR BUSWAY.

3 REUSE EXISTING WALL PENETRATION FOR CONTROL CIRCUITS. MODIFY THE EXISTING OPENINGS AS REQUIRED FOR INSTALLATION OF NEW BUSWAYS.

4 TRANSITION FROM BUSWAY INTO TOP OF SWITCHGEAR.

5 ADJUST OR RELOCATE LIGHTS AS REQUIRED FOR INSTALLATION OF NEW BUSDUCT TO GENERATOR #4.

6 ADJUST EXISTING UNIT HEATER AS REQUIRED FOR INSTALLATION OF NEW BUSDUCT TO GENERATOR #4.

7 PROVIDE NEW CABLE TRAY FOR CABLES SERVING T-36.

8 PROVIDE NEW WIREWAY FOR CONDUITS ROUTING TO/FROM T-36. REMOVE AND REPLACE CABLES FROM T-36 ACCORDING TO SHEET 092E601. COORDINATE WIREWAY SIZE WITH INSTALLED CONDUITS. EXTEND ROOF EAVE TO COVER NEW WIREWAY.

9 SWITCHGEAR TOP-HATS AS REQUIRED FOR TRANSITION TO CABLE TRAY.

10 APPROXIMATE LOCATION OF EXISTING GROUND RING. FIELD-VERIFY EXACT LOCATION.

11 PROVIDE AND INSTALL REPLACEMENT EXTERIOR WIREWAY FOR CABLES TO TRANSFORMER T-45.

12 COORDINATE PAD SIZE WITH SIZE OF SUPPLIED EQUIPMENT.

13 NEC-REQUIRED WORKING AND EGRESS SPACE. KEEP SPACE CLEAR.

REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL) CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 04/17/2026 ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. S-DELGADO	092 E102	POWER DISTRIBUTION CENTER GENERATOR BUILDING FLOOR PLAN	113 182752
	CADD A S-DELGADO			PMIS/PKG NO. 318719
	TECH. REVIEW: P. DAVILA			SHEET 42 OF 51
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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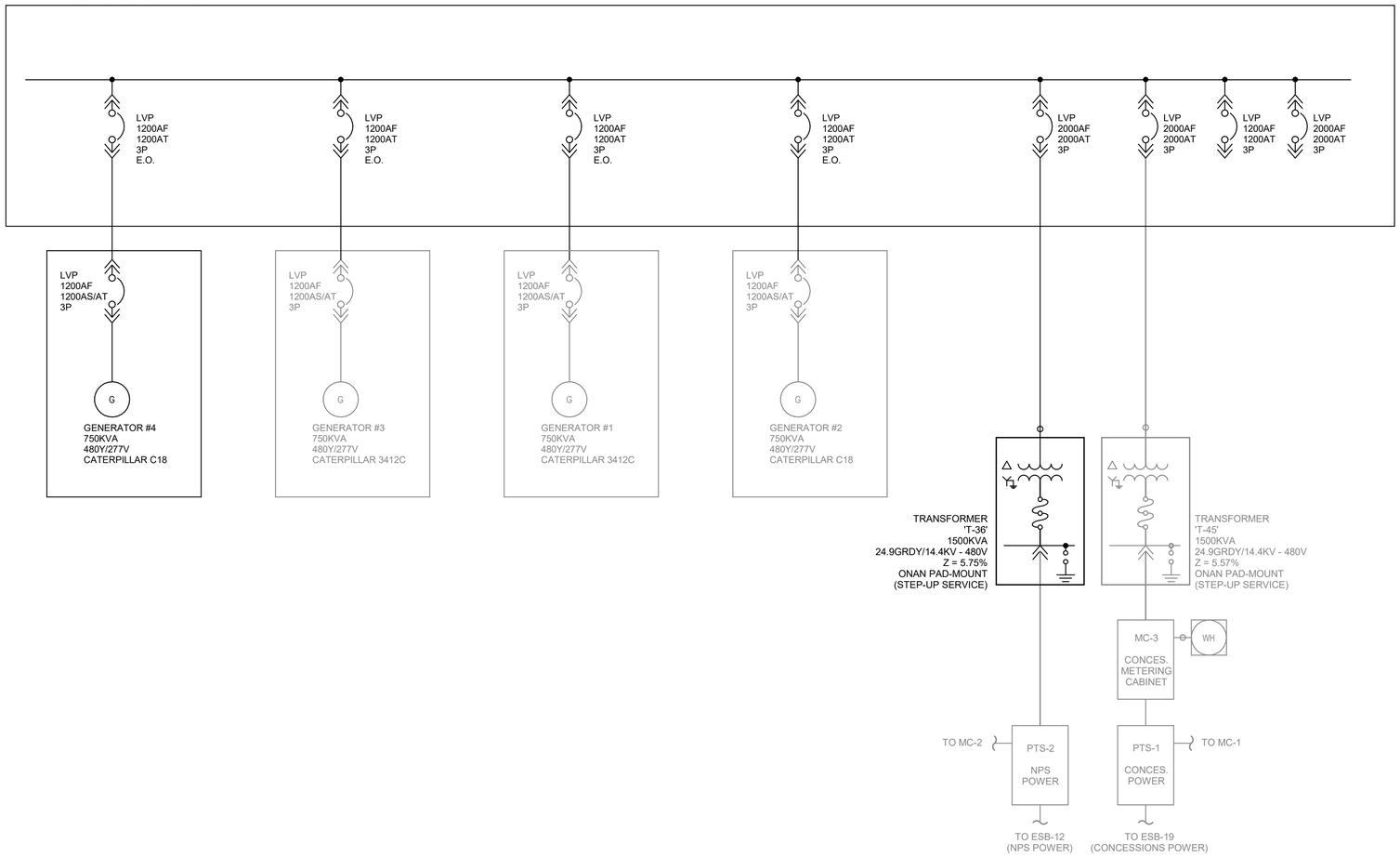
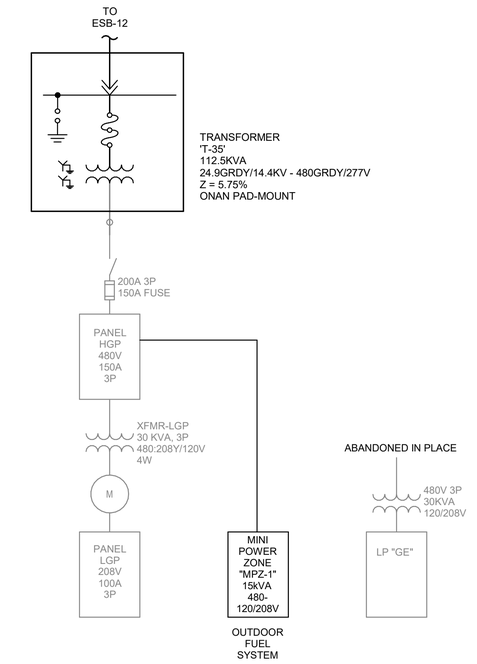
SWGR-1 4000A, 480V, 3PH, 3W, 100 KAIC, NEMA 1 ENCLOSURE





REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL)
CERTIFICATE NO.
58037
TRAVIS T.
MOORE
Date signed 1/14/2026
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO DRAWN: A S-DELGADO TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 092 E600	TITLE OF SHEET POWER DISTRIBUTION CENTER GENERATOR BUILDING POWER DISTRIBUTION FUNCTIONAL DIAGRAM NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 43 OF 51

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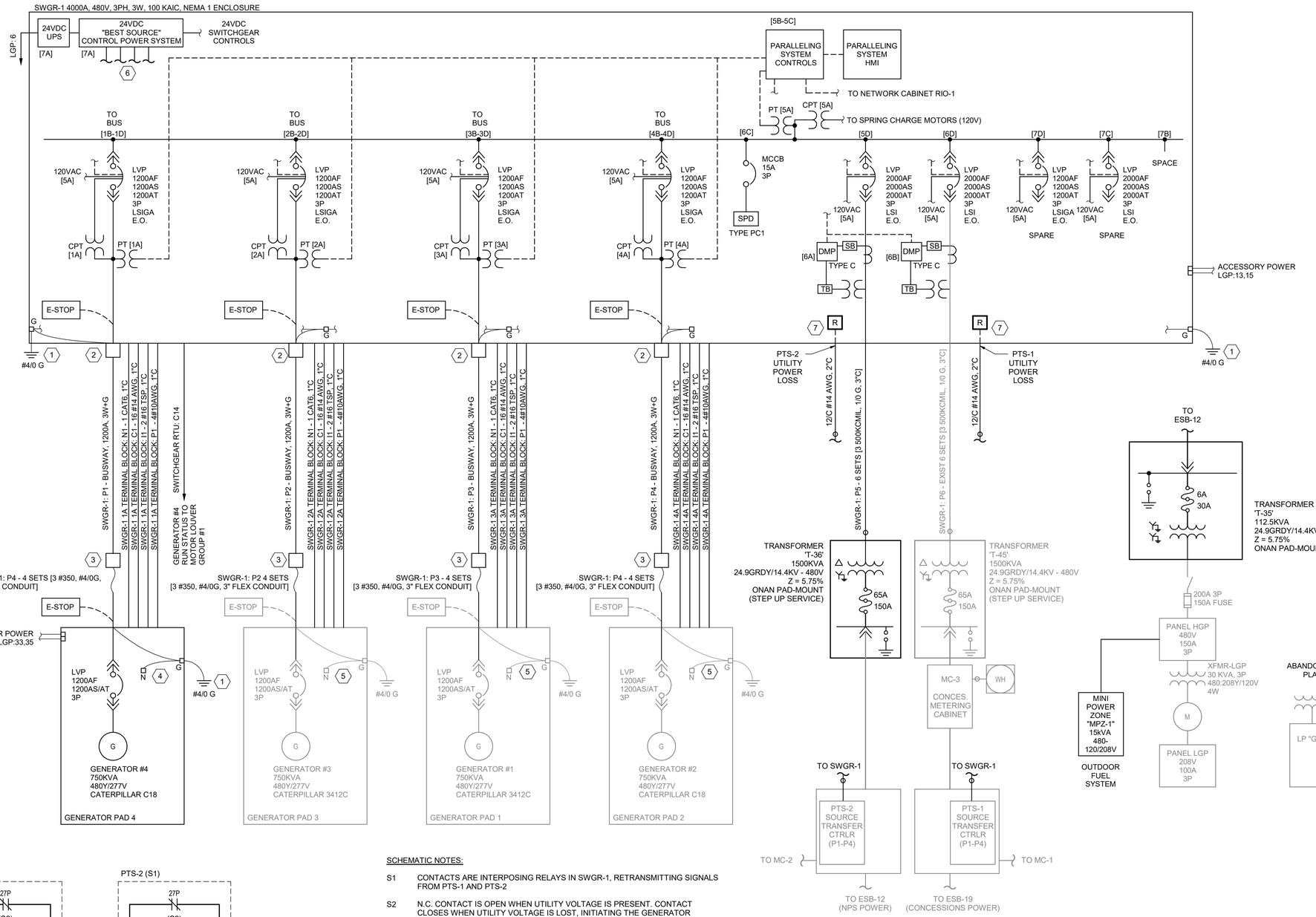

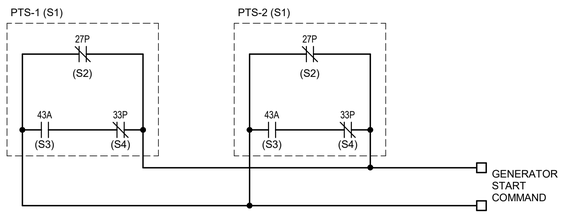

2 092E601 GENERATOR START SIGNALS SCHEMATIC
NO SCALE

SCHEMATIC NOTES:

S1 CONTACTS ARE INTERPOSING RELAYS IN SWGR-1, RETRANSMITTING SIGNALS FROM PTS-1 AND PTS-2

S2 N.C. CONTACT IS OPEN WHEN UTILITY VOLTAGE IS PRESENT. CONTACT CLOSES WHEN UTILITY VOLTAGE IS LOST, INITIATING THE GENERATOR SYSTEM ON LOSS OF UTILITY VOLTAGE. SIGNAL ORIGINATES IN THE REMOTE INDICATION SYSTEM.

S3 N.O. CONTACT IS OPEN WHEN THE SOURCE TRANSFER CONTROLLER IS IN 'MANUAL' POSITION. CONTACT IS CLOSED WHEN TRANSFER CONTROLLER IS IN 'AUTOMATIC' POSITION. THIS PREVENTS A 'START' COMMAND IF THE UTILITY SWITCH IS OPENED MANUALLY WHILE THE TRANFER CONTROLLER IS IN 'MANUAL'. SIGNAL ORIGINATES IN THE REMOTE INDICATION SYSTEM.

S4 N.C. CONTACT IS OPEN WHEN UTILITY SOURCE SWITCH IS CLOSED. CONTACT CLOSES WHEN UTILITY SOURCE IS OPEN, INITIATING THE GENERATOR SYSTEM WHEN THE SWITCH IS OPEN. SIGNAL ORIGINATES AT THE "C9" SWITCH POSITION CONTACTS.

GENERAL NOTES:

A. REFER TO SHEET 092E603 FOR ADDITIONAL SYSTEM CONTROL, INSTRUMENT, AND NETWORK WIRING.

B. REFER TO SHEET 092Y601 FOR SCADA NETWORK BLOCK DIAGRAM.

KEYNOTES

1 BOND TO GROUND BAR.

2 TRANSITION / END CABLE TAP BOX AS REQUIRED FOR BUSWAY ENTRY INTO SWITCHGEAR.

3 TRANSITION / END CABLE TAP BOX AS REQUIRED FOR BUSWAY CONNECTION TO GENERATOR.

4 BOND NEUTRAL TO GROUND AT GENERATOR.

5 CONFIRM NEUTRAL IS BONDED TO GROUND AT GENERATOR. INSTALL NEUTRAL-GROUND BOND IF NOT PRESENT.

6 24VDC CIRCUIT FROM EACH GENERATOR BATTERY SYSTEM TO "BEST SOURCE" CONTROL POWER SYSTEM IN SWITCHGEAR SWGR-1.

7 INSTALL ONE 4PDT INTERPOSING RELAY FOR EACH OF THE FOLLOWING DISCRETE STATUS SIGNALS FROM EACH PRIMARY TRANSFER SWITCH: LEFT SOURCE VOLTAGE, RIGHT SOURCE VOLTAGE, OVERCURRENT LOCKOUT, AUTO-TRANSFER SYSTEM "READY", CONTROLLER MANUAL/AUTOMATIC STATUS, LEFT SOURCE SWITCH POSITION ('b' CONTACT), RIGHT SOURCE SWITCH POSITION ('b' CONTACT). WIRE ONE CONTACT FROM EACH RELAY INTO THE PARALLELING CONTROL SYSTEM FOR STATUS MONITORING AND DISPLAY ON THE SYSTEM HMI. WIRE ADDITIONAL RELAYS FOR GENERATOR "START" CONTROLS AS INDICATED ON THE DIAGRAM ON THIS PAGE.

PRIMARY TRANSFER SWITCH (PTS-1, PTS-2) NOTES:

P1 PRIMARY TRANFER SWITCHES PTS-1 AND PTS-2 ARE S&C ELECTRIC "PMH-9" SWITCHGEAR, EQUIPPED WITH S&C "MICRO-AT" SOURCE TRANSFER CONTROL. SEE PHOTOS #3 AND #4 ON 092E701.

P2 CONTRACTOR SHALL ENGAGE MANUFACTURER FIELD SERVICES TO INSPECT, CLEAN, TEST, AND REPAIR ANY BROKEN OR DEFECTIVE PARTS ON THE PTS-1 AND PTS-2 SWITCHGEAR ASSEMBLIES AND MICRO-AT TRANSFER CONTROLLERS.

P3 MANUFACTURER FIELD SERVICES SHALL VERIFY PROGRAMMING AND CONFIGURATION OF THE MICRO-AT TRANSFER CONTROLLER TO ONLY ALLOW OPEN-TRANSITION SWITCHING. FIELD SERVICES TEAM SHALL ASSIST IN GENERATOR/TRANSFER SYSTEM STARTUP, TESTING, AND COMMISSIONING.

P4 BOTH UNITS ARE EQUIPPED WITH THE "Y4" REMOTE INDICATION FEATURE WHICH INCLUDES FORM-C DRY CONTACTS TO REPORT THESE STATUS CONDITIONS: LEFT SOURCE VOLTAGE, RIGHT SOURCE VOLTAGE, OVERCURRENT LOCKOUT, AUTO-TRANSFER SYSTEM "READY", CONTROLLER MANUAL/AUTOMATIC STATUS. ALL STATUS SIGNALS IN BOTH UNITS SHALL BE WIRED INTO AUXILIARY INTERPOSING RELAYS IN SWITCHGEAR SWGR-1.

SUMMARY SYSTEM SEQUENCE OF OPERATIONS

1. UTILITY POWER FAILS.
2. PTS UNITS SIGNAL GENERATOR SYSTEM TO START.
3. SWGR-1 STARTS AND SYNCHRONIZES ALL AVAILABLE GENERATORS.
4. PTS UNITS SWITCH TO GENERATOR POWER.
5. SWGR-1 MONITORS POWER DEMAND, AND MODULATES QUANTITY OF GENERATORS OPERATING TO FOLLOW SYSTEM POWER DEMAND.
6. UTLITY POWER RESTORED.
7. AFTER A TIME DELAY, PTS UNITS RETRANSFER TO UTILITY POWER USING OPEN-TRANSITION SWITCHING.
8. OPERATING GENERATORS FOLLOW COOL-DOWN SEQUENCE.

REGISTERED PROFESSIONAL ENGINEER OF ARIZONA, U.S.A. CERTIFICATE NO. 58037 TRAVIS T. MOORE Date signed 8/13/26

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO CADD A S-DELGADO TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 092 E601	TITLE OF SHEET POWER DISTRIBUTION CENTER GENERATOR BUILDING ONE-LINE DIAGRAM NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 44 OF 51

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	1	2	3	4	5	6	7
	GENERATOR #4 CONTROL WIRING TERMINAL BLOCK	GENERATOR #3 CONTROL WIRING TERMINAL BLOCK	GENERATOR #1 CONTROL WIRING TERMINAL BLOCK	GENERATOR #2 CONTROL WIRING TERMINAL BLOCK			
A	GENERATOR #4 CPT AND PT	GENERATOR #3 CPT AND PT	GENERATOR #1 CPT AND PT	GENERATOR #2 CPT AND PT	BUS AUTOMATION CPT/PT	N.P.S. METERING PACKAGE	BREAKER OPERATION UPS SYSTEM
B	GENERATOR #4 HMI GEN #4 E-STOP	GENERATOR #3 HMI GEN #3 E-STOP	GENERATOR #1 HMI GEN #1 E-STOP	GENERATOR #2 HMI GEN #2 E-STOP	PARALLELING SYSTEM HMI AND CONTROLS	CONCESSIONS METERING PACKAGE	SPACE
C	GENERATOR #4 AUTOMATION CONTROLS HARDWARE	GENERATOR #3 AUTOMATION CONTROLS HARDWARE	GENERATOR #1 AUTOMATION CONTROLS HARDWARE	GENERATOR #2 AUTOMATION CONTROLS HARDWARE		SURGE PROTECTION DEVICE	SPARE 2000AF
D	GENERATOR #4 CIRCUIT BREAKER INDICATION LIGHTS	GENERATOR #3 CIRCUIT BREAKER INDICATION LIGHTS	GENERATOR #1 CIRCUIT BREAKER INDICATION LIGHTS	GENERATOR #2 CIRCUIT BREAKER INDICATION LIGHTS	N.P.S. FEEDER CIRCUIT BREAKER (TRANSFORMER T-36)	CONCESSIONS FEEDER CIRCUIT BREAKER (TRANSFORMER T-45)	SPARE 1200AF
	22 IN.	22 IN.	22 IN.	22 IN.	22 IN.	22 IN.	22 IN.

1 / 092E602 SWITCHGEAR SWGR-1 FRONT ELEVATION

NO SCALE

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REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO.
58037
TRAVIS T.
MOORE
ARIZONA, U.S.A.

A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

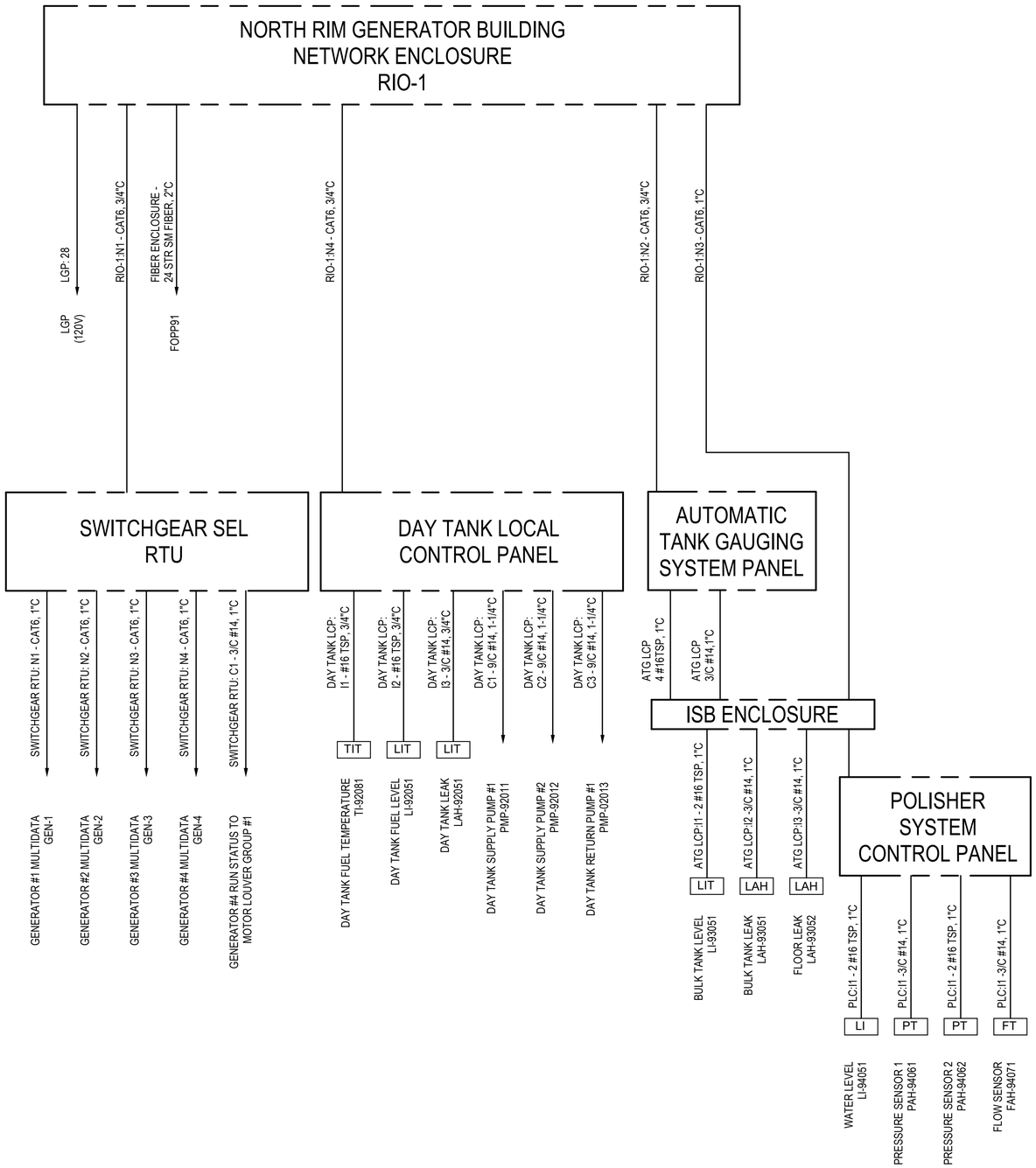
DESIGNED: A. S-DELGADO
DRAWN: A S-DELGADO
TECH. REVIEW: P. DAVILA
DATE: AUG 2026

SUB SHEET NO. 092 E602

TITLE OF SHEET
POWER DISTRIBUTION CENTER
GENERATOR BUILDING
SWITCHGEAR ELEVATION

NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 45 OF 51



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REGISTERED PROFESSIONAL ENGINEER, ELECTRICAL
CERTIFICATE NO.
58037
TRAVIS T. MOORE
Date Signed 4/16/2026
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO DRAWN: A S-DELGADO TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 092 E603	TITLE OF SHEET POWER DISTRIBUTION CENTER GENERATOR BUILDING PLC ONE-LINE DIAGRAM NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 46 OF 51

BRANCH PANEL: HGP

LOCATION: SWITCHGEAR & SHOP ROOM 101
SUPPLY FROM: T-35
MOUNTING: SURFACE
ENCLOSURE: NEMA 1

VOLTS: 480Y/277
PHASES: 3
WIRES: 4
INTEGRAL SPD:

A.I.C. RATING:
MAINS TYPE: MCB
MAINS RATING: 150 A
SERVICE ENTRANCE LABEL:

WIRE & CONDUIT	CKT	CIRCUIT DESCRIPTION	CLASS	TRIP	P	A		B		C		P	TRIP	CLASS	CIRCUIT DESCRIPTION	CKT	WIRE & CONDUIT
	1					3333.3	8333.3									2	
3 #12, #12G, 3/4"C	3	UNIT HEATER 2	Heating	15 A	3			3333.3	8333.3			3	40 A	Heating	UNIT HEATER 1	4	3 #8, #10G, 1"C
	5									3333.3	8333.3					6	
	7					8333.3	--									8	
3 #8, #10G, 1"C	9	UNIT HEATER 3	Heating	40 A	3			8333.3	--			3	--	--	SPACE	10	
	11									8333.3	--					12	
	13					333.3	8526.7									14	
3 #10, #10G, 2"C	15	XFMR-MPZ-1	Spare; HTG; FUEL	25 A	3			333.3	7126.7			3	40 A	Spare; HTG; HVAC; LTS;...	XFMR-LGP	16	4 #2, #8G, 1-1/2"C
	17									333.3	8076.7					18	
	19	SPACE	--	--	1	--	--					1	--	--	SPACE	20	
	21	SPACE	--	--	1			--	--			1	--	--	SPACE	22	
	23	SPACE	--	--	1					--	--	1	--	--	SPACE	24	
	25	SPACE	--	--	1	--	--					1	--	--	SPACE	26	
	27	SPACE	--	--	1			--	--			1	--	--	SPACE	28	
	29	SPACE	--	--	1					--	--	1	--	--	SPACE	30	
	31	SPACE	--	--	1	--	--					1	--	--	SPACE	32	
	33	SPACE	--	--	1			--	--			1	--	--	SPACE	34	
	35	SPACE	--	--	1					--	--	1	--	--	SPACE	36	
	37	SPACE	--	--	1	--	--					1	--	--	SPACE	38	
	39	SPACE	--	--	1			--	--			1	--	--	SPACE	40	
	41	SPACE	--	--	1					--	--	1	--	--	SPACE	42	
				TOTAL LOAD:		27605 VA		26351 VA		27200 VA							
				TOTAL AMPS:		100 A		95 A		99 A							

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND
HTG	3500 VA	100.00%	3500 VA
HVAC	1980 VA	100.00%	1980 VA
LTS	4550 VA	125.00%	5688 VA
GEN	10500 VA	97.62%	10250 VA
REC	2700 VA	100.00%	2700 VA
IC	500 VA	100.00%	500 VA
PWR	1000 VA	100.00%	1000 VA
Heating	60000 VA	100.00%	60000 VA
FUEL	0 VA	0.00%	0 VA

PANEL TOTALS	
TOTAL CONNECTED LOAD:	**81152 VA**
TOTAL ESTIMATED DEMAND:	**81948 VA**
TOTAL CONNECTED:	**98 A**
TOTAL ESTIMATED DEMAND	**99 A**
SPARE AMPERAGE:	**25.00%**
TOTAL ESTIMATED DEMAND + SPARE:	**123 A**

NOTES:

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REGISTERED PROFESSIONAL ENGINEER ELECTRONICALLY
CERTIFICATE NO.
58037
TRAVIS T.
MOORE
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO CADD A S-DELGADO TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 092 E604	TITLE OF SHEET POWER DISTRIBUTION CENTER GENERATOR BUILDING PANEL SCHEDULES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 47 OF 51

BRANCH PANEL: LGP

LOCATION: SWITCHGEAR & SHOP ROOM 101
SUPPLY FROM: XFMR LGP
MOUNTING: WALL MOUNTED
ENCLOSURE: NEMA 1

VOLTS: 208Y/120
PHASES: 3
WIRES: 4
INTEGRAL SPD: No

A.I.C. RATING:
MAINS TYPE: MCB
MAINS RATING: 100 A
SERVICE ENTRANCE LABEL: No

WIRE & CONDUIT	CKT	CIRCUIT DESCRIPTION	CLASS	TRIP	P	A		B		C		P	TRIP	CLASS	CIRCUIT DESCRIPTION	CKT	WIRE & CONDUIT
2 #12, #12G, 3/4"C	1	SWITCHGEAR LIGHTING	LTS	20 A	1	1200.0	900.0					1	20 A	HVAC; REC	BATHROOM RECEPTACLES, EF-1	2	2 #12, #12G, 3/4"C
2 #12, #12G, 3/4"C	3	GENERATOR LIGHTING	LTS	20 A	1			1350.0	0.0			1	20 A	HVAC	WH-1	4	2 #12, #12G, 3/4"C
2 #12, #12G, 3/4"C	5	PLENUM LIGHTING	LTS	20 A	1					1200.0	1000.0	1	20 A	PWR	SWGR-1 CONTROL POWER	6	2 #12, #12G, 3/4"C
2 #12, #12G, 3/4"C	7	REC	REC	20 A	1	900.0	0.0					1	20 A	--	SPARE	8	
2 #12, #12G, 3/4"C	9	REC	REC	20 A	1			900.0	1250.0			2	20 A	GEN	GENERATOR #1 HTR AND CHARGER	10	2 #12, #12G, 3/4"C
2 #12, #12G, 3/4"C	11	EF-2	HVAC	20 A	1					0.0	1250.0					12	
	13	SWGR-1 AUX POWER 1	--	20 A	1	0.0	1250.0					2	20 A	GEN	GENERATOR #2 HTR AND CHARGER	14	2 #12, #12G, 3/4"C
	15	SWGR-1 AUX POWER 2	--	20 A	1			0.0	1250.0							16	
2 #12, #12G, 3/4"C	17	LOUVER GROUP 1	HVAC	15 A	1					0.0	1250.0	2	20 A	GEN	GENERATOR #3 HTR AND CHARGER	18	2 #12, #12G, 3/4"C
2 #12, #12G, 3/4"C	19	LOUVER GROUP 2	HVAC	15 A	1	0.0	1250.0									20	
2 #12, #12G, 3/4"C	21	LOUVER GROUP 3	HVAC	15 A	1			0.0	0.0			1	20 A	--	REC	22	2 #12, #12G, 3/4"C
2 #12, #12G, 3/4"C	23	FUEL DISPENSER	HVAC	20 A	2					50.0	200.0	1	20 A	LTS	EMERGENCY LIGHTING	24	2 #12, #12G, 3/4"C
	25					50.0	600.0					1	20 A	LTS	OUTDOOR FLOODLIGHTS	26	2 #12, #12G, 3/4"C
3 #12, #12G, 3/4"C	27							626.7	500.0			1	20 A	IC	RIO-1	28	2 #12, #12G, 3/4"C
	29	FUEL PUMP (*)	HVAC	20 A	3					626.7	0.0	3	20 A	--	SPARE	30	
	31					626.7	0.0									32	
2 #12, #12G, 3/4"C	33	GENERATOR #4 HTR AND CHARGER	GEN	20 A	2			1250.0	0.0							34	
	35									1250.0	1250.0	2	20 A	HTG	TANK WEATER HEATER	36	2 #12, #12G, 3/4"C
2 #12, #12G, 3/4"C	37	GENERATOR #1 BATTERY CHARGER	GEN	15 A	1	500.0	1250.0									38	
2 #12, #12G, 3/4"C	39	UNL	--	20 A	1			0.0	0.0			1	20 A	--	SPARE	40	
2 #12, #12G, 3/4"C	41	STP	--	20 A	1					0.0	0.0	1	20 A	--	SPARE	42	
				TOTAL LOAD:		8527 VA		7127 VA		8077 VA							
				TOTAL AMPS:		72 A		59 A		69 A							

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND
HTG	2500 VA	100.00%	2500 VA
HVAC	1980 VA	100.00%	1980 VA
LTS	4550 VA	125.00%	5688 VA
GEN	10500 VA	97.62%	10250 VA
REC	2700 VA	100.00%	2700 VA
IC	500 VA	100.00%	500 VA
PWR	1000 VA	100.00%	1000 VA

PANEL TOTALS	
TOTAL CONNECTED LOAD:	23730 VA
TOTAL ESTIMATED DEMAND:	24618 VA
TOTAL CONNECTED:	66 A
TOTAL ESTIMATED DEMAND	68 A
SPARE AMPERAGE:	25.00%
TOTAL ESTIMATED DEMAND + SPARE:	85 A

NOTES:
(*) SWITCHED NEUTRAL

BRANCH PANEL: MPZ-1

LOCATION: EXTERIOR
SUPPLY FROM: XFMR-MPZ-1
MOUNTING: SURFACE
ENCLOSURE: NEMA 4X

VOLTS: 208Y/120
PHASES: 3
WIRES: 4
INTEGRAL SPD: No

PRIMARY A.I.C. RATING: 65kAIC
PRIMARY BREAKER: MCB
PRIMARY BREAKER RATING: 25A*
TRANSFORMER RATING: 15kVA

SECONDARY A.I.C. RATING: 10 kAIC
SECONDARY BREAKER: MCB
SECONDARY BREAKER RATING 60 A
SERVICE ENTRANCE LABEL: No

WIRE & CONDUIT	CKT	CIRCUIT DESCRIPTION	CLASS	TRIP	P	A		B		C		P	TRIP	CLASS	CIRCUIT DESCRIPTION	CKT	WIRE & CONDUIT
2#12, #12 G, 3/4"C	1	AUTOMATIC TANK GAUGE PNL	FUEL	20 A	1	0.0	0.0					1	20 A	FUEL	DAY TANK CONTROL PANEL	2	2#12, #12 G, 3/4"C
2#12, #12 G, 1"C	3	POLISHER SYSTEM CONTROL PNL	FUEL	20 A	1			0.0	0.0			1	20 A	FUEL	DAY TANK SUPPLY PUMP #2	4	2#12, #12 G, 3/4"C
2#12, #12 G, 3/4"C	5	DAY TANK SUPPLY PUMP #3	FUEL	20 A	1					0.0	333.3					6	
2#12, #12 G, 3/4"C	7	DAY TANK SUPPLY PUMP #1	FUEL	20 A	1	0.0	333.3					3	20 A	HTG	HEAT TRACE PANEL	8	3#12, #12 G, 3/4"C
	9	SPARE	--	20 A	1			0.0	333.3							10	
	11	SPARE	--	20 A	1					0.0	0.0	1	20 A	--	SPARE	12	
	13	SPARE	--	20 A	1	0.0	0.0					1	20 A	--	SPARE	14	
	15	SPARE	--	20 A	1			0.0	0.0			1	20 A	--	SPARE	16	
	17	SPARE	--	20 A	1					0.0	0.0	1	20 A	--	SPARE	18	
	19	SPARE	--	20 A	1	0.0	0.0					1	20 A	--	SPARE	20	
	21	SPARE	--	20 A	1			0.0	0.0			1	20 A	--	SPARE	22	
	23	SPARE	--	20 A	1					0.0	0.0	1	20 A	--	SPARE	24	
				TOTAL LOAD:		333 VA		333 VA		333 VA							
				TOTAL AMPS:		3 A		3 A		3 A							

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND
HTG	1000 VA	100.00%	1000 VA
FUEL	0 VA	0.00%	0 VA

PANEL TOTALS	
TOTAL CONNECTED LOAD:	1000 VA
TOTAL ESTIMATED DEMAND:	1000 VA
TOTAL CONNECTED:	3 A
TOTAL ESTIMATED DEMAND	3 A
SPARE AMPERAGE:	25.00%
TOTAL ESTIMATED DEMAND + SPARE:	3 A

NOTES:
NOT ALL EQUIPMENT SHOWN. UTILIZE SPARE BREAKERS AS NECESSARY FOR OTHER MANUFACTURER-SUPPLIED EQUIPMENT.

*ESTIMATED PRIMARY AND SECONDARY BREAKER SIZES. FINAL SIZES TO BE DETERMINED BY MPZ MANUFACTURER.

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO. 58037
TRAVIS T. MOORE
DATE SIGNED 1/AUG/2026
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO DRAWN: A S-DELGADO TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 092 E605	TITLE OF SHEET POWER DISTRIBUTION CENTER GENERATOR BUILDING PANEL SCHEDULES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 48 OF 51

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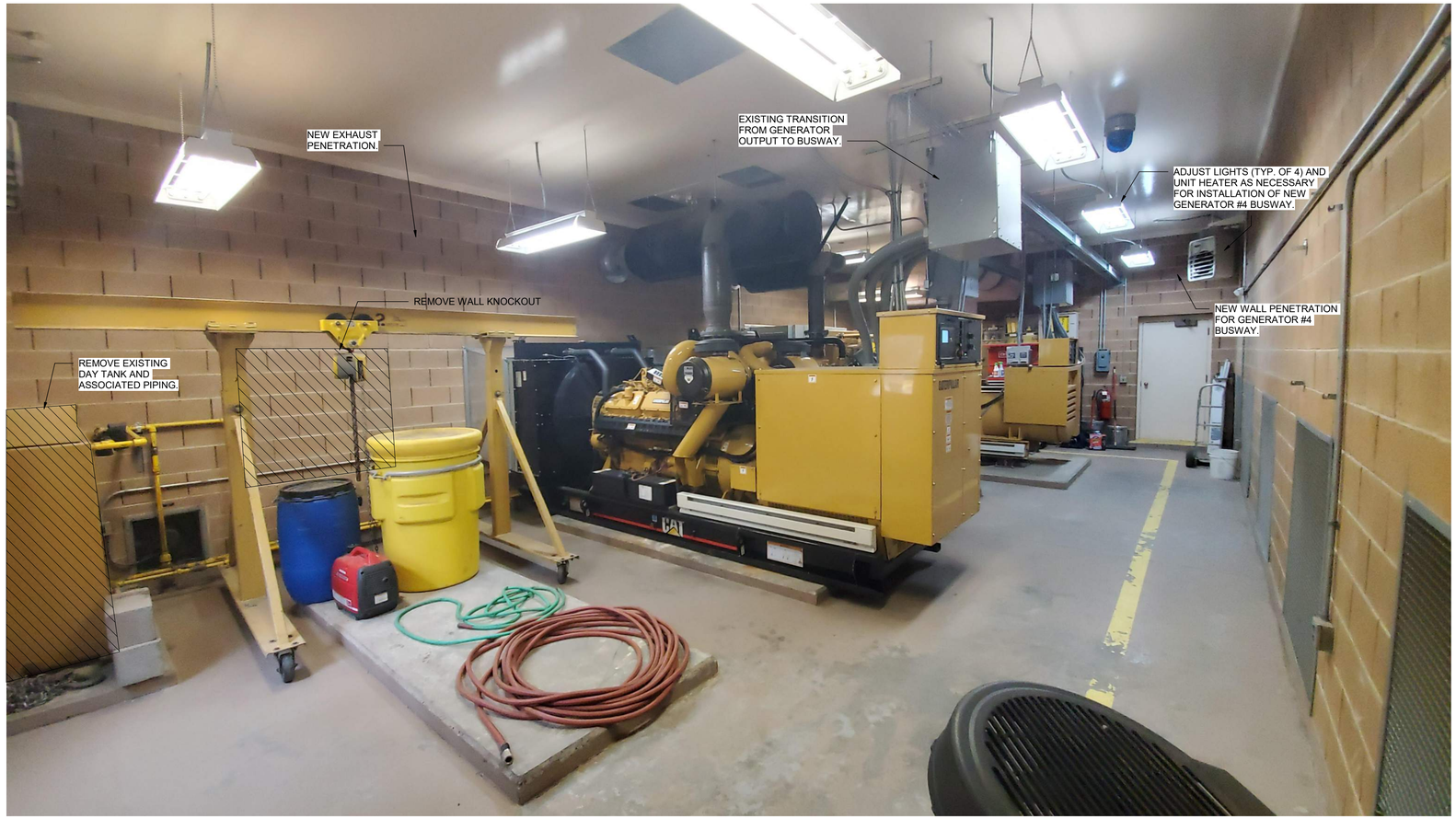
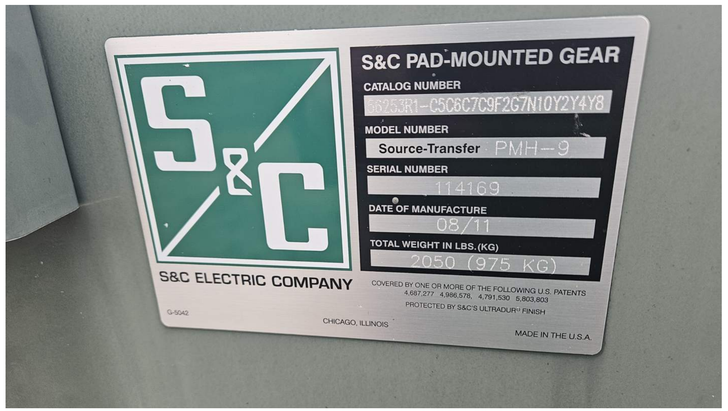


1 092E701 PHOTOGRAPH - GENERATOR ROOM
NO SCALE

2 092E701 PHOTOGRAPH - WIREWAY
NO SCALE



3 092E701 PHOTOGRAPH - PTS-2 NAMEPLATE (SIMILAR FOR PTS-1)
NO SCALE

4 092E701 PHOTOGRAPH - PTS-2 TRANSFER EQUIPMENT (SIMILAR FOR PTS-1)
NO SCALE

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 58037 — TRAVIS T. MOORE — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO DRAWN: A S-DELGADO TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 092 E701	TITLE OF SHEET POWER DISTRIBUTION CENTER GENERATOR BUILDING PHOTOGRAPHS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 49 OF 51

Autodesk Docs://10350545_NPS_GRCA_N_Rim_Utility_2026/10350545_092_GENBLDG_E.rvt
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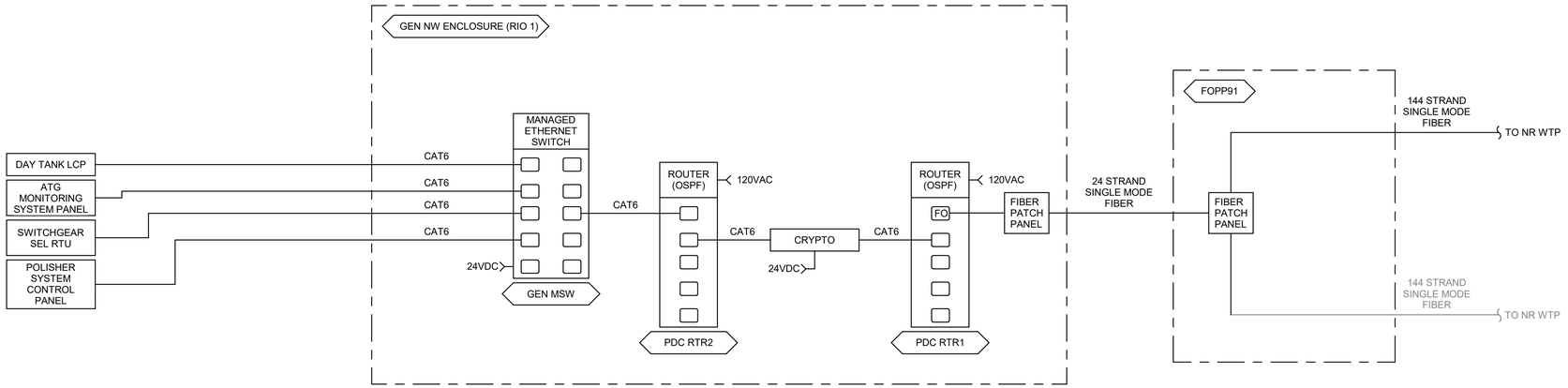

Registered Professional Engineer (Electrical)
CERTIFICATE NO.
85305
JOSEPHINE A BRAINARD
Date Signed 4/17/26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: P. KEMPF TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 092 Y601	TITLE OF SHEET POWER DISTRIBUTION CENTER SCADA NETWORK BLOCK DIAGRAM NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 50 OF 51

Autodesk Docs://10350545_NPS_GRCA_N_Rim_Utility_2026/10350545_092_GENBLDG_E.rvt
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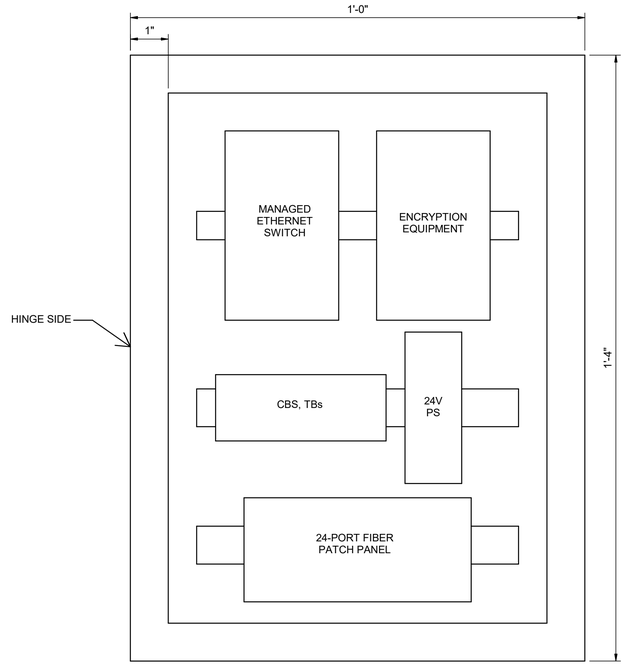
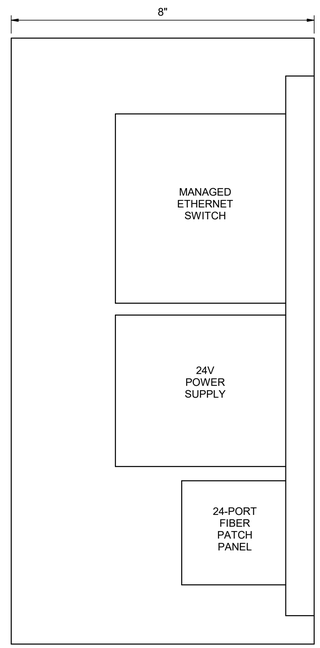


1 - **GEN NW ENCLOSURE FRONT INTERIOR ELEVATION**
NOT TO SCALE



2 - **GEN NW ENCLOSURE SIDE INTERIOR ELEVATION**
NOT TO SCALE

Autodesk Docs://10350545_NPS_GRCA_N_Rim_Utility_2026/10350545_092_GENBLDG_E.rvt
4/17/2026 10:40:07 AM

Registered Professional Engineer (Electrical)
CERTIFICATE NO.
85305
JOSEPHINE A BRAINARD
Date Signed 4/17/26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: P. KEMPF TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 092 Y620	TITLE OF SHEET POWER DISTRIBUTION CENTER GENERATOR NETWORK ENCLOSURE LAYOUT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 51 OF 51